SHARP LOCAL PROPAGATION OF CHAOS FOR DIFFUSIVE LOG/RIESZ GAS DYNAMICS IN THE HILBERT–SCHMIDT REGIME

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Abstract. We prove sharp bounds on the correlations of diffusive particle systems with repulsive logarithmic and Riesz interactions in $\mathbb{R}^{\mathsf{d}}$, $\mathsf{d} \geq 2$, in the Hilbert–Schmidt regime $0 \leq \mathsf{s} < \mathsf{d}/2$. Starting from a modulated Gibbs measure at any fixed inverse temperature greater than half that of the dynamics, the relative entropy and chi-square divergence of the k-particle marginal from the mean-field product measure are bounded by Ck^2/N^2, and its total variation distance by Ck/N. We obtain finite-time estimates under one-body regularity and Poincaré bounds and a sufficient bound on the time integral of $\|\nabla u_t\|_{L^\infty}$, where u_t is the mean-field continuity velocity. For small logarithmic perturbations of a confined thermal equilibrium measure, the estimates are uniform in time at every fixed positive inverse temperature. The confinement may be anisotropic and need not be globally convex. The rates are optimal, as shown by the stationary Gibbs subclass. Self-similar coordinates give corresponding unconfined results for dilated thermal equilibrium measures and sufficiently diffuse Gaussian perturbations. The proof combines partition-function estimates with a decreasing-exponent estimate relative to a moving Gibbs measure and weighted estimates for the first marginal. We also construct a four-dimensional whole-space example with a stationary mean-field measure and independent initial data. The full particle measure has bounded relative entropy but a chi-square divergence that tends to infinity with the particle number at a fixed positive time, despite zero integrability expenditure. Finally, for the four-dimensional periodic logarithmic gas, we construct a smooth trajectory along which the full relative entropy remains bounded but the terminal full chi-square divergence tends to infinity on a fixed interval of time. Thus, even when the particle system starts from the modulated Gibbs measure associated with the initial mean-field measure and the mean-field trajectory is smooth, an N-uniform full-particle chi-square bound need not hold on an arbitrary fixed time interval.

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1. Introduction and main results

The purpose of this paper is to quantify the correlations generated by the overdamped dynamics of a logarithmic or Riesz gas. The mean-field equation describes the limiting one-particle density. We ask how accurately its product law approximates a labeled group of particles, with an estimate that keeps track of both the size of the group and the elapsed time. A sharp dependence on the group size determines how many particles remain approximately independent. Its persistence in time depends on the evolution of the mean-field measure and on the correlations present initially. We establish quantitative estimates under conditions on both, and construct two examples showing why those conditions matter when the estimate includes the full particle law.

The particle dynamics combine singular pairwise repulsion, an external force, and independent Brownian noise. Fix $\mathsf{d} \geq 2$, $\beta > 0$, and $0 \leq \mathsf{s} < \mathsf{d}/2$, and set

$$\mathsf{g}(x) = \begin{cases} -\log|x|, & \mathsf{s} = 0, \\ |x|^{-\mathsf{s}}/\mathsf{s}, & \mathsf{s} > 0. \end{cases} \tag{1.1}$$

Logarithmic and Riesz interactions provide basic models of long-range particle systems. The Coulomb cases describe electrostatic repulsion, while the logarithmic interaction also appears in

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random matrix theory and two-dimensional vortex dynamics. Varying the Riesz exponent permits one to study how the strength of the singularity affects correlations and the mean-field limit. We refer to [Ser26, RS23, RS24] for further background and references.

For $N \geq 1$, write $[N] := \{1, \dots, N\}$ and $X_N := (x_1, \dots, x_N)$ for a configuration. The mean-field Hamiltonian is

$$\mathcal{H}_N(X_N) = \sum_{i=1}^{N} V(x_i) + \frac{1}{N} \sum_{i<j} \mathsf{g}(x_i - x_j). \tag{1.2}$$

We consider its overdamped Langevin dynamics,

$$\mathrm{d}x_i = -\left(\nabla V(x_i) + \frac{1}{N} \sum_{j \neq i} \nabla \mathsf{g}(x_i - x_j) \right) \mathrm{d}t + \sqrt{2/\beta}\, \mathrm{d}B_i, \qquad 1 \leq i \leq N, \tag{1.3}$$

where $(B_i)_{i \in [N]}$ are independent Brownian motions. We write $X_N(t) := (x_1(t), \dots, x_N(t))$ for the full random configuration. The inverse temperature β is independent of N. In this scaling, the mean-field interaction and Brownian diffusion both contribute to the limiting equation; this is the diffusive temperature regime of [DGR26a]. The class includes the planar Coulomb gas and the three-dimensional Coulomb gas. More generally, the restriction $\mathsf{s} < \mathsf{d}/2$ is the local square-integrability condition on the interaction potential. Square integrability of the force is not assumed.

The corresponding mean-field equation is

$$\partial_t \mu = \beta^{-1} \Delta \mu + \operatorname{div}(\mu(\nabla V + \nabla \mathsf{g} * \mu)). \tag{1.4}$$

It replaces the empirical interaction force in (1.3) by the force generated by the one-particle density. Qualitative propagation of chaos asks whether each fixed labeled marginal converges to the corresponding product of mean-field measures as $N \to \infty$. Quantitative estimates must also describe the residual dependence at finite N and the time intervals on which it remains small; see, for example, [Lac23]. Our main results concern confined whole-space dynamics, and a self-similar change of variables gives explicit unconfined classes. The sharp local estimates also carry over, mutatis mutandis, to the periodic setting, where the absence of spatial tails simplifies the proof; see remark 6.5. The hypotheses on V and the solution of (1.4) are stated with each result.

Initial preparation. For the sharp local chaos estimates, we choose an initial measure that incorporates the singular pair repulsion around a prescribed probability density. This permits nonstationary mean-field trajectories while preserving quantitative control of the initial correlations. The measure is a modulated Gibbs measure; at a matching thermal equilibrium it reduces to the canonical Gibbs ensemble. See the work of Serfaty and the third author [RS25] for its role in the modulated-free-energy method, and [Ser26] for a broader account of modulated Gibbs measures and the modulated-energy framework. Its normalization and equilibrium correlations are studied in [DGR26a, DGR26b].

For a probability density μ, denote its interaction potential and energy, the modulated energy, and the modulated Gibbs measure, respectively, by

$$h_\mu = \mathsf{g} * \mu, \qquad I_\mu = \frac{1}{2} \int h_\mu \, \mathrm{d}\mu, \tag{1.5}$$

$$N\mathsf{F}_N(X_N, \mu) = \frac{1}{N} \sum_{i<j} \mathsf{g}(x_i - x_j) - \sum_i h_\mu(x_i) + N I_\mu, \tag{1.6}$$

$$\mathrm{d}\mathbb{Q}_{N,b}(\mu) = G_{N,b}(\mu)\, \mathrm{d}\mu^{\otimes N}, \qquad G_{N,b}(\mu) = \mathsf{K}_{N,b}(\mu)^{-1} e^{-bN\mathsf{F}_N(X_N,\mu)}, \qquad b \geq 0. \tag{1.7}$$

The normalizing constant $\mathsf{K}_{N,b}(\mu)$ is the modulated partition function. The subtracted one-body and constant terms in (1.6) remove the mean-field contribution of the prescribed background. The normalizations are those of [DGR26a, DGR26b].

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For an initial inverse temperature $\beta_0 \geq 0$, consider the preparation

$$P_{N,0} = \mathbb{Q}_{N,\beta_0}(\mu_0). \tag{1.8}$$

The parameter β_0 controls only the initial correlations; both the particle and mean-field dynamics are at inverse temperature β. The choice $\beta_0 = \beta$ is allowed but is not required. The first marginal of (1.8) need not equal μ_0, and proposition 5.1 supplies the required initial one-particle estimate. The actual law at positive time need not be $\mathbb{Q}_{N,\beta}(\mu_t)$: the evolving mean-field measure determines a reference Gibbs law, not the solution of the particle forward equation. The proof estimates the difference between these laws and then its effect on the marginals. Proposition 1.6 explains why the initial correlations matter when the conclusion includes $k = N$; that example starts from independent and identically distributed (iid) samples.

The initial configuration is independent of the Brownian motions. It is almost surely pairwise distinct because (1.8) is absolutely continuous with respect to a product density. Under the bounded-Hessian assumption on V used below, proposition B.1 gives a pathwise unique global strong solution of (1.3), with no collisions on finite time intervals. We write $P_{N,t}$ for its law and $P_{N,t}^{(k)}$ for its labeled k-particle marginal.

1.1. **Local, mesoscopic, and full-particle information.** We measure the difference between $P_{N,t}^{(k)}$ and $\mu_t^{\otimes k}$ in relative entropy, chi-square divergence, and total variation. For probability measures $P \ll Q$, these are defined by

$$\mathsf{H}(P \mid Q) := \int \log\left(\frac{\mathrm{d}P}{\mathrm{d}Q}\right) \mathrm{d}P, \tag{1.9}$$

$$\chi^2(P \mid Q) := \int \left|\frac{\mathrm{d}P}{\mathrm{d}Q} - 1\right|^2 \mathrm{d}Q, \tag{1.10}$$

$$\mathrm{TV}(P, Q) := \sup_A |P(A) - Q(A)| = \frac{1}{2}\int \left|\frac{\mathrm{d}P}{\mathrm{d}Q} - 1\right| \mathrm{d}Q, \tag{1.11}$$

where the supremum is over measurable sets. When $P \not\ll Q$, the first two quantities are $+\infty$, while the supremum still defines total variation. Relative entropy is unnormalized. A bound on the full relative entropy does not by itself give the sharp marginal rate. Indeed, entropy subadditivity and exchangeability give

$$\mathsf{H}(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) \leq \frac{2k}{N}\mathsf{H}(P_{N,t} \mid \mu_t^{\otimes N}), \qquad 1 \leq k \leq N. \tag{1.12}$$

Thus, an order-one full entropy bound yields only an order-k/N estimate through (1.12). The estimates below have the sharper order k^2/N^2 in relative entropy and chi-square divergence, and order k/N in total variation.

For fixed k, these are local size-of-chaos estimates. They also imply convergence for every mesoscopic sequence $k = k_N \to \infty$ with $k_N/N \to 0$. At $k = N$, the same inequalities have a different meaning: they bound the full density in $L^2(\mu_t^{\otimes N})$, without asserting convergence of the full law to a product. Our sharp local estimates hold for all $1 \leq k \leq N$ under their stated hypotheses.

The full-particle conclusion makes the role of the initial law and the time restriction particularly visible. Proposition 1.6 shows that independent initialization can lead to an unbounded full chi-square at a fixed later time even when the mean-field measure is stationary. Theorem 1.7 preserves the Gibbs initialization but shows that smoothness of a nonstationary trajectory does not suffice on an arbitrary finite interval. Both examples have bounded full relative entropy. They concern the stronger full-particle estimate, not a failure of fixed-marginal propagation of chaos.

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1.2. **Finite-time control along a prescribed trajectory.** We first give a finite-time criterion in terms of the prescribed mean-field solution and the static partition and transport-commutator estimates proved in [DGR26a]. Set

$$B_t = -\nabla V - \nabla \mathsf{g} * \mu_t \tag{1.13}$$

and

$$u_t = -\nabla V - \nabla \mathsf{g} * \mu_t - \beta^{-1} \nabla \log \mu_t. \tag{1.14}$$

Thus, B_t is the drift of the nonlinear diffusion, and (1.4) is also the continuity equation with velocity u_t.

Theorem 1.1 (Finite-time sharp local estimate under an integrated condition)**.** *Fix $\beta > 0$ and $\beta_0 > \beta/2$. Let $V \in C^3_{\mathrm{loc}}(\mathbb{R}^{\mathsf{d}})$ satisfy*

$$\sup_{x \in \mathbb{R}^{\mathsf{d}}} \frac{|\nabla V(x)|}{1+|x|} + \|\nabla^2 V\|_{L^\infty(\mathbb{R}^{\mathsf{d}})} + \|\nabla^3 V\|_{L^\infty(\mathbb{R}^{\mathsf{d}})} < \infty. \tag{1.15}$$

Let μ_t be a smooth positive solution of (1.4) on $[0,T]$. Assume the following bounds, with constants independent of $t \in [0,T]$.

(i) Density and logarithmic Hessian. *There are $M, a > 0$ and $L < \infty$ such that*

$$0 < \mu_t(x) \le M e^{-a|x|^2}, \qquad x \in \mathbb{R}^{\mathsf{d}}, \tag{1.16}$$

and, writing $U_t = -\log \mu_t$,

$$-L\mathrm{Id} \le \nabla^2 U_t(x) \le L\mathrm{Id}, \qquad x \in \mathbb{R}^{\mathsf{d}}. \tag{1.17}$$

(ii) One-body Poincaré inequality. *There is $C_{\mathrm{P}} < \infty$ such that*

$$\int \left| \varphi - \int \varphi \, \mathrm{d}\mu_t \right|^2 \mathrm{d}\mu_t \le C_{\mathrm{P}} \int |\nabla \varphi|^2 \, \mathrm{d}\mu_t, \qquad \varphi \in C_c^\infty(\mathbb{R}^{\mathsf{d}}). \tag{1.18}$$

(iii) Drift derivatives. *There is $C_B < \infty$ such that*

$$\|\nabla B_t\|_{L^\infty} + \|\nabla^2 B_t\|_{L^\infty} \le C_B. \tag{1.19}$$

(iv) Static estimates. *The estimates of proposition 2.1 hold uniformly along μ_t. Let $\kappa > 0$ be a common commutator coefficient in (2.7).*

If the integrability expenditure along the mean-field trajectory *satisfies*

$$A_\mu(T) := \frac{\beta}{\kappa} \int_0^T \|\nabla u_t\|_{L^\infty} \, \mathrm{d}t < \log\left(2 \min\left\{ 1, \frac{\beta_0}{\beta} \right\} \right), \tag{1.20}$$

then the solution of (1.3) with preparation (1.8) satisfies, for every $N \ge 2$ and $1 \le k \le N$,

$$\sup_{0 \le t \le T} \left\{ \chi^2(P^{(k)}_{N,t} \mid \mu_t^{\otimes k}) + \mathsf{H}(P^{(k)}_{N,t} \mid \mu_t^{\otimes k}) \right\} \le C_T \frac{k^2}{N^2}, \tag{1.21}$$

$$\sup_{0 \le t \le T} \mathrm{TV}(P^{(k)}_{N,t}, \mu_t^{\otimes k}) \le C_T \frac{k}{N}. \tag{1.22}$$

The constant depends only on the preceding one-body bounds, the static constants, β_0, the model parameters, and the positive gap in (1.20).

Assumption (iv) follows from the Gaussian upper bound in (i) by proposition 2.1; we state it separately to identify the commutator coefficient entering (1.20). We have not sought optimal regularity or decay assumptions on the mean-field solution. Extensions under weaker one-body hypotheses are left to the interested reader.

The right-hand side of (1.20) is positive precisely because $\beta_0 > \beta/2$. When $\beta_0 \ge \beta$, it reduces to $\log 2$. This sufficient condition is not claimed sharp. Its origin is the loss of integrability in

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the decreasing-exponent estimate of Section 3, together with the comparison between the initial Gibbs measure and the moving reference. The theorem does not require that μ_t remain close to an equilibrium. We next give a class of initial measures for which the required bounds hold on $[0,\infty)$.

1.3. **Uniform-in-time control near a confined thermal equilibrium measure.** We say that the confinement force is *asymptotically linear with a positive-definite limiting matrix* if there exist $A = A^{\mathsf{T}} > 0$ and $b \in \mathbb{R}^{\mathsf{d}}$ such that

$$\widetilde{V}(x) := V(x) - \tfrac{1}{2} x \cdot Ax - b \cdot x \tag{1.23}$$

has a smooth gradient, all derivatives of $\nabla\widetilde{V}$ are bounded, and

$$\lim_{R\to\infty} \sup_{|x|\geq R} |\nabla^{j+1}\widetilde{V}(x)| = 0 \qquad \text{for every } j \geq 0. \tag{1.24}$$

Writing λ_A for the smallest eigenvalue of A, these assumptions imply

$$V(x) = \frac{1}{2} x \cdot Ax + b \cdot x + o(|x|), \tag{1.25}$$

$$\nabla^2 V(x) \longrightarrow A, \tag{1.26}$$

$$\nabla^2 V(x) \geq \frac{\lambda_A}{2}\mathrm{Id} \qquad \text{for } |x| \geq R_V \tag{1.27}$$

for some $R_V < \infty$. The equations are invariant under a common spatial translation, and adding a constant to V changes neither equation. We may therefore take $b = 0$ and $V(0) = 0$ in the proofs and continue to write V for the resulting confinement. The class includes anisotropic positive quadratic confinements, arbitrary smooth compactly supported perturbations of them, and unbounded sublinear perturbations such as a multiple of $\log(1+|x|^2)$. No global convexity assumption is imposed. The finite-time theorem uses less: the asymptotically linear-force condition is used only to verify uniform-in-time logarithmic stability by comparison with the anisotropic Ornstein–Uhlenbeck semigroup.

For a confinement satisfying (1.23)–(1.24), the thermal probability density is the unique solution

$$\pi(x) = Z_\pi^{-1} \exp\left[-\beta\left(V(x) + (\mathsf{g} * \pi)(x)\right)\right]. \tag{1.28}$$

Its existence, smoothness, Gaussian tail bounds, and one-body Poincaré inequality are recorded with proofs in lemma A.1. At the matching inverse temperature, $\mathbb{Q}_{N,\beta}(\pi)$ is the usual canonical Gibbs ensemble.

The logarithmic perturbation class. A logarithmic perturbation is normalized as

$$\mu_0 = \frac{\pi e^{z_0}}{\int \pi e^{z_0}}. \tag{1.29}$$

The simplest class covered below is

$$z_0(x) = \ell \cdot x + \varphi(x), \qquad \ell \in \mathbb{R}^{\mathsf{d}}, \quad \varphi \in C_c^\infty(\mathbb{R}^{\mathsf{d}}). \tag{1.30}$$

In particular, we allow unbounded logarithmic relative densities and do not assume two-sided boundedness of μ_0/π.

We also use the completion of this class. For an integer $m \geq 3$, let $\mathcal{X}^m$ be the completion of smooth affine functions plus C_c^∞, modulo constants, in the norm

$$\|z\|_{\mathcal{X}^m} = \sum_{j=1}^{m} \|\nabla^j z\|_\infty. \tag{1.31}$$

A representative is fixed, when needed, by $\int z \, d\pi = 0$. The completion and its use in the one-body stability argument are discussed in Section 6.1.

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Theorem 1.2 (Uniform-in-time sharp local chaos near a confined thermal equilibrium measure)**.** *Let V satisfy (1.23)–(1.24), and fix $\mathsf{d}, \mathsf{s}, \beta$, an initial inverse temperature $\beta_0 > \beta/2$, and an integer $m \geq 3$. There are $\delta > 0$ and $C < \infty$ with the following property.*

Let z_0 be of the form (1.30), and assume

$$\|z_0\|_{\mathcal{X}^m} \leq \delta. \tag{1.32}$$

Define μ_0 by (1.29). Then (1.4) has a global smooth positive solution, and the solution of (1.3) with preparation (1.8) satisfies, for every $N \geq 2$ and $1 \leq k \leq N$,

$$\sup_{t \geq 0} \left\{ \chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) + \mathsf{H}(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) \right\} \leq C \frac{k^2}{N^2}, \tag{1.33}$$

$$\sup_{t \geq 0} \mathrm{TV}(P_{N,t}^{(k)}, \mu_t^{\otimes k}) \leq C \frac{k}{N}. \tag{1.34}$$

The constants depend only on $\mathsf{d}, \mathsf{s}, \beta, \beta_0, m$, the matrix A, and finitely many uniform bounds and decay moduli for $\nabla \tilde{V}$ through an order determined by m. They are independent of N, k, t and of higher derivatives of z_0. The same conclusion holds for smooth data in the completion for which the higher norms needed for classical regularity are finite.

The dynamical inverse temperature β is arbitrary and fixed, and the initial inverse temperature may be any fixed $\beta_0 > \beta/2$; the allowed size δ depends on both. Positivity of the centered interaction operator is used to prove stability of the mean-field solution and of the homogeneous equation for the first-marginal error. The initial first-marginal estimate instead follows from a partition bound at a larger inverse temperature; see proposition 5.1.

Corollary 1.3 (Short-time sharp local chaos)**.** *Fix $\beta > 0$ and $\beta_0 > \beta/2$, and let the confinement force satisfy (1.23)–(1.24). For every z_0 of the form (1.30), without a smallness assumption, there is $T_* > 0$ such that the conclusions of theorem 1.1 hold on $[0, T_*]$.*

Remark 1.4 (Sharpness through equilibrium). The admissible class contains the matched choice $\beta_0 = \beta$. For the thermal equilibrium measure (1.28), the corresponding preparation is

$$\mathrm{d}\mathbb{Q}_{N,\beta}(\pi) \;\propto\; \exp\left[-\beta \left(\sum_{i=1}^N V(x_i) + \frac{1}{N} \sum_{i<j} \mathsf{g}(x_i - x_j) \right) \right] \mathrm{d}X_N. \tag{1.35}$$

Both this canonical Gibbs measure and the mean-field equilibrium measure are stationary. The two-sided equilibrium estimates of [DGR26b, Theorem 2.1] therefore establish optimality of the present rates. In proposition 6.4, we include a proof of the required stationary lower bounds using the estimates already available in this paper and the static fluctuation formula of [DGR26a], in an effort to make the sharpness argument self-contained. Together with theorem 1.2, these bounds show that, for $2 \leq k \leq N$, the chi-square divergence and relative entropy have order k^2/N^2, and total variation has order k/N. Consequently, the rates in (1.21) and (1.33) cannot be improved uniformly over the stated classes. The lower bounds start at $k = 2$; the upper estimates also hold for $k = 1$.

Remark 1.5 (Unconfined consequences). Theorem 7.1 gives two explicit classes through the self-similar transformation (7.1). For the logarithmic interaction, the initial measures are dilates of small logarithmic perturbations of the thermal equilibrium measure for $V(x) = |x|^2/4$. For positive Riesz exponents, they are sufficiently diffuse dilates of small Gaussian perturbations. The lower bound on the dilation scale is part of the Riesz assertion. The full statements and their proofs are given in Section 7; remark 6.5 records the corresponding periodic result.

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1.4. **The initial condition and the full-particle chi-square.** The first counterexample concerns the initial law of the N-particle system. Independent samples have zero chi-square divergence from the product reference at time zero. Nevertheless, for a suitable confined system, the interaction generates fluctuations that make the full chi-square unbounded in N at a fixed later time of order one with respect to N. The mean-field measure in this example is stationary, so its continuity velocity and integrability expenditure vanish identically.

Proposition 1.6 (Stationary counterexample from independent initial data)**.** *Let* $\mathsf{d} = 4$ *and* $0 \le \mathsf{s} < 1$. *There are families of inverse temperatures* $\beta_\delta > 0$, *confinements* $V_\delta \in C^\infty_{\mathrm{loc}}(\mathbb{R}^4)$, *smooth positive probability densities* μ_δ, *and times* $T_\delta > 0$, *indexed by* $0 < \delta < 1/2$, *with the following properties.*

(i) *The confinement satisfies* (1.23)–(1.24) *with* $A = (4\beta_\delta)^{-1}\mathrm{Id}$ *and* $b = 0$. *More explicitly, set*

$$\widetilde{V}_\delta(x) := V_\delta(x) - \frac{|x|^2}{8\beta_\delta}, \qquad \lim_{R\to\infty} \sup_{|x|\ge R} |\nabla^{j+1}\widetilde{V}_\delta(x)| = 0 \quad (j \ge 0), \tag{1.36}$$

and every derivative of $\nabla\widetilde{V}_\delta$ *is bounded. The density* μ_δ *is the thermal equilibrium measure for* V_δ *and* β_δ. *Its mean-field trajectory satisfies*

$$\mu_t = \mu_\delta, \qquad u_t = 0 \quad (t \ge 0), \qquad A_{\mu_\delta}(T) = 0 \quad (T < \infty). \tag{1.37}$$

(ii) *For every* $N \ge 2$, *consider* (1.3) *with confinement* V_δ, *inverse temperature* β_δ, *and initial law*

$$P_{N,0} = \mu_\delta^{\otimes N}, \tag{1.38}$$

independent of the Brownian motions. The system has a pathwise unique global noncolliding strong solution. Its law satisfies

$$\sup_{N\ge 2} \sup_{t\ge 0} \mathsf{H}(P_{N,t} \mid \mu_\delta^{\otimes N}) < \infty. \tag{1.39}$$

(iii) *For every fixed* $N \ge 2$ *and* $t < \infty$, *the full chi-square is finite, and it is zero at* $t = 0$. *For every sufficiently small fixed* δ,

$$\lim_{N\to\infty} \chi^2(P_{N,T_\delta} \mid \mu_\delta^{\otimes N}) = +\infty. \tag{1.40}$$

All parameters, the confinement, the initial density, and the terminal time are fixed before $N \to \infty$. *The bound in* (1.39) *may depend on* s *and* δ.

The construction and proof are given in Appendix C. We have chosen a dimension and range of Riesz exponents that simplify the construction, without seeking the greatest generality. Obtaining such a counterexample throughout the dimension and exponent range of theorem 1.1 is left to the interested reader. Stationarity here concerns μ_δ, not the particle law initialized by $\mu_\delta^{\otimes N}$. The stationary trajectory satisfies the intrinsic one-body and static hypotheses of theorem 1.1, and it is the zero logarithmic perturbation of its own thermal equilibrium measure. Thus, replacing the preparation required in theorems 1.1 and 1.2 by iid initialization while keeping their full-particle conclusion is impossible under these trajectory conditions alone. This conclusion does not result from setting $\beta_0 = 0$ in (1.20), and it does not determine the optimal range of initial Gibbs temperatures. The confinement is asymptotically isotropic but not exactly quadratic.

The divergence in (1.40) does not preclude local or mesoscopic chaos. In fact, remark C.8 gives a marginal entropy bound of order k/N, uniform for all time, and hence convergence whenever $k = o(N)$. The sharp local iid rates remain undecided by this example. The next result concerns the time restriction: even the matched Gibbs initial law does not give unrestricted full-particle chi-square control along every smooth mean-field trajectory.

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1.5. **Failure of unrestricted full-particle chi-square control.** Within this subsection, we recycle g for the mean-zero spectral logarithm on the unit torus $\mathbb{T}^4$, normalized by

$$\widehat{\mathsf{g}}(n) = \frac{1}{2\pi^2|n|^4}, \qquad n \in \mathbb{Z}^4 \setminus \{0\}. \tag{1.41}$$

It has local singularity $-\log|x|$. Consider the periodic versions of (1.3) and (1.4), with $V = 0$ and preparation $P_{N,0} = \mathbb{Q}_{N,\beta}(\mu_0)$ at the same inverse temperature as the dynamics.

Theorem 1.7 (Failure of unrestricted full-particle chi-square control)**.** *For every sufficiently large fixed β, there is $\mu_0 \in C^\infty(\mathbb{T}^4)$ with $\inf \mu_0 > 0$ such that the corresponding mean-field trajectory is smooth and strictly positive on $[0,5]$. The laws $P_{N,t}$ of the corresponding particle system satisfy*

$$\lim_{N\to\infty} \chi^2(P_{N,5} \mid \mu_5^{\otimes N}) = +\infty, \tag{1.42}$$

while each finite-N divergence is finite, the initial full chi-square divergences are uniformly bounded in N, and

$$\sup_N \sup_{0\le t\le 5} \mathsf{H}(P_{N,t} \mid \mu_t^{\otimes N}) < \infty. \tag{1.43}$$

Moreover, there are a real-valued $\psi \in C^\infty(\mathbb{T}^4)$ and $v > 2\,\mathrm{Var}_{\mu_5}(\psi) > 0$ such that, under $P_{N,5}$,

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \left(\psi(x_i) - \int \psi \, \mathrm{d}\mu_5 \right) \underset{N\to\infty}{\longrightarrow} \mathcal{N}(0, v) \quad \textit{in law.} \tag{1.44}$$

Consequently, an estimate of the form

$$\chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) + \mathsf{H}(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) \le C_T \frac{k^2}{N^2}, \qquad 1 \le k \le N, \tag{1.45}$$

cannot hold for every smooth positive periodic log/Riesz trajectory throughout the Hilbert–Schmidt class.

The mean-field measure and inverse temperature are fixed before N tends to infinity. Unlike proposition 1.6, this example starts from a modulated Gibbs measure whose inverse temperature $\beta_0 = \beta$ agrees with that of the dynamics. Its mechanism is a collective translation. Diffusion continually injects fluctuations into a collective translation mode that is invisible to the relative-coordinate interaction, while the product reference continues to resolve the much narrower one-particle scale. The accumulated Brownian variance is too large for a uniform relative second moment. The proof is given in Section 8. Together, the two examples show that neither stationarity of the mean-field measure with independent initialization nor Gibbs initialization along an arbitrary smooth trajectory suffices for the full-particle bound. The bounded relative entropies in (1.39) and (1.43) do not control these relative second moments.

We carry out this Gibbs-initialized collective-translation construction in the periodic setting to simplify the presentation. Preliminary calculations suggest an analogous construction for the confined whole-space systems of theorem 1.2, using a collective Ornstein–Uhlenbeck translation mode in the isotropic case. We leave this extension to the interested reader. The stationary example in proposition 1.6 uses independent initial data and a different fluctuation mechanism.

1.6. **Relation to previous work.** Sharp local entropy estimates for mean-field diffusions were developed by Lacker [Lac23] through a hierarchy of marginal relative entropies. Lacker and Le Flem [LLF23] obtained uniform-in-time estimates under one-body logarithmic Sobolev inequalities and quantitative interaction and temperature conditions. These results distinguish the local entropy scale from the weaker estimate obtained by subadditivity. Hess-Childs and Rowan [HCR25, Theorem 1.4 and Remarks 1.6 and 1.8] prove L^2 expansions for bounded interaction forces, including the sharp leading marginal estimate and higher-order corrections on a quantitative growing-marginal range.

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Their stated tensorized initialization can be relaxed when the initial laws satisfy the corresponding expansion bounds. Thus, correlated initialization is not itself a distinction from that method.

For singular interactions, relative-entropy and modulated-energy methods provide mean-field estimates without requiring a Lipschitz pair force; see [JW18, BJW19, BJW23, NRS22, RS23, RS25, CdCRS25a, CdCRS25b]. These are global approaches: relative entropy and modulated free energy are evaluated on the full N-particle measure, while modulated energy compares the empirical measure with the mean-field measure. For more details on these approaches, we refer the reader to the third author's recent survey [Ros25].

Wang [Wan26, Theorems 1–2 and Remark 2] treats periodic forces with a divergence-free $W^{-1,\infty}$ part and a bounded remainder. His entropy theorem gives the sharp k^2/N^2 dependence under a smallness condition, with a uniform-in-time conclusion under additional decay assumptions. His short-time weighted L^2 theorem gives the sharp N^{-2} order for fixed k, with exponential rather than quadratic dependence on k. These force hypotheses differ from those of the repulsive whole-space gradient systems considered here.

Duerinckx and Jabin [DJ25, Corollary 1.3(i) and Remark 1.2] obtain short-time order-N^{-1} estimates for every fixed marginal in a weighted density norm, starting from independent particles, and analyze further correlation corrections. Their overdamped result assumes square integrability of the force and a lower bound on its divergence. The square-integrability requirement alone restricts a Riesz force to $\mathsf{s} < \mathsf{d}/2 - 1$ at the collision singularity, whereas our estimates reach the full Hilbert–Schmidt range $\mathsf{s} < \mathsf{d}/2$ for the potential. In particular, they include the planar logarithm and three-dimensional Coulomb potential. The comparisons therefore concern the interaction assumptions, marginal-size dependence, initial laws, and time regime, rather than the fixed-k power of N alone. In their subsequent work [DJ26, Theorem 1.1 and Appendix A], Duerinckx and Jabin derive the linearized Vlasov equation from signed infinitesimal perturbations of a stationary kinetic Gibbs equilibrium. That result concerns linear response around equilibrium, whereas the present theorems estimate the complete marginal probability densities along a nonlinear mean-field trajectory.

We also mention that, for equilibrium Gibbs measures, Lacker [Lac22] obtains sharp marginal Fisher-information and corresponding relative-entropy bounds under suitable functional inequalities, using a perturbative force-level hierarchy. Ren and Wang [RW25] derive sharp relative-entropy estimates for all marginal sizes from a hierarchy of conditional entropies, under functional-inequality, transport, and force-integrability hypotheses.

The estimates on singular Gibbs measures used below come from the authors' two companion works. The partition-function paper [DGR26a, Theorems 2.2 and 2.8] supplies the partition and transport-commutator estimates, and also treats diffusive mean-field limits and fluctuations. See also the concurrent work of Wang and Zhao [WZ26] for overlapping partition-function, commutator, and fluctuation results, and the earlier high-temperature partition bounds and limiting formulas of Duerinckx and Jabin [DJ26, Theorem 2.1 and Proposition 2.2]. The static fluctuation formula and finite-particle adjoint identity [DGR26a, Propositions 2.11 and 8.2] enter the periodic counterexample. The iid counterexample uses the same adjoint identity and commutator estimates, but its initial covariance is the ordinary iid covariance. Here, these inputs are combined with estimates for the evolution of complete marginal densities and with two obstructions to unrestricted full-particle chi-square control.

The authors' equilibrium paper [DGR26b, Lemma 5.2 and Lemma 6.1] also develops the screened inverse and orthogonal product-space decomposition used here; we give proofs in lemmas 2.4 and 5.4. Its quantitative marginal-density expansion identifies the first Bogolyubov correction and, together with the associated divergence-limit theorem, gives the leading constants for fixed and growing sublinear marginals under the separate hypotheses of [DGR26b, Theorems 2.3 and 2.4]. Neither its cavity expansion nor these asymptotic results are needed below. We instead use a weighted elliptic argument for the initial first marginal, since the same weighted estimates enter its

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propagation, and give the stationary lower-bound deduction in proposition 6.4. The Hilbert–Schmidt threshold obstruction in [DGR26b, Theorem 2.5] concerns the collision singularity. Both dynamical counterexamples here lie below that threshold: they concern amplification of initial iid fluctuations or accumulation of Brownian fluctuations, rather than loss of square integrability of the potential.

The attractive logarithmic interaction is treated in our companion work [DGR26c]. In dimension two, for example, we prove there the corresponding sharp local chaos estimates throughout the full subcritical range $0 < \beta < 4$.

Remark 1.8 (Kinetic dynamics). Extending the sharp local estimates of this paper to underdamped Langevin dynamics is an interesting problem. Feng and Wang [FW26] recently introduced a kinetic version of the modulated Gibbs formalism popularized by Serfaty and the third author [RS25]. Together with the weighted marginal estimates of Bresch, Jabin, and Soler [BJS25], this yields a short-time mean-field limit for singular kinetic systems, including the three-dimensional Vlasov–Poisson–Fokker–Planck equation. Obtaining the sharp dependence on the number of observed particles in relative entropy and chi-square divergence requires additional estimates.

1.7. **Proof strategy.** The proof of theorem 1.1 reduces to two estimates: a bounded full-density second moment and an order-N^{-1} first-marginal error in the weighted L^2 norm. Indeed, the product-space projection of lemma 5.4 gives

$$\chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) \le k\chi^2(P_{N,t}^{(1)} \mid \mu_t) + \frac{k(k-1)}{N(N-1)}\chi^2(P_{N,t} \mid \mu_t^{\otimes N}), \qquad 1 \le k \le N. \tag{1.46}$$

The full moment is also used inside the first-marginal argument, where it controls the centered two-particle source through (5.15). Figure 1 records these dependencies.

Set

$$\Gamma_{N,t} = \mathbb{Q}_{N,\beta}(\mu_t), \quad G_{N,t} = \frac{\mathrm{d}\Gamma_{N,t}}{\mathrm{d}\mu_t^{\otimes N}}, \quad q_{N,t} = \frac{\mathrm{d}P_{N,t}}{\mathrm{d}\Gamma_{N,t}}, \quad f_{N,t} = q_{N,t}G_{N,t}. \tag{1.47}$$

The reference $\Gamma_{N,t}$ incorporates the singular pair repulsion. Its failure to solve the particle forward equation is the centered transport commutator computed in (2.32), whose exponential moment under $\Gamma_{N,t}$ is controlled by [DGR26a]. The Donsker–Varadhan lemma converts that estimate into a relative-entropy term in the equation for the norm of $q_{N,t}$. Choosing a decreasing exponent in the $L^{p(t)}(\Gamma_{N,t})$ estimate cancels this term. As long as $p(t) > 2$, Hölder's inequality and static moments of $G_{N,t}$ give the full $L^2(\mu_t^{\otimes N})$ bound. The initial norm is estimated by comparing $\mathbb{Q}_{N,\beta_0}(\mu_0)$ and $\mathbb{Q}_{N,\beta}(\mu_0)$ in lemma 3.2; it equals one in the matched case. This comparison and the subsequent loss of integrability give the temperature-dependent sufficient condition (1.20).

We estimate the initial first marginal directly from the modulated Gibbs measure. Differentiation of its density and weighted elliptic duality convert the static second-moment estimate into an order-N^{-1} first-marginal bound; see proposition 5.1. We then propagate this estimate using only the exact first-marginal equation (5.18). The full second moment controls its centered two-particle source through (5.15), so no evolution equation for a higher marginal is required. Testing this source against the backward mean-field propagator produces a difference of two test-function gradients. The weighted commutator estimate of lemma 4.1 combines gradient and Hessian bounds while leaving a factor of the background density along the segment joining the two points. For $|x - y| > 1$, the force bound and the mass of the background suffice. Backward smoothing gives a time singularity that is integrable throughout $\mathsf{s} < \mathsf{d}/2$. This proves the first-marginal estimate needed in (1.46), without square integrability of the force or a positive lower bound on the whole-space density.

The hierarchy argument of Duerinckx and Jabin [DJ25, Sections 3–4] estimates linear correlations of all orders with time-dependent Gibbs-type weights. Their later equilibrium-fluctuation argument [DJ26, Sections 3–4] already uses global density bounds and orthogonal product-space projections, followed by the first-marginal equation. This reduction is also central here. The additional task is to control a Gibbs reference moving along a nonlinear mean-field trajectory and to obtain a

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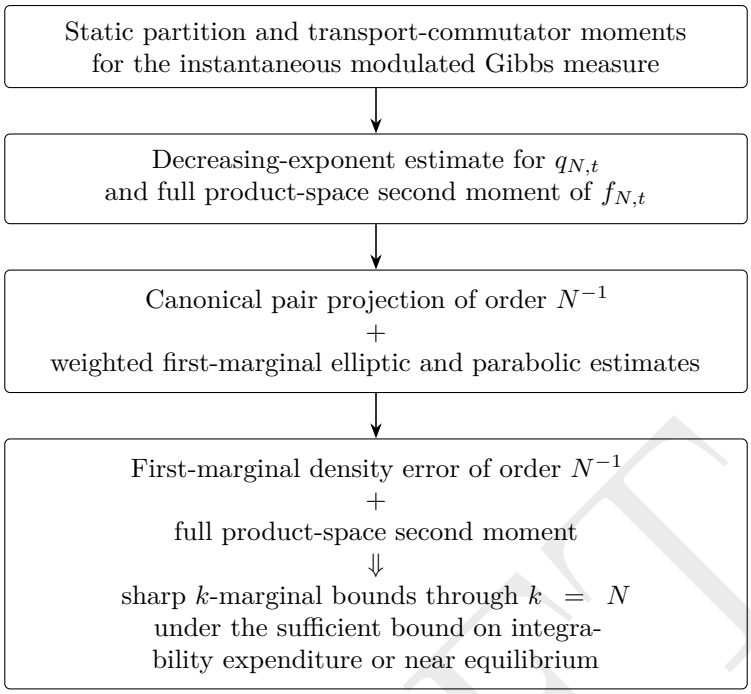


FIGURE 1. Proof of the sharp local estimates through the full-particle scale. The full moment, supplied under the sufficient bound on integrability expenditure or near equilibrium, controls the pair source inside the first-marginal estimate and is used again in the final product-space projection.

quantitative first-marginal error. The decreasing-exponent estimate controls the reference defect, and the weighted commutator estimate controls the singular source in that equation.

The moving Gibbs formalism is part of the modulated-free-energy method [BJW19, RS25]. Varying-exponent differentiation is classical in hypercontractivity and smoothing [Gro75, CL95]; the decreasing exponent used here controls the reference defect without a many-particle logarithmic Sobolev inequality. Previous works also use time-dependent norms; for example, in [BJS25], the time dependence is in an energy weight, while the Lebesgue exponent remains fixed.

For theorem 1.2, two stability arguments complement the finite-time estimates. We estimate the homogeneous first-marginal equation in the quadratic form associated with the mean-field linearization, with the finite-particle factor $(N-1)/N$ included in the interaction term; see (5.35). This form combines the weighted L^2 norm of the signed density error with its interaction energy. Near equilibrium, contraction on fixed time intervals, combined with the source estimate above, prevents accumulation of the first-marginal error. Separately, the mean-field stability argument transfers the $L^2(\pi)$ spectral gap of the linearized equation to the logarithmic derivative norm by comparison with the anisotropic Ornstein–Uhlenbeck semigroup. This gives the uniform background bounds and integrable velocity used in the proof. The first argument is given in proposition 5.3; the mean-field estimates are stated in Section 6.1 and proved in Appendix A. The unconfined results follow through the self-similar transformation of Serfaty and the third author [RS24], with an additional decreasing-coupling argument for $\mathsf{s} > 0$.

The counterexamples compare a terminal linear statistic under the actual particle law and under the product reference. A Gaussian limit with variance greater than twice the product variance has infinite chi-square divergence from the product Gaussian limit. Data processing and weak lower semicontinuity then exclude a bounded full-particle chi-square divergence. The two constructions obtain this variance separation in different ways. For independent initialization, the backward

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linearized evolution amplifies the initial product fluctuation, while the Brownian contribution is lower order; see (C.44) and proposition C.7. For the periodic Gibbs preparation, the Brownian contribution alone exceeds twice the product variance; see proposition 8.8. The terminal statistic is, up to a vanishing error, the sum of an initial fluctuation and an asymptotically independent Gaussian Brownian fluctuation. The Brownian variance already forces infinite chi-square divergence of every subsequential limit from the product Gaussian, irrespective of the initial limiting distribution. The static central limit theorem for the initial modulated Gibbs measure then identifies the initial fluctuation as Gaussian and gives the explicit terminal Gaussian limit in (1.44).

1.8. **Organization of the paper.** Sections 2 and 3 give the static inputs, the moving-reference defect, and the decreasing-exponent estimate. Sections 4 and 5 prove the weighted first-marginal estimates and the passage to all marginals. Section 6 verifies the confined background hypotheses, justifies approximation, and completes the confined proofs. Section 7 proves the unconfined results. The proof of theorem 1.7 is in Section 8. The required one-body stability and finite-particle well-posedness arguments are in Appendices A and B, respectively. The proof of proposition 1.6, including the weaker marginal entropy estimate, is in Appendix C.

1.9. **Statement on AI use.** This working manuscript was generated with the assistance of generative AI tools, including OpenAI's ChatGPT and Codex and Anthropic's Claude, for literature identification, the development and testing of mathematical arguments, exposition, and consistency checks. The authors have not yet reviewed and independently verified all AI-generated content. They are in the process of formalizing the results in Lean and revising the natural-language presentation. The manuscript is therefore being shared as a draft, and its statements, proofs, references, and exposition remain subject to correction. Any completed formal verification applies only to the statements and versions explicitly identified in the accompanying verification records.

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2. Static inputs and the moving-reference defect

This section supplies the exponential estimate for the moving-reference defect used in Section 3. We first recall the partition and commutator estimates, including their uniformity under heat truncation. We then compute the defect and estimate its exponential moment. The screened inverse recorded below will also enter the initial one-mark estimate in proposition 5.1.

To include the self-similar applications in the same calculation, we allow a prescribed C^1_{loc} nonincreasing interaction coefficient $c(t) \in [0, c_*]$. The particle force and mean-field force are both multiplied by $c(t)$, and the reference is $\mathbb{Q}_{N,\beta c(t)}(\mu_t)$. The constant case $c = 1$ gives the confined theorems; decreasing coefficients enter theorem 7.1. Set

$$E_{N,t} = N\mathsf{F}_N(X_N, \mu_t), \qquad C_N[v] = \frac{N}{2}\mathsf{A}_1(v, X_N, \mu_t), \tag{2.1}$$

where

$$\mathsf{A}_1(v, X_N, \mu) = \iint_{x\neq y} (v(x) - v(y)) \cdot \nabla \mathsf{g}(x-y)\, \mathrm{d}\left(\frac{1}{N}\sum_i \delta_{x_i} - \mu\right)^{\otimes 2}(x, y). \tag{2.2}$$

Thus, C_N denotes the transport commutator, with precisely half the normalization of $N\mathsf{A}_1$.

2.1. Static and heat-truncation estimates.

All backgrounds in the applications satisfy a common bound

$$0 < \mu(x) \le Me^{-a|x|^2}, \qquad a > 0. \tag{2.3}$$

No positive lower bound is asserted. The static results require substantially less; (2.3) conveniently supplies every logarithmic moment used here.

We collect four inputs and indicate their later roles. Proposition 2.1 gives the partition and commutator estimates used in Section 3. Lemma 2.2 records the potential estimates that make those static bounds uniform under heat truncation. Lemma 2.3 supplies the cutoff-uniform force estimates used in lemmas 4.1, 6.2 and 6.3. Finally, lemma 2.4 gives the inverse bound used in the initial one-mark estimate proposition 5.1.

Proposition 2.1 (Partition and commutator estimates)**.** *For the kernels (1.1), every fixed $B < \infty$ admits constants, uniform over (2.3), such that*

$$e^{bI_\mu} \le \mathsf{K}_{N,b}(\mu) \le C_B, \quad 0 \le b \le B, \qquad |I_\mu| + \|\widetilde{\mathsf{g}}\|_{L^2(\mu^{\otimes 2})} + \|h_\mu\|_{L^2(\mu)} \le C. \tag{2.4}$$

Here, $\widetilde{\mathsf{g}}$ is the centered kernel

$$\widetilde{\mathsf{g}}(x, y) = \mathsf{g}(x-y) - h_\mu(x) - h_\mu(y) + 2I_\mu. \tag{2.5}$$

In particular,

$$\int G^p_{N,b}\, \mathrm{d}\mu^{\otimes N} = \frac{\mathsf{K}_{N,pb}(\mu)}{\mathsf{K}_{N,b}(\mu)^p} \le C_{p,B}, \qquad 0 \le b \le B, \tag{2.6}$$

for every fixed $1 \le p < \infty$.

For fixed $\beta, c_* > 0$, there are $\kappa, K > 0$ such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[e^{\sigma\kappa c C_N[v]}] \le K, \qquad 0 \le c \le c_*, \quad \|\nabla v\|_{L^\infty} \le 1, \quad \sigma = \pm 1. \tag{2.7}$$

These estimates hold uniformly for the heat truncations defined in (2.12).

Proof. The partition estimate is [DGR26a, Theorem 2.2], and the commutator estimate is [DGR26a, Theorem 2.8 and Sections 7.3–7.4]. Appendix A of that reference makes their background dependence explicit. The bound (2.3) controls the logarithmic energy and all fixed polynomial moments; the local singularity is square integrable exactly when $2\mathsf{s} < \mathsf{d}$. The lower bound in (2.4) is Jensen's inequality, since the product expectation of E_N is $-I_\mu$. In the whole-space logarithmic case I_μ need not be nonnegative: the useful uniform lower bound is $e^{-B|I_\mu|}$, not one. Uniformity for $0 \le b \le B$

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also follows by Hölder from the estimate at $b = B$, and (2.6) follows directly from the definition of $G_{N,b}$.

It remains to explain how the local numerator estimate of [DGR26a, Lemma 7.2] yields the coefficient proportional to c in (2.7). With $C_N = N\mathsf{A}_1/2$, the cited estimate has the form

$$\mathbb{E}_{\mu^{\otimes N}}[e^{\sigma a_0 b C_N[v]-bE_N}] \le C, \qquad 0 < b \le b_0, \tag{2.8}$$

where $a_0, b_0 > 0$ are common to the background class and $\|\nabla v\|_{L^\infty} \le 1$. For $b = \beta c > 0$, set $b_1 = \min\{b, b_0\}$. Cauchy–Schwarz gives

$$\mathbb{E}_{\mu^{\otimes N}}[e^{\sigma a_0 b_1 C_N[v]/2-bE_N}] \le \mathbb{E}_{\mu^{\otimes N}}[e^{\sigma a_0 b_1 C_N[v]-b_1 E_N}]^{1/2}\mathsf{K}_{N,2b-b_1}(\mu)^{1/2} \le C. \tag{2.9}$$

The constants are uniform for $0 < b \le \beta c_*$, since $0 \le 2b - b_1 \le 2\beta c_*$. Dividing by $\mathsf{K}_{N,b}(\mu)$ and using its lower bound in (2.4) gives the corresponding Gibbs moment. Finally,

$$\kappa \le \frac{a_0}{2}\min\left\{\beta, \frac{b_0}{c_*}\right\} \quad \Longrightarrow \quad \kappa c \le \frac{a_0 b_1}{2}. \tag{2.10}$$

Hölder's inequality gives (2.7) at the smaller coefficient. At $c = 0$, its integrand is one. The cutoff-uniformity follows from lemma 2.2 and the corresponding proofs in the cited reference. □

For the heat truncation, set

$$p_t(x) := (4\pi t)^{-\mathsf{d}/2} e^{-|x|^2/(4t)}, \qquad a_{\mathsf{d},\mathsf{s}} := \frac{(4\pi)^{\mathsf{d}/2}}{2^{\mathsf{s}+1}\Gamma(1+\mathsf{s}/2)}. \tag{2.11}$$

For $\mathsf{s} > 0$, define

$$\mathsf{g}_\varepsilon(x) := a_{\mathsf{d},\mathsf{s}} \int_\varepsilon^\infty t^{(\mathsf{d}-\mathsf{s})/2-1} p_t(x)\,\mathrm{d}t, \qquad \varepsilon > 0. \tag{2.12}$$

For $\mathsf{s} = 0$, replace $p_t(x)$ by $p_t(x) - p_t(e)$, where e is a fixed unit vector. This fixes $\mathsf{g}_\varepsilon(e) = 0$ and the limit $-\log|x|$. We set $\mathsf{g}_0 := \mathsf{g}$. The parameter ε is a heat time: the corresponding length scale in [DGR26a, Section 4.1] is $\sqrt{\varepsilon}$. In the logarithmic case our normalization differs from that reference by an additive constant, which cancels from the normalized Gibbs density. Each auxiliary particle equation and its modulated Gibbs reference use the same truncated kernel.

For $\eta > 0$ and $\varepsilon \ge 0$, set

$$\mathsf{f}_{\eta,\varepsilon}(x) := a_{\mathsf{d},\mathsf{s}} \int_\varepsilon^{\max\{\varepsilon,\eta^2\}} t^{(\mathsf{d}-\mathsf{s})/2-1} p_t(x)\,\mathrm{d}t. \tag{2.13}$$

Lemma 2.2 (Heat-truncation estimates for the potential)**.** *Let μ satisfy (2.3), let $0 \le \varepsilon \le 1$, and let $0 < \eta \le 1$. Then the following estimates hold with constants independent of ε and η.*

(i) The short-range part satisfies

$$\|\mathsf{f}_{\eta,\varepsilon}\|_{L^1} \le C\eta^{\mathsf{d}-\mathsf{s}}, \qquad \|\mathsf{f}_{\eta,\varepsilon} * \mu\|_{L^\infty} \le CM\eta^{\mathsf{d}-\mathsf{s}}, \tag{2.14}$$

and, for every Lipschitz vector field v,

$$|(v(x) - v(y)) \cdot \nabla \mathsf{f}_{\eta,\varepsilon}(x-y)| \le C\|\nabla v\|_{L^\infty} \mathsf{f}_{\sqrt{2}\eta,2\varepsilon}(x-y). \tag{2.15}$$

(ii) The remaining potential $\mathsf{g}_{\max\{\varepsilon,\eta^2\}}$ has Fourier multiplier

$$0 \le \widehat{\mathsf{g}_{\max\{\varepsilon,\eta^2\}}}(\xi) \le C|\xi|^{\mathsf{s}-\mathsf{d}}, \qquad \xi \ne 0. \tag{2.16}$$

If q is obtained from this potential by a smooth Fourier cutoff bounded by one and vanishing on $|\xi| \le K_0$, where $K_0 \ge 1$, then

$$\|q\|^2_{L^2(\mathbb{R}^{\mathsf{d}})} \le C\int_{|\xi|>K_0} |\xi|^{2\mathsf{s}-2\mathsf{d}}\,\mathrm{d}\xi \le CK_0^{2\mathsf{s}-\mathsf{d}}, \tag{2.17}$$

$$\sup_x \int |q(x-y)|^2\,\mathrm{d}\mu(y) \le M\|q\|_2^2, \qquad \|q*(\mu\,\cdot)\|_{L^2(\mu)\to L^2(\mu)} \le \|q(x-y)\|_{L^2(\mu^{\otimes 2})} \le \sqrt{M}\|q\|_2. \tag{2.18}$$

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Centering $q(x-y)$ preserves these bounds up to a numerical factor. Moreover,

$$\left|\mathsf{g}_{\max\{\varepsilon,\eta^2\}}(0)\right| \le \begin{cases} C\eta^{-\mathsf{s}}, & \mathsf{s}>0, \\ C(1+|\log\eta|), & \mathsf{s}=0, \end{cases} \tag{2.19}$$

and, in the logarithmic case,

$$t^{\mathsf{d}/2-1}|p_t(z)-p_t(e)| \le C(1+|z|^2)t^{-2}, \qquad t\ge 1. \tag{2.20}$$

(iii) The centered interaction operator associated with g_ε is nonnegative on $L^2_0(\mu)$.

Proof. Integrating the heat kernel in space gives the first estimate in (2.14); the second follows from $\|\mu\|_\infty \le M$. For (2.15), use $|v(x)-v(y)| \le \|\nabla v\|_{L^\infty}|x-y|$ and the Gaussian bound $|z||\nabla p_t(z)| \le Cp_{2t}(z)$, followed by the change of heat-time variable $r=2t$.

Taking Fourier transforms in the heat representation gives (2.16). Plancherel's theorem then gives (2.17). The density bound in (2.3) yields the first estimate in (2.18), while the second is the Hilbert–Schmidt bound for the integral operator with kernel $q(x-y)$. Subtracting either one-variable average, and then the total average, changes these estimates only by a numerical factor.

Evaluating the heat representation at the origin gives (2.19). For the logarithm, the subtraction at e makes the large-time integral convergent; the mean-value theorem applied to the Gaussian exponential gives (2.20). Hence, changing the lower heat cutoff does not affect the large-time cancellation, and the moments supplied by (2.3) control the resulting one-body terms.

Finally, the Fourier multiplier in (2.16) is nonnegative. The additive constant in the logarithmic representation does not contribute to the quadratic form on zero-mass functions. Centering the kernel likewise leaves that quadratic form unchanged, which proves the last assertion. □

The estimates in lemma 2.2 are the short-range, high-frequency, row, and diagonal bounds used in the cutoff-uniform partition and commutator arguments of [DGR26a, Sections 4, 5.1, and 7]. The partition argument also uses nonnegativity of the centered quadratic form, whereas the commutator argument retains the deterministic factor N in the diagonal bound.

Lemma 2.3 (Heat-truncation estimates for the force)**.** *For $0\le\varepsilon\le 1$,*

$$|\nabla \mathsf{g}_\varepsilon(x)| \le C|x|^{-\mathsf{s}-1}, \qquad \sup_{0\le\varepsilon\le 1}\left(\|\nabla \mathsf{g}_\varepsilon\|_{L^1(B_1)} + \|\nabla \mathsf{g}_\varepsilon\|_{L^\infty(B_1^c)}\right) \le C. \tag{2.21}$$

As $\varepsilon\downarrow 0$, the force converges in $L^1(B_1)$ and with all derivatives locally away from the origin. Moreover,

$$\sup_{0\le\varepsilon\le 1}\int_{|z|\le 1}|z|^{2\alpha}|\nabla \mathsf{g}_\varepsilon(z)|^2\,\mathrm{d}z \le C\int_0^1 r^{\mathsf{d}-1+2\alpha-2\mathsf{s}-2}\,\mathrm{d}r<\infty, \qquad \alpha>\mathsf{s}+1-\mathsf{d}/2. \tag{2.22}$$

Proof. Differentiating (2.12) and using the scaling of ∇p_t gives the pointwise estimate in (2.21). Its local L^1 and exterior L^∞ consequences follow because $\mathsf{s}+1<\mathsf{d}$. The difference $\nabla \mathsf{g}-\nabla \mathsf{g}_\varepsilon$ is the integral over $0<t<\varepsilon$; dominated convergence gives convergence in $L^1(B_1)$, while differentiation under the integral gives convergence of every derivative on compact sets separated from the origin. Finally, the pointwise bound in (2.21) and radial integration give (2.22). □

Lemma 2.4 (Screened inverse)**.** *Let T_μ be the integral operator on $L^2_0(\mu)$ with kernel $\widetilde{\mathsf{g}}$. Suppose that the bounds in (2.4) hold up to coupling $2b$, where $b\ge 0$. Then*

$$\left\|\left(\mathrm{Id}+b\frac{N-1}{N}T_\mu\right)^{-1}\right\|_{L^2_0(\mu)\to L^2_0(\mu)} \le \frac{2N}{N+1} \le 2. \tag{2.23}$$

The conclusion is uniform under the heat truncation and does not require positive definiteness of the interaction.

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Proof. We give a short proof of the estimate in [DGR26b, Lemma 5.2] using the partition bound already stated here. If $b = 0$ or $N = 1$, the inverse is the identity. Suppose that $b > 0$ and $N \geq 2$. Fix a bounded real $\varphi \in L^2_0(\mu)$ and set $\nu = (1 + \eta\varphi)\mu$, where $|\eta|\|\varphi\|_\infty < 1/2$. The bounds in (2.4) imply that $(x, y) \mapsto \mathsf{g}(x - y)$ belongs to $L^2(\mu^{\otimes 2})$, so all the following integrals are finite. By (1.6),

$$\mathbb{E}_{\nu^{\otimes M}}[M\mathsf{F}_M(X_M, \mu)] = \frac{M\eta^2}{2}\langle\varphi, T_\mu\varphi\rangle_{L^2(\mu)} - I_\nu. \tag{2.24}$$

Changing the product reference from $\mu^{\otimes M}$ to $\nu^{\otimes M}$ and applying Jensen's inequality gives

$$\log \mathsf{K}_{M,2b}(\mu) \geq -M\big(\mathsf{H}(\nu \mid \mu) + b\eta^2\langle\varphi, T_\mu\varphi\rangle_{L^2(\mu)}\big) + 2bI_\nu. \tag{2.25}$$

For fixed η, divide by M and let $M \to \infty$, using the uniform partition bound. Then

$$\mathsf{H}((1 + \eta\varphi)\mu \mid \mu) + b\eta^2\langle\varphi, T_\mu\varphi\rangle_{L^2(\mu)} \geq 0. \tag{2.26}$$

Since φ is bounded and centered, the entropy is $\eta^2\|\varphi\|^2_{L^2(\mu)}/2 + o(\eta^2)$. Dividing by η^2 and then letting $\eta \to 0$ yields

$$\langle\varphi, T_\mu\varphi\rangle_{L^2(\mu)} \geq -\frac{1}{2b}\|\varphi\|^2_{L^2(\mu)}. \tag{2.27}$$

Bounded centered functions are dense in $L^2_0(\mu)$, and T_μ is bounded and self-adjoint. Thus, (2.27) holds throughout that space, and

$$\mathrm{Id} + b\frac{N-1}{N}T_\mu \geq \frac{N+1}{2N}\mathrm{Id}. \tag{2.28}$$

The spectral theorem proves (2.23). The argument applies at each heat cutoff with the same final constant. For the repulsive kernels, lemma 2.2(iii) gives $T_\mu \geq 0$ and hence the stronger inverse bound 1. □

2.2. **The exact defect, including a varying coupling.** Set $\Gamma_{N,t} = \mathbb{Q}_{N,\beta c(t)}(\mu_t)$, and let $L_{N,t}$ be the particle generator with interaction coefficient $c(t)$. The continuity velocity is

$$u_t = -\nabla V - c(t)\nabla h_t - \beta^{-1}\nabla \log \mu_t, \qquad \partial_t\mu_t + \mathrm{div}(\mu_t u_t) = 0. \tag{2.29}$$

The score identity is

$$b^N_i = u_t(x_i) + \beta^{-1}\nabla_i \log \Gamma_{N,t}. \tag{2.30}$$

Consequently, the failure of $\Gamma_{N,t}$ to solve the particle forward equation is

$$\partial_t\Gamma_{N,t} - L^*_{N,t}\Gamma_{N,t} = R_{N,t}\Gamma_{N,t}, \tag{2.31}$$

$$R_{N,t} = -\beta\left(c(t)C_N[u_t] + c'(t)E_{N,t} - \mathbb{E}_{\Gamma_{N,t}}[c(t)C_N[u_t] + c'(t)E_{N,t}]\right). \tag{2.32}$$

Indeed, the continuity equation and integration by parts give

$$(\partial_t + u_t(x)\cdot\nabla)h_t(x) = \int (u_t(x) - u_t(y))\cdot\nabla\mathsf{g}(x - y)\,\mathrm{d}\mu_t(y), \tag{2.33}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t}I_{\mu_t} = \frac{1}{2}\iint (u_t(x) - u_t(y))\cdot\nabla\mathsf{g}(x - y)\,\mathrm{d}\mu_t(x)\,\mathrm{d}\mu_t(y). \tag{2.34}$$

Applying these identities to the pair, one-body, and constant terms in (1.6), with their original coefficients, yields

$$\left(\partial_t + \sum_{i=1}^N u_t(x_i)\cdot\nabla_i\right)E_{N,t} = C_N[u_t]. \tag{2.35}$$

The product reference satisfies the corresponding continuity equation. Integrating the resulting equation for the unnormalized Gibbs density therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\log \mathsf{K}_{N,\beta c(t)}(\mu_t) = -\beta\mathbb{E}_{\Gamma_{N,t}}[c(t)C_N[u_t] + c'(t)E_{N,t}]. \tag{2.36}$$

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Dividing the density $e^{-\beta c(t)E_{N,t}}\mu_t^{\otimes N}$ by its integral $\mathsf{K}_{N,\beta c(t)}(\mu_t)$ gives $\Gamma_{N,t}$. Using (2.30), its logarithmic source is

$$\frac{\partial_t\Gamma_{N,t}-L_{N,t}^*\Gamma_{N,t}}{\Gamma_{N,t}}=-\beta(c(t)C_N[u_t]+c'(t)E_{N,t})-\frac{\mathrm{d}}{\mathrm{d}t}\log\mathsf{K}_{N,\beta c(t)}(\mu_t)=R_{N,t}. \tag{2.37}$$

This proves (2.32). These calculations hold at every smooth heat cutoff, where Gaussian tails justify the integrations by parts; Section 6.2 gives the passage to the singular particle laws.

For $c=1$, Jensen's inequality with both signs in (2.7) gives

$$\log\mathbb{E}_{\Gamma_{N,t}}[e^{-R_{N,t}/\lambda_0(t)}]\le C_0,\qquad \lambda_0(t)=\frac{\beta}{\kappa}\|\nabla u_t\|_{L^\infty}. \tag{2.38}$$

Indeed, if $\|\nabla u_t\|_{L^\infty}>0$, (2.32) and Jensen's inequality give

$$\begin{aligned}\log\mathbb{E}_{\Gamma_{N,t}}[e^{-R_{N,t}/\lambda_0(t)}]&=\log\mathbb{E}_{\Gamma_{N,t}}[e^{\kappa C_N[u_t]/\|\nabla u_t\|_{L^\infty}}]-\frac{\kappa}{\|\nabla u_t\|_{L^\infty}}\mathbb{E}_{\Gamma_{N,t}}[C_N[u_t]]\\&\le\log\mathbb{E}_{\Gamma_{N,t}}[e^{\kappa C_N[u_t]/\|\nabla u_t\|_{L^\infty}}]+\log\mathbb{E}_{\Gamma_{N,t}}[e^{-\kappa C_N[u_t]/\|\nabla u_t\|_{L^\infty}}]\le 2\log K.\end{aligned} \tag{2.39}$$

If $\|\nabla u_t\|_{L^\infty}=0$, the velocity is spatially constant, so $C_N[u_t]=R_{N,t}=0$.

For nonincreasing coupling an additional one-sided energy moment suffices. For every fixed $b>0$,

$$\sup_{\substack{N\ge2,\ 0\le c\le c_*\\ \mu\text{ satisfies (2.3)}}}\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[e^{-b(E_N-\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[E_N])}]\le C_b<\infty. \tag{2.40}$$

To estimate the mean subtracted here, the centered degree-two decomposition gives

$$E_N+I_\mu=\frac{1}{N}\sum_{i<j}\widetilde{\mathsf{g}}(x_i,x_j)-\frac{1}{N}\sum_{i=1}^N(h_\mu(x_i)-2I_\mu). \tag{2.41}$$

Orthogonality under $\mu^{\otimes N}$ then yields

$$\|E_N\|_{L^2(\mu^{\otimes N})}^2=I_\mu^2+\frac{N-1}{2N}\|\widetilde{\mathsf{g}}\|_{L^2(\mu^{\otimes2})}^2+\frac{1}{N}\|h_\mu-2I_\mu\|_{L^2(\mu)}^2\le C. \tag{2.42}$$

Consequently, Cauchy–Schwarz and (2.6) give

$$\sup_{\substack{N\ge2,\ 0\le c\le c_*\\ \mu\text{ satisfies (2.3)}}}\left|\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[E_N]\right|\le\sup_{N,c,\mu}\|G_{N,\beta c}\|_{L^2(\mu^{\otimes N})}\|E_N\|_{L^2(\mu^{\otimes N})}\le C. \tag{2.43}$$

The second supremum is over the same N,c,μ. The exact identity

$$\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[e^{-b(E_N-\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[E_N])}]=e^{b\mathbb{E}_{\mathbb{Q}_{N,\beta c}(\mu)}[E_N]}\frac{\mathsf{K}_{N,\beta c+b}(\mu)}{\mathsf{K}_{N,\beta c}(\mu)} \tag{2.44}$$

and (2.4) prove (2.40).

Set

$$\lambda_0(t):=\frac{\beta}{\kappa}\|\nabla u_t\|_{L^\infty}+\frac{\beta}{b}|c'(t)|. \tag{2.45}$$

For $\lambda_0(t)>0$, Hölder's inequality, (2.7), and (2.40) give

$$\log\mathbb{E}_{\Gamma_{N,t}}[e^{-R_{N,t}/\lambda_0(t)}]\le\frac{2\beta\|\nabla u_t\|_{L^\infty}\log K}{\kappa\lambda_0(t)}+\frac{\beta|c'(t)|\log C_b}{b\lambda_0(t)}\le\max\{2\log K,\log C_b\}. \tag{2.46}$$

Terms with zero coefficient are omitted; if $\lambda_0(t)=0$, then $R_{N,t}=0$. The sign $c'\le0$ is important: the required energy exponential has the repulsive, negative sign. Its coefficient b can be arbitrarily large and fixed, because the static partition estimate holds at every fixed inverse temperature. Thus, decreasing the coupling creates no uncontrolled positive-energy exponential.

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3. Full-particle integrability with a decreasing exponent

The aim of this section is to bound the full density in $L^2(\mu_t^{\otimes N})$. We first estimate the density relative to the moving Gibbs reference, allowing the exponent to decrease. We then compare the initial Gibbs preparation with that reference and pass to the product reference using the static density moments. The three steps are proposition 3.1, lemma 3.2, and corollary 3.3. The whole-space integrations by parts at a fixed heat cutoff are justified in lemma 6.3.

Proposition 3.1 (A varying-power estimate for a moving reference)**.** *Let P_t solve the diffusion forward equation and let a positive moving probability density Γ_t satisfy (2.31). Suppose*

$$\log \mathbb{E}_{\Gamma_t}[e^{-R_t/\lambda_0(t)}] \le C_0, \tag{3.1}$$

where λ_0 is nonnegative and locally integrable and $R_t = 0$ when $\lambda_0(t) = 0$. Work either with smooth positive data on a compact manifold or with the smooth whole-space approximations of lemma 6.3. For finite $p_0 > 1$, define

$$A(t) = \int_0^t \lambda_0(r)\,\mathrm{d}r, \qquad 1 - \frac{1}{p(t)} = \left(1 - \frac{1}{p_0}\right) e^{-A(t)}. \tag{3.2}$$

Then $q_t = \mathrm{d}P_t/\,\mathrm{d}\Gamma_t$ satisfies

$$\log\|q_t\|_{L^{p(t)}(\Gamma_t)} \le \log\|q_0\|_{L^{p_0}(\Gamma_0)} + C_0\left(1 - \frac{1}{p_0}\right)(1 - e^{-A(t)}). \tag{3.3}$$

Proof. Write b for the diffusion drift, retain the diffusion coefficient β^{-1}, and suppress the time variable when no confusion can arise. Substituting $P = q\Gamma$ into the forward equation gives

$$\partial_t q = \beta^{-1}\Delta q + (2\beta^{-1}\nabla \log \Gamma - b)\cdot\nabla q - Rq. \tag{3.4}$$

For fixed $p > 1$, multiply by $pq^{p-1}\Gamma$ and differentiate the moving measure. Integration by parts cancels the transport terms. The defect contributes $-pR + R$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\int q^p\,\mathrm{d}\Gamma = -\frac{4(p-1)}{\beta p}\int |\nabla(q^{p/2})|^2\,\mathrm{d}\Gamma - (p-1)\int Rq^p\,\mathrm{d}\Gamma. \tag{3.5}$$

Set $M_p = \int q^p\,\mathrm{d}\Gamma$ and introduce the normalized power tilt

$$\mathrm{d}\nu_p = \frac{q^p}{M_p}\,\mathrm{d}\Gamma = \frac{q^{p-1}}{M_p}\,\mathrm{d}P. \tag{3.6}$$

The measure ν_p identifies the contribution from the varying exponent as a relative entropy. Indeed, differentiating with $p = p(t)$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\|q\|_{L^{p(t)}(\Gamma_t)} = -\frac{4(p-1)}{\beta p^2}\frac{\int |\nabla(q^{p/2})|^2\,\mathrm{d}\Gamma}{M_p} + \frac{p-1}{p}\mathbb{E}_{\nu_p}[-R] + \frac{p'}{p^2}\mathsf{H}(\nu_p \mid \Gamma). \tag{3.7}$$

The term from differentiating the exponent is $p'p^{-2}(p\mathbb{E}_{\nu_p}[\log q] - \log M_p)$, and $p\mathbb{E}_{\nu_p}[\log q] - \log M_p = \mathsf{H}(\nu_p \mid \Gamma)$ by (3.6). For $\lambda_0(t) > 0$, apply the Donsker–Varadhan lemma, as in [DGR26a, Section 7.1], to $-R/\lambda_0$. Equivalently, use nonnegativity of relative entropy with respect to the probability measure whose density relative to Γ is proportional to e^{-R/λ_0}. Together with (3.1), this gives

$$\mathbb{E}_{\nu_p}[-R] \le \lambda_0(t)(\mathsf{H}(\nu_p \mid \Gamma) + C_0). \tag{3.8}$$

On the other hand, differentiating (3.2) gives

$$p' = -\lambda_0(t)p(p-1). \tag{3.9}$$

Thus, after applying (3.8) in (3.7), the two coefficients of $\mathsf{H}(\nu_p \mid \Gamma)$ cancel. Discarding the nonpositive Dirichlet term, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\|q\|_{L^{p(t)}(\Gamma_t)} \le C_0\lambda_0(t)\left(1 - \frac{1}{p(t)}\right) = -C_0\frac{\mathrm{d}}{\mathrm{d}t}\left(1 - \frac{1}{p(t)}\right). \tag{3.10}$$

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Integration proves the assertion. When $\lambda_0 = 0$, both the defect and exponent derivative vanish. The same computation holds almost everywhere for an absolutely continuous exponent. □

The cost of discarding diffusion. At a fixed exponent $p > 1$, the entropy contribution can instead be absorbed into diffusion when

$$\lambda_0 \operatorname{Ent}_\Gamma(q^p) \le \frac{4}{\beta p} \int |\nabla(q^{p/2})|^2 \, \mathrm{d}\Gamma. \tag{3.11}$$

A logarithmic Sobolev inequality with a sufficiently small constant would provide this estimate, but the static density moments do not supply it. Positive definiteness of the interaction operator is not convexity of the particle Hamiltonian: the tangential Hessian of a repulsive Riesz potential is negative and singular. The varying exponent cancels the entropy directly. Its price is the monotone loss of integrability, with $p(t) \downarrow 1$ if $A(t) \to \infty$.

Lemma 3.2 (Comparison of Gibbs preparations)**.** *Let $b_0, b_1 \ge 0$ and $p > 1$. Then*

$$\int \left(\frac{\mathrm{d}\mathbb{Q}_{N,b_0}(\mu)}{\mathrm{d}\mathbb{Q}_{N,b_1}(\mu)} \right)^p \mathrm{d}\mathbb{Q}_{N,b_1}(\mu) = \frac{\mathsf{K}_{N,b_1}(\mu)^{p-1} \mathsf{K}_{N,pb_0-(p-1)b_1}(\mu)}{\mathsf{K}_{N,b_0}(\mu)^p}, \tag{3.12}$$

whenever the right-hand side is finite. The same identity holds for the heat truncations. If the static estimates of proposition 2.1 hold and

$$pb_0 - (p-1)b_1 \ge 0, \tag{3.13}$$

then the left-hand side of (3.12) is bounded uniformly in N and in the heat cutoff. For fixed p, the bound is uniform when b_0 and b_1 range over a compact interval and (3.13) holds, with a constant depending only on that interval, p, and the static background class.

Proof. Substitution of the two Gibbs densities gives (3.12) directly. Under (3.13), every partition function on its right-hand side has a nonnegative fixed coupling. The upper bounds in (2.4), together with the Jensen lower bound for $\mathsf{K}_{N,b_0}(\mu)$, prove the uniform estimate. The cutoff statement is identical. □

Corollary 3.3 (Full-particle second moment)**.** *Assume the static inputs hold uniformly for a family of consistent approximations and the scalar exponential estimate holds with a common λ_0. Suppose that, for some finite $p_0 > 2$,*

$$\sup_{N,\varepsilon} \|q^\varepsilon_{N,0}\|_{L^{p_0}(\Gamma^\varepsilon_{N,0})} < \infty, \qquad \left(1 - \frac{1}{p_0}\right) e^{-A(T)} > \frac{1}{2}, \qquad A(T) = \int_0^T \lambda_0(t) \, \mathrm{d}t. \tag{3.14}$$

Then

$$\sup_{N,\varepsilon,\, 0 \le t \le T} \|f^\varepsilon_{N,t}\|^2_{L^2((\mu^\varepsilon_t)^{\otimes N})} < \infty. \tag{3.15}$$

The assertion is time-uniform when (3.14) holds with $T = \infty$.

Proof. We suppress the approximation parameter and set $p_* = p(T) > 2$, with $p(t)$ defined by (3.2). Then $p(t) \ge p_*$ on $[0, T]$. Since Γ_t is a probability measure, proposition 3.1 and the $L^{p(t)}(\Gamma_t)$ to $L^{p_*}(\Gamma_t)$ inclusion bound $\|q_t\|_{L^{p_*}(\Gamma_t)}$ uniformly over the family. Using $f_{N,t} = q_{N,t} G_{N,t}$ and Hölder with conjugate exponents $p_*/2$ and $p_*/(p_* - 2)$, we obtain

$$\int f^2_{N,t} \, \mathrm{d}\mu_t^{\otimes N} = \int q^2_{N,t} G_{N,t} \, \mathrm{d}\Gamma_{N,t} \le \|q_{N,t}\|^2_{L^{p_*}(\Gamma_{N,t})} \left(\int G_{N,t}^{\frac{2(p_*-1)}{p_*-2}} \, \mathrm{d}\mu_t^{\otimes N} \right)^{\frac{p_*-2}{p_*}}. \tag{3.16}$$

The last factor is uniformly bounded by (2.6); its exponent is finite because $p_* > 2$. This proves (3.15).

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In the matched case $P_{N,0} = \Gamma_{N,0}$, one may choose p_0 after fixing any $A(T) < \log 2$. For example, taking $p_0 = 8$ and $A(T) \le \log(7/6)$ gives $p(t) \ge 4$ and $\log\|q_t\|_{L^4(\Gamma_t)} \le C_0/8$. Therefore,

$$\int f_{N,t}^2 \, \mathrm{d}\mu_t^{\otimes N} \le e^{C_0/4} \left(\int G_{N,t}^3 \, \mathrm{d}\mu_t^{\otimes N} \right)^{1/2}. \tag{3.17}$$

The constants do not depend on time when the corresponding assumptions hold on the whole half-line. □

For the constant-coupling dynamics, take $b_0 = \beta_0$ and $b_1 = \beta$ in lemma 3.2. Writing $r = 1 - 1/p_0$, the following strict inequalities are sufficient for (3.14):

$$\frac{1}{2} e^{A(T)} < r < \min\left\{1, \frac{\beta_0}{\beta}\right\}. \tag{3.18}$$

Such an r exists exactly under the sufficient condition in (1.20). In the matched case, the resulting threshold is $\log 2$; it records the need to leave an exponent above two in (3.16). Neither this value nor its different-temperature counterpart is claimed sharp. One cannot restart the calculation with relative density one at a later time without replacing the actual particle law by the instantaneous Gibbs law. The periodic counterexample in theorem 1.7 shows that some restriction is genuinely needed if the target is a uniform full-particle second moment, but it does not identify the optimal hypothesis or a sharp numerical threshold.

4. Weighted two-point, elliptic, and parabolic estimates

The first-marginal argument in Section 5 uses three estimates. The two-point commutator is bounded by a mixed norm involving both the gradient and Hessian of the test function. Weighted elliptic duality controls that norm in the initial preparation argument, while backward smoothing controls it along the mean-field trajectory. The latter two estimates follow from the one-body Bochner calculus and nonautonomous Kolmogorov theory; we give the proofs in the weighted form needed here.

4.1. **A weighted two-point commutator.**

For a symmetric $\eta \in L^2(\mu^{\otimes 2})$ and $0 \le \varepsilon \le 1$, define

$$\langle S_\eta^\varepsilon, \psi \rangle := -\frac{1}{2} \iint \eta(x,y)(\nabla\psi(x) - \nabla\psi(y)) \cdot \nabla \mathsf{g}_\varepsilon(x-y) \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y), \qquad S_\eta := S_\eta^0. \tag{4.1}$$

For $\varepsilon > 0$, symmetry of η and oddness of $\nabla \mathsf{g}_\varepsilon$ identify this functional with the distribution

$$S_\eta^\varepsilon = \mathrm{div}_x \left(\mu(x) \int \nabla \mathsf{g}_\varepsilon(x-y) \eta(x,y) \, \mathrm{d}\mu(y) \right). \tag{4.2}$$

We do not assume that $\int \eta(x,y) \, \mathrm{d}\mu(y) = 0$ for μ-almost every x; only symmetry of η is used below.

Lemma 4.1 (Weighted two-point commutator estimate)**.** *Let $\mu = e^{-U}$ be a probability density satisfying*

$$\|\mu\|_\infty \le M, \qquad \nabla^2 U \ge -K \mathrm{Id}, \qquad K \ge 0. \tag{4.3}$$

Fix $R > 0$ and $0 \le \alpha \le 1$, and assume

$$J_{\alpha,R}^2 := \sup_{0 \le \varepsilon \le 1} \int_{|z| \le R} |z|^{2\alpha} |\nabla \mathsf{g}_\varepsilon(z)|^2 \, \mathrm{d}z < \infty, \tag{4.4}$$

$$B_R := \sup_{0 \le \varepsilon \le 1} \operatorname*{ess\,sup}_{|z| > R} |\nabla \mathsf{g}_\varepsilon(z)| < \infty. \tag{4.5}$$

Then, for every $0 \le \varepsilon \le 1$,

$$|\langle S_\eta^\varepsilon, \psi \rangle| \le \|\eta\|_{L^2(\mu^{\otimes 2})} \Big(C_\alpha \sqrt{M} \, e^{\alpha K R^2/16} J_{\alpha,R} \|\nabla\psi\|_{L^2(\mu)}^{1-\alpha} \|\nabla^2 \psi\|_{L^2(\mu)}^{\alpha} + B_R \|\nabla\psi\|_{L^2(\mu)} \Big), \qquad \forall \psi \in C_c^\infty(\mathbb{R}^{\mathsf{d}}). \tag{4.6}$$

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Consequently, S_η^ε extends uniquely to a bounded linear functional on the closure of $C_c^\infty(\mathbb{R}^\mathsf{d})$ in $H^2(\mu)$, with the same estimate and constants independent of ε. Here, $H^2(\mu)$ carries the norm obtained by summing the weighted L^2 norms of derivatives of orders zero, one, and two.

Proof. The lower Hessian bound on U gives, for $w = (1-\theta)x + \theta y$,

$$\mu(x)\mu(y) \le e^{K\theta(1-\theta)|x-y|^2/2}\mu(w)\mu(x)^\theta\mu(y)^{1-\theta} \le Me^{K|x-y|^2/8}\mu(w). \tag{4.7}$$

We estimate the gradient difference at each displacement z. Splitting its square into the two gradient terms and bounding the unused density by M, we obtain

$$\int \mu(x)\mu(x-z)|\nabla\psi(x) - \nabla\psi(x-z)|^2\,\mathrm{d}x \le 4M\|\nabla\psi\|_{L^2(\mu)}^2. \tag{4.8}$$

Alternatively, integrate the Hessian along the segment from x to $x-z$, apply Jensen's inequality in θ, and use (4.7). The change of variables $x \mapsto x - \theta z$ then gives

$$\begin{aligned}&\int \mu(x)\mu(x-z)|\nabla\psi(x) - \nabla\psi(x-z)|^2\,\mathrm{d}x\\ &\quad\le |z|^2\int_0^1\int \mu(x)\mu(x-z)|\nabla^2\psi(x-\theta z)|^2\,\mathrm{d}x\,\mathrm{d}\theta \le Me^{K|z|^2/8}|z|^2\|\nabla^2\psi\|_{L^2(\mu)}^2.\end{aligned} \tag{4.9}$$

The left-hand sides of (4.8) and (4.9) are the same. Raising their right-hand sides to the powers $1-\alpha$ and α, respectively, and integrating against $|\nabla \mathsf{g}_\varepsilon(z)|^2$ on $|z| \le R$, yields

$$\begin{aligned}&\iint_{|x-y|\le R} |\nabla\psi(x) - \nabla\psi(y)|^2|\nabla \mathsf{g}_\varepsilon(x-y)|^2\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y)\\ &\quad\le 4^{1-\alpha}Me^{\alpha KR^2/8}J_{\alpha,R}^2\|\nabla\psi\|_{L^2(\mu)}^{2(1-\alpha)}\|\nabla^2\psi\|_{L^2(\mu)}^{2\alpha}.\end{aligned} \tag{4.10}$$

On the complementary region $|x-y| > R$, we use $|\nabla \mathsf{g}_\varepsilon(x-y)| \le B_R$ and $\mu(\mathbb{R}^\mathsf{d}) = 1$:

$$\begin{aligned}&\iint_{|x-y|>R} |\nabla\psi(x) - \nabla\psi(y)|^2|\nabla \mathsf{g}_\varepsilon(x-y)|^2\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y)\\ &\quad\le 2B_R^2\iint (|\nabla\psi(x)|^2 + |\nabla\psi(y)|^2)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) = 4B_R^2\|\nabla\psi\|_{L^2(\mu)}^2.\end{aligned} \tag{4.11}$$

Cauchy–Schwarz in (4.1), followed by (4.10) and (4.11), proves (4.6). For $\alpha = 0$ or $\alpha = 1$, use (4.8) or (4.9) directly. The extension follows by density in the indicated completion. □

By lemma 2.3, the log/Riesz kernels (1.1) verify the force assumptions with $R = 1$ whenever

$$\max\{0, \mathsf{s} + 1 - \mathsf{d}/2\} < \alpha < 1. \tag{4.12}$$

This interval is nonempty throughout $0 \le \mathsf{s} < \mathsf{d}/2$. Choosing $\alpha < 1$ will make the time singularity in the application to proposition 5.2 integrable.

4.2. **Weighted elliptic duality.** Let

$$\mathcal{A}_\mu := -\mu^{-1}\operatorname{div}(\mu\nabla) = -\Delta + \nabla U\cdot\nabla \tag{4.13}$$

denote the nonnegative operator associated with the closed form $\int|\nabla\psi|^2\,\mathrm{d}\mu$ on $L^2(\mu)$.

Lemma 4.2 (One-body elliptic estimate)**.** *Suppose $\mu = e^{-U}$ is a smooth probability density satisfying*

$$\|\nabla^2 U\|_{L^\infty} < \infty, \qquad \nabla^2 U \ge -K\mathrm{Id}, \qquad K \ge 0. \tag{4.14}$$

Assume also the Poincaré inequality

$$\int |\psi - \mu(\psi)|^2\,\mathrm{d}\mu \le C_P\int|\nabla\psi|^2\,\mathrm{d}\mu. \tag{4.15}$$

For each $\zeta \in L_0^2(\mu)$, the centered solution $\mathcal{A}_\mu\psi = \zeta$ satisfies

$$\|\nabla\psi\|_{L^2(\mu)} \le \sqrt{C_P}\|\zeta\|_{L^2(\mu)}, \qquad \|\nabla^2\psi\|_{L^2(\mu)} \le \sqrt{1+KC_P}\|\zeta\|_{L^2(\mu)}. \tag{4.16}$$

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Proof. The Poincaré inequality makes $\int \nabla\psi \cdot \nabla\varphi \, \mathrm{d}\mu$ a Hilbert-space inner product on the centered form domain, and $\varphi \mapsto \int \zeta\varphi \, \mathrm{d}\mu$ is continuous for the resulting norm. The Riesz–Fréchet representation theorem (see, e.g., [Bre11, Theorem 5.5]) therefore gives the centered weak solution. Testing its equation with ψ proves the first estimate in (4.16).

The integrated Bochner identity for the weighted diffusion operator is

$$\int |\mathcal{A}_\mu \psi|^2 \, \mathrm{d}\mu = \int |\nabla^2 \psi|^2 \, \mathrm{d}\mu + \int \nabla^2 U(\nabla\psi, \nabla\psi) \, \mathrm{d}\mu. \tag{4.17}$$

This is the standard Γ_2 identity; see [BGL14, Chapters 1 and 4]. We justify its extension to the weak solution. Local elliptic regularity gives $\psi \in H^2_{\mathrm{loc}}$. Choose $\chi \in C_c^\infty$ equal to one on B_1 and zero outside B_2, and set $\chi_R(x) := \chi(x/R)$. Then

$$\mathcal{A}_\mu(\chi_R\psi) = \chi_R\zeta - 2\nabla\chi_R \cdot \nabla\psi + \psi\mathcal{A}_\mu\chi_R. \tag{4.18}$$

The bounded Hessian of U gives at most linear growth of ∇U. Thus, $\mathcal{A}_\mu\chi_R$ is uniformly bounded for $R \geq 1$ and supported in $B_{2R} \setminus B_R$. Since $\psi, \nabla\psi \in L^2(\mu)$, we obtain

$$\|\chi_R\psi - \psi\|_{L^2(\mu)} + \|\mathcal{A}_\mu(\chi_R\psi) - \zeta\|_{L^2(\mu)} \longrightarrow 0 \qquad (R \to \infty). \tag{4.19}$$

For each fixed R, mollify the compactly supported H^2 function $\chi_R\psi$, extended by zero to $\mathbb{R}^{\mathsf{d}}$. On a compact neighborhood of its support, the weight is bounded above and below and ∇U is bounded. Convergence of these mollifications in unweighted H^2 therefore implies convergence in the graph norm in (4.19). Choosing $R = n$ and a sufficiently small mollification scale gives $\psi_n \in C_c^\infty(\mathbb{R}^{\mathsf{d}})$ such that

$$\|\psi_n - \psi\|_{L^2(\mu)} + \|\mathcal{A}_\mu\psi_n - \zeta\|_{L^2(\mu)} \longrightarrow 0. \tag{4.20}$$

The Dirichlet-form identity gives convergence of $\nabla\psi_n$ in $L^2(\mu)$. Applying (4.17) to $\psi_n - \psi_m$ gives

$$\|\nabla^2(\psi_n - \psi_m)\|^2_{L^2(\mu)} \leq \|\mathcal{A}_\mu(\psi_n - \psi_m)\|^2_{L^2(\mu)} + K\|\nabla(\psi_n - \psi_m)\|^2_{L^2(\mu)} \longrightarrow 0. \tag{4.21}$$

Local positivity of μ identifies the limit as the distributional Hessian of ψ. The upper bound on $\nabla^2 U$ then permits passage to the limit in (4.17). Using its lower bound and the first estimate, we conclude that

$$\|\nabla^2\psi\|^2_{L^2(\mu)} \leq \|\zeta\|^2_{L^2(\mu)} + K\|\nabla\psi\|^2_{L^2(\mu)} \leq (1 + KC_P)\|\zeta\|^2_{L^2(\mu)}. \tag{4.22}$$

□

Combining the two lemmas gives the estimate used in proposition 5.1:

$$\sup_{0\leq\varepsilon\leq 1} |\langle S^\varepsilon_\eta, \mathcal{A}_\mu^{-1}\zeta\rangle| \leq \left(C_\alpha\sqrt{M}e^{\alpha K R^2/16} J_{\alpha,R} C_P^{(1-\alpha)/2}(1 + KC_P)^{\alpha/2} + B_R\sqrt{C_P}\right) \times \|\eta\|_{L^2(\mu^{\otimes 2})}\|\zeta\|_{L^2(\mu)}, \qquad \zeta \in L^2_0(\mu). \tag{4.23}$$

4.3. **Backward smoothing in the moving weight.** Let μ_t be a smooth positive probability density satisfying

$$\mathcal{L}_t := \beta^{-1}\Delta + B_t \cdot \nabla, \qquad \partial_t\mu_t = \mathcal{L}_t^*\mu_t. \tag{4.24}$$

Under the spatial Lipschitz and linear-growth bounds below, the stochastic differential equation with drift B_t and diffusion coefficient $\sqrt{2/\beta}$ has a unique nonexplosive strong solution from each point. This is the standard existence and uniqueness theorem (see, e.g., [Øks03, Theorem 5.2.1]). Its transition operators are

$$P_{r,t}\psi(x) := \mathbb{E}[\psi(x_t^{r,x})], \qquad \mathrm{d}x_v^{r,x} = B_v(x_v^{r,x})\,\mathrm{d}v + \sqrt{2/\beta}\,\mathrm{d}W_v, \quad x_r^{r,x} = x. \tag{4.25}$$

The Markov property, also for time-dependent coefficients, follows by restarting the solution with independent Brownian increments; see [Øks03, Section 7.1]. Thus, $P_{r,t} = P_{r,v}P_{v,t}$ for $r \leq v \leq t$, and uniqueness of the forward equation gives $\mu_r P_{r,t} = \mu_t$. In proposition 5.2, the drift is $B_t = -\nabla V - c(t)\nabla h_t$.

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Lemma 4.3 (Weighted backward derivative estimates)**.** *Assume that*

$$B \in C([0,T]; C^2_{\mathrm{loc}}(\mathbb{R}^{\mathsf{d}}; \mathbb{R}^{\mathsf{d}})), \qquad \sup_{t\in[0,T]} \sup_{x\in\mathbb{R}^{\mathsf{d}}} \frac{|B_t(x)|}{1+|x|} < \infty, \tag{4.26}$$

where continuity into C^2_{loc} means locally uniform continuity of the spatial derivatives through order two. Suppose also that

$$\sup_{t\in[0,T]} \|\nabla B_t\|_{L^\infty} < \infty, \qquad \sup_{t\in[0,T]} \|\nabla^2 B_t\|_{L^\infty} < \infty. \tag{4.27}$$

For $0 \le r < t \le T$ and $\psi \in L^2(\mu_t)$,

$$\|P_{r,t}\psi\|_{L^2(\mu_r)} \le \|\psi\|_{L^2(\mu_t)}, \tag{4.28}$$

$$\|\nabla P_{r,t}\psi\|_{L^2(\mu_r)} \le C_T (t-r)^{-1/2} \|\psi\|_{L^2(\mu_t)}, \tag{4.29}$$

$$\|\nabla^2 P_{r,t}\psi\|_{L^2(\mu_r)} \le C_T (t-r)^{-1} \|\psi\|_{L^2(\mu_t)}. \tag{4.30}$$

The constants depend only on T, β, d and the two bounds in (4.27).

Proof. Jensen's inequality and $\mu_r P_{r,t} = \mu_t$ give

$$\int |P_{r,t}\psi|^2 \,\mathrm{d}\mu_r \le \int P_{r,t}(|\psi|^2)\,\mathrm{d}\mu_r = \int |\psi|^2 \,\mathrm{d}\mu_t. \tag{4.31}$$

For the derivatives, we use the Bernstein–Bochner calculation for the nonautonomous propagator; see [KLL10, AL16, Lun95] for the evolution-family and derivative estimates, and [BGL14] for the diffusion calculus.

First let $\psi \in C_c^\infty(\mathbb{R}^{\mathsf{d}})$ and set $F_r := P_{r,t}\psi$. Its backward equation and (4.24) give

$$\partial_r F_r = -\mathcal{L}_r F_r, \qquad \frac{\mathrm{d}}{\mathrm{d}r} \int a_r \,\mathrm{d}\mu_r = \int (\partial_r a_r + \mathcal{L}_r a_r)\,\mathrm{d}\mu_r. \tag{4.32}$$

The second identity is initially applied with a smooth spatial cutoff. Set

$$E(r) := \int F_r^2 \,\mathrm{d}\mu_r, \qquad G(r) := \int |\nabla F_r|^2 \,\mathrm{d}\mu_r, \qquad H(r) := \int |\nabla^2 F_r|^2 \,\mathrm{d}\mu_r. \tag{4.33}$$

The product rule for $\mathcal{L}_r$, applied to F_r^2, $|\nabla F_r|^2$, and $|\nabla^2 F_r|^2$, yields

$$E'(r) = \frac{2}{\beta} G(r), \tag{4.34}$$

$$G'(r) = \frac{2}{\beta} H(r) - 2 \sum_{i,j=1}^{\mathsf{d}} \int \partial_i F_r \, \partial_i B_r^j \, \partial_j F_r \,\mathrm{d}\mu_r \ge \frac{2}{\beta} H(r) - C G(r), \tag{4.35}$$

and

$$\begin{aligned} H'(r) = \frac{2}{\beta} \int |\nabla^3 F_r|^2 \,\mathrm{d}\mu_r &- 4 \sum_{i,j,\ell=1}^{\mathsf{d}} \int \partial_{ij} F_r \, \partial_i B_r^\ell \, \partial_{j\ell} F_r \,\mathrm{d}\mu_r \\ &- 2 \sum_{i,j,\ell=1}^{\mathsf{d}} \int \partial_{ij} F_r \, \partial_{ij} B_r^\ell \, \partial_\ell F_r \,\mathrm{d}\mu_r \\ &\ge -C(H(r) + G(r)). \end{aligned} \tag{4.36}$$

The first two derivatives of B_r bound the drift terms, and the nonnegative third-derivative integral in (4.36) has been discarded.

We justify the integrated equality in (4.34) and the integrated inequalities in (4.35) and (4.36) without a decay assumption on μ_r. For smooth compactly supported ψ, the equations for the first two spatial derivatives of the stochastic flow give bounded F_r, ∇F_r, and $\nabla^2 F_r$ on $0 \le r \le t$. Apply Itô's formula along the diffusion stopped on balls. Nonexplosion and these bounds permit passage

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to the limit in the terminal evaluations and drift terms; Fatou's lemma applies to the nonnegative derivative integrals.

The computation may first be made after spatial mollification of B. To pass to the regularity in (4.26), fix $r < t$, start each mollified diffusion with law μ_r, and use its own law at each intermediate time in the preceding calculation. The first two drift-derivative bounds are preserved by mollification. The stochastic flows and their first two derivatives converge after localization, while the evaluations of ψ and its derivatives remain bounded uniformly in the mollification parameter. Dominated convergence therefore gives the stated integrated relations with the original laws μ_v, $r \le v \le t$.

Since (4.35) implies $G'(r) + CG(r) \ge 0$, the function $e^{Cr}G(r)$ is nondecreasing. Hence $G(r) \le e^{C(v-r)}G(v)$ for $r \le v \le t$. Setting $h := t - r$ and integrating (4.34), we obtain

$$\int_r^t G(v)\,\mathrm{d}v \le \frac{\beta}{2}E(t), \tag{4.37}$$

$$G(r) \le \frac{e^{CT}}{h}\int_r^t G(v)\,\mathrm{d}v \le C_T h^{-1}E(t). \tag{4.38}$$

Apply the latter estimate also at $r + h/2$. Integrating (4.35) over $[r, r + h/2]$ gives

$$\int_r^{r+h/2} H(v)\,\mathrm{d}v \le \frac{\beta}{2}\left(G(r + h/2) - G(r) + C\int_r^{r+h/2} G(v)\,\mathrm{d}v\right) \le C_T h^{-1}E(t). \tag{4.39}$$

By (4.36), for $r \le v \le r + h/2$,

$$H(r) \le e^{C(v-r)}H(v) + C\int_r^v e^{C(a-r)}G(a)\,\mathrm{d}a. \tag{4.40}$$

Averaging in v and using (4.37) and (4.39), we conclude that

$$H(r) \le \frac{C_T}{h}\int_r^{r+h/2} H(v)\,\mathrm{d}v + C_T\int_r^t G(v)\,\mathrm{d}v \le C_T h^{-2}E(t), \tag{4.41}$$

where $h \le T$ was used in the last inequality. Taking square roots proves the derivative estimates for $\psi \in C_c^\infty$.

For a general $\psi \in L^2(\mu_t)$, choose $\psi_n \in C_c^\infty(\mathbb{R}^{\mathsf{d}})$ with $\psi_n \to \psi$ in $L^2(\mu_t)$. Applying (4.28)–(4.30) to $\psi_n - \psi_m$ gives, for $j = 0, 1, 2$,

$$\|\nabla^j P_{r,t}(\psi_n - \psi_m)\|_{L^2(\mu_r)} \le C_T(t - r)^{-j/2}\|\psi_n - \psi_m\|_{L^2(\mu_t)} \longrightarrow 0. \tag{4.42}$$

For $j = 0$, the limit is $P_{r,t}\psi$ by (4.31). Positivity and local smoothness of μ_r identify the other two limits as its weak derivatives. Passing to the limit proves the lemma. □

5. One-mark control and the global all-marginal enhancement

The full second moment from Section 3 controls the two-particle projection that appears as a source in the first-marginal equation (5.18). We first estimate the initial marginal by weighted elliptic duality, then propagate this estimate by the backward bounds of Section 4. For the uniform-in-time result, we combine a homogeneous energy estimate with the finite-time source estimate on successive intervals. Finally, lemma 5.4 transfers the first-marginal estimate and the full second moment to every $1 \le k \le N$.

We first work with the heat-truncated interaction. In this section, we suppress the cutoff from g, the corresponding background, and the particle densities; the constants in the estimates are independent of the cutoff. The interaction coefficient is the function c introduced in Section 2. On a finite interval $[0, T]$, the common background assumptions are (2.3), the static estimates of proposition 2.1, and

$$\sup_{0\le t\le T}\left(\|\nabla^2 U_t\|_{L^\infty} + \|\nabla B_t\|_{L^\infty} + \|\nabla^2 B_t\|_{L^\infty}\right) < \infty, \qquad U_t := -\log \mu_t. \tag{5.1}$$

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Here, $B_t = -\nabla V - c(t)\nabla h_t$ is the nonlinear diffusion drift, as in (1.13) for $c \equiv 1$, and $\partial_t \mu_t = \mathcal{L}_t^* \mu_t$ with the generator (4.24). The Poincaré inequality for μ_t is assumed where the elliptic and uniform-in-time estimates require it. The concrete verification of these hypotheses is in Section 6.1 and Appendix A; the passage to the singular particle law is in Section 6.2.

We will use the following consequence of lemma 2.3. If $\|\mu\|_{L^\infty} \le M$, then

$$\sup_x \int |\nabla \mathsf{g}(x-y)| \, \mathrm{d}\mu(y) \le M \|\nabla \mathsf{g}\|_{L^1(B_1)} + \|\nabla \mathsf{g}\|_{L^\infty(B_1^c)} \le C_M. \tag{5.2}$$

Schur's test applied to the kernel $|\nabla \mathsf{g}(x-y)|$ gives

$$\|\nabla \mathsf{g} * (\mu a)\|_{L^2(\mu)} \le C_M \|a\|_{L^2(\mu)}, \qquad \|\nabla h_\mu\|_\infty \le C_M. \tag{5.3}$$

Only local integrability of $\nabla \mathsf{g}$, supplied by $\mathsf{s} + 1 < \mathsf{d}$, is used here; square integrability of the force is not required.

5.1. **The initial marginal.** The next estimate uses the screened inverse of lemma 2.4 and does not require positive definiteness of the interaction.

Proposition 5.1 (Weighted initial one-mark estimate)**.** *Let $\mu_0 = e^{-U_0}$ satisfy the assumptions of lemma 4.2 and the Gaussian upper bound (2.3). For every fixed $b \ge 0$, the initial law $\mathbb{Q}_{N,b}(\mu_0)$ satisfies*

$$\|G_{N,b}^{(1)} - 1\|_{L^2(\mu_0)} \le \frac{C}{N}. \tag{5.4}$$

The constant depends on b, the one-body Poincaré and Hessian bounds, the constants in (2.3), the static second moment, and the inverse bound in lemma 2.4. It is uniform when b ranges over a fixed compact interval.

Proof. For $b = 0$, the initial law is the product law and there is nothing to prove. For $b > 0$, suppress the initial time and the preparation parameter in the densities, and set

$$a := G_N^{(1)} - 1, \qquad \eta(x,y) := G_N^{(2)}(x,y) - 1 - a(x) - a(y), \qquad M_0 := \sup_{N \ge 2} \int G_N^2 \, \mathrm{d}\mu^{\otimes N}. \tag{5.5}$$

The moment M_0 is finite by (2.6). Both a and η have zero average in each of their coordinates. The orthogonal decomposition in lemma 5.4 gives

$$\|\eta\|_{L^2(\mu^{\otimes 2})} \le \left(\frac{2(M_0 - 1)}{N(N-1)} \right)^{1/2} \le \frac{2\sqrt{M_0 - 1}}{N}, \tag{5.6}$$

$$\|a\|_{L^2(\mu)} \le \sqrt{M_0 - 1}. \tag{5.7}$$

Differentiating the Gibbs density before removing the cutoff, and then integrating the other $N-1$ coordinates, gives

$$\nabla a = \frac{b}{N}(1+a)\nabla h - b\frac{N-1}{N}\left(\nabla \mathsf{g} * (\mu a) + \int \nabla \mathsf{g}(x-y)\eta(x,y) \, \mathrm{d}\mu(y) \right). \tag{5.8}$$

In fact, the two-particle term in the differentiated marginal is the integral of $\nabla \mathsf{g}(x-y)$ against $\mu(y)[1 + a(x) + a(y) + \eta(x,y)]$. Its first two terms are $(1 + a(x))\nabla h(x)$, which leave precisely the coefficient b/N in (5.8).

Since a is centered, $\nabla T_\mu a = \nabla \mathsf{g} * (\mu a)$. Hence the centered function

$$z := \left(\mathrm{Id} + b\frac{N-1}{N} T_\mu \right) a \tag{5.9}$$

satisfies the distributional equation

$$-\operatorname{div}(\mu \nabla z) = -\frac{b}{N} \operatorname{div}(\mu(1+a)\nabla h) + b\frac{N-1}{N} S_\eta. \tag{5.10}$$

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For $\zeta \in L^2_0(\mu)$, let ψ be the centered solution of $\mathcal{A}_\mu \psi = \zeta$ from lemma 4.2. Pairing (5.10) with ψ yields

$$\int z\zeta \,\mathrm{d}\mu = \frac{b}{N}\int (1+a)\nabla h \cdot \nabla \psi \,\mathrm{d}\mu + b\frac{N-1}{N}\langle S_\eta, \psi\rangle. \tag{5.11}$$

The first term is controlled by (5.3). For the second, apply lemma 4.1 with $\alpha = 1$. Thus,

$$\begin{aligned}\left|\int z\zeta \,\mathrm{d}\mu\right| &\le \frac{C}{N}(1+\|a\|_{L^2(\mu)})\|\nabla\psi\|_{L^2(\mu)} + C\|\eta\|_{L^2(\mu^{\otimes 2})}\left(\|\nabla^2\psi\|_{L^2(\mu)} + \|\nabla\psi\|_{L^2(\mu)}\right)\\ &\le C\left(\frac{1+\|a\|_{L^2(\mu)}}{N} + \|\eta\|_{L^2(\mu^{\otimes 2})}\right)\|\zeta\|_{L^2(\mu)}.\end{aligned} \tag{5.12}$$

The second inequality is lemma 4.2. Its graph-core approximation justifies (5.11): the left side is continuous in the operator graph norm, and the two sources are continuous in the weighted gradient and Hessian norms. Taking the supremum over centered ζ of unit norm, using (5.6)–(5.7), and applying lemma 2.4 gives (5.4). □

5.2. **The exact first-marginal equation.** Write $f^{(k)}_{N,t} := \frac{\mathrm{d}P^{(k)}_{N,t}}{\mathrm{d}\mu_t^{\otimes k}}$, and set

$$\begin{aligned} a_{N,t} &:= f^{(1)}_{N,t} - 1, \qquad w_{N,t} := \mu_t a_{N,t}, \qquad \theta_N := \frac{N-1}{N},\\ \eta_{N,t}(x,y) &:= f^{(2)}_{N,t}(x,y) - 1 - a_{N,t}(x) - a_{N,t}(y).\end{aligned} \tag{5.13}$$

The normalization and marginal identities imply

$$\int a_{N,t}\,\mathrm{d}\mu_t = 0, \qquad \int \eta_{N,t}(x,y)\,\mathrm{d}\mu_t(y) = \int \eta_{N,t}(x,y)\,\mathrm{d}\mu_t(x) = 0 \quad \text{a.e.} \tag{5.14}$$

If the full moment is at most M_*, lemma 5.4 gives

$$\|\eta_{N,t}\|_{L^2(\mu_t^{\otimes 2})} \le \frac{2\sqrt{M_*-1}}{N}. \tag{5.15}$$

The two-particle law has Lebesgue density

$$\frac{\mathrm{d}P^{(2)}_{N,t}}{\mathrm{d}x\,\mathrm{d}y}(x,y) = \mu_t(x)\mu_t(y) + w_{N,t}(x)\mu_t(y) + \mu_t(x)w_{N,t}(y) + \mu_t(x)\mu_t(y)\eta_{N,t}(x,y). \tag{5.16}$$

Since the one-body part of the particle drift is $B_t + c(t)\nabla h_t$, the first equation of the marginal hierarchy is

$$\partial_t(\mu_t f^{(1)}_{N,t}) = \mathcal{L}_t^*(\mu_t f^{(1)}_{N,t}) + c(t)\,\mathrm{div}\left(\theta_N \mu_t(x)\int \nabla \mathsf{g}(x-y) f^{(2)}_{N,t}(x,y)\,\mathrm{d}\mu_t(y) - \mu_t f^{(1)}_{N,t}\nabla h_t\right), \tag{5.17}$$

where $\mathcal{L}_t^* = \beta^{-1}\Delta - \mathrm{div}(B_t\,\cdot)$. Substituting (5.16) into (5.17) and subtracting $\partial_t\mu_t = \mathcal{L}_t^*\mu_t$ gives

$$\partial_t w_{N,t} = \mathcal{L}_t^* w_{N,t} + \theta_N c(t)\,\mathrm{div}(\mu_t \nabla \mathsf{g} * w_{N,t}) - \frac{c(t)}{N}\,\mathrm{div}(w_{N,t}\nabla h_t) - \frac{c(t)}{N}\,\mathrm{div}(\mu_t\nabla h_t) + \theta_N c(t) S_{\eta_{N,t}}. \tag{5.18}$$

The terms proportional to ∇h_t in (5.17) have the combined coefficient $c(t)(\theta_N - 1) = -c(t)/N$. They therefore contribute $-\frac{c(t)}{N}\,\mathrm{div}((\mu_t + w_{N,t})\nabla h_t)$ to (5.18). The part linear in $w_{N,t}$ remains in the homogeneous equation; the last two terms of (5.18) are the prescribed sources.

Proposition 5.2 (Finite-time weighted one-mark estimate)**.** *Suppose the bounds for μ_t and B_t at the beginning of this section hold on $[0,T]$, and assume*

$$\sup_{N\ge 2}\sup_{0\le t\le T}\|f_{N,t}\|^2_{L^2(\mu_t^{\otimes N})} \le M_* < \infty, \tag{5.19}$$

$$\|a_{N,0}\|_{L^2(\mu_0)} \le \frac{C_0}{N}. \tag{5.20}$$

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Then

$$\sup_{0\le t\le T}\|a_{N,t}\|_{L^2(\mu_t)} \le \frac{C_T}{N}. \tag{5.21}$$

More generally, for an interval length $L>0$, whenever the same bounds for μ_t and B_t hold on $[r_0, r_0+L]$, the linear equation (5.18) with a prescribed measurable symmetric two-particle source satisfying (5.15) satisfies

$$\sup_{r_0\le t\le r_0+L}\|a_{N,t}\|_{L^2(\mu_t)} \le C_L\left(\|a_{N,r_0}\|_{L^2(\mu_{r_0})}+\frac{1}{N}\right). \tag{5.22}$$

The constant C_L depends on the interval length L, the common bounds for μ_t and B_t, β, c_*, and M_*. The constant C_T has the same dependencies with $L=T$ and also depends on C_0. In particular, the constant in (5.22) is independent of r_0 when those bounds are uniform in time. The zero-initial-data response to the last two terms of (5.18) is therefore bounded by C_L/N.

Proof. Fix $t\in(0,T]$ and $\psi\in L^2(\mu_t)$. Testing with the backward propagator $P_{r,t}\psi$ from lemma 4.3 gives

$$\begin{aligned}\int a_{N,t}\psi\,\mathrm{d}\mu_t = &\int a_{N,0}P_{0,t}\psi\,\mathrm{d}\mu_0\\ &-\int_0^t c(r)\int\left(\theta_N\nabla \mathsf{g}*(\mu_r a_{N,r})-\frac{1+a_{N,r}}{N}\nabla h_r\right)\cdot\nabla P_{r,t}\psi\,\mathrm{d}\mu_r\,\mathrm{d}r\\ &+\theta_N\int_0^t c(r)\langle S_{\eta_{N,r}},P_{r,t}\psi\rangle\,\mathrm{d}r.\end{aligned} \tag{5.23}$$

The contraction in (4.28) bounds the first term by $\|a_{N,0}\|_{L^2(\mu_0)}\|\psi\|_{L^2(\mu_t)}$. For the flux term, (5.3) and (4.29) give

$$\left|\int\left(\theta_N\nabla\mathsf{g}*(\mu_r a_{N,r})-\frac{1+a_{N,r}}{N}\nabla h_r\right)\cdot\nabla P_{r,t}\psi\,\mathrm{d}\mu_r\right| \le C_T\frac{\|a_{N,r}\|_{L^2(\mu_r)}+N^{-1}}{\sqrt{t-r}}\|\psi\|_{L^2(\mu_t)}. \tag{5.24}$$

Choose α as in (4.12), so in particular $\alpha<1$. Combining lemmas 4.1 and 4.3 with (5.15) bounds the two-particle integrand by

$$\frac{C_T}{N}\left((t-r)^{-(1+\alpha)/2}+(t-r)^{-1/2}\right)\|\psi\|_{L^2(\mu_t)}. \tag{5.25}$$

Indeed, the mixed term in the commutator estimate is bounded by $(t-r)^{-(1-\alpha)/2-\alpha}\|\psi\|_{L^2(\mu_t)}$. Both time singularities in (5.25) are integrable. After integrating the source terms and taking the supremum over $\|\psi\|_{L^2(\mu_t)}\le 1$, we obtain

$$y(t)\le\frac{C_T}{N}+C_T\int_0^t(t-r)^{-1/2}y(r)\,\mathrm{d}r,\qquad y(t):=\|a_{N,t}\|_{L^2(\mu_t)}. \tag{5.26}$$

The constant includes the initial bound C_0/N. Substituting this inequality into its integral term and using

$$\int_s^t(t-r)^{-1/2}(r-s)^{-1/2}\,\mathrm{d}r=\pi,\qquad 0\le s<t, \tag{5.27}$$

gives

$$y(t)\le\frac{C_T(1+2C_T\sqrt{T})}{N}+\pi C_T^2\int_0^t y(s)\,\mathrm{d}s. \tag{5.28}$$

Gronwall's inequality proves (5.21). Keeping the initial norm rather than replacing it by C_0/N, and translating the interval, proves (5.22).

The same argument constructs the auxiliary linear solution for a prescribed source. At fixed cutoff, the right side of (5.23) defines a bounded functional of ψ and hence an element of $L^2(\mu_t)$. On an interval of length δ, the part depending linearly on a has operator norm at most $2C_T\sqrt{\delta}$ in the norm $\sup_t\|a_t\|_{L^2(\mu_t)}$. Choosing this number below one gives a contraction. Iterating on

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successive intervals constructs the solution, and the same argument proves uniqueness. The source integral is well defined by (5.25); its mass is zero because $P_{r,t}1 = 1$ and both source distributions annihilate constants. No probability density on the full configuration space is required for this auxiliary solution. The testing of the actual cutoff law is justified in lemma 6.3. □

5.3. **A time-uniform closure by homogeneous energy and fixed intervals.** We now separate the homogeneous evolution in (5.18) from its last two source terms. The quadratic form (5.35), associated with the positive operator $\mathrm{Id}+\beta\theta_N c(t)T_{\mu_t}$, gives contraction of the homogeneous evolution over a sufficiently long fixed interval. The estimate (5.22) bounds the response to the sources on the same interval. Iterating these two estimates will control the first marginal for all times, always retaining its actual value at the beginning of the next interval.

Proposition 5.3 (Uniform weighted one-mark estimate)**.** *Assume the common bounds for μ_t and B_t hold uniformly for $t \geq 0$, and assume*

$$\sup_{N\geq 2}\sup_{t\geq 0}\|f_{N,t}\|^2_{L^2(\mu_t^{\otimes N})} \leq M_* < \infty. \tag{5.29}$$

Suppose the interaction coefficient satisfies

$$c \in C^1_{\mathrm{loc}}([0,\infty)), \qquad 0 \leq c(t) \leq c_*, \qquad c'(t) \leq 0, \qquad t \geq 0. \tag{5.30}$$

Assume that each μ_t satisfies the Poincaré inequality with a common constant C_P:

$$\int \left|\varphi - \int \varphi \,\mathrm{d}\mu_t\right|^2 \mathrm{d}\mu_t \leq C_P \int |\nabla\varphi|^2 \,\mathrm{d}\mu_t, \qquad \varphi \in C_c^\infty(\mathbb{R}^{\mathsf{d}}), \quad t \geq 0. \tag{5.31}$$

Suppose also that

$$\int_0^\infty \|\nabla u_t\|_{L^\infty} \,\mathrm{d}t < \infty \tag{5.32}$$

and

$$\|a_{N,0}\|_{L^2(\mu_0)} \leq C_0/N. \tag{5.33}$$

Then, for the repulsive kernels (1.1),

$$\sup_{t\geq 0}\|a_{N,t}\|_{L^2(\mu_t)} \leq C/N. \tag{5.34}$$

Proof. For a signed density $w = \mu_t a$ of mass zero, set

$$\mathcal{Q}_{N,t}(w) := \int a^2 \,\mathrm{d}\mu_t + \beta\theta_N c(t) \iint \mathsf{g}(x-y)w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y. \tag{5.35}$$

On $L^2_0(\mu_t)$, the centered interaction operator T_{μ_t} is nonnegative and uniformly bounded, by lemma 2.2. Consequently,

$$\|a\|^2_{L^2(\mu_t)} \leq \mathcal{Q}_{N,t}(w) \leq C_B\|a\|^2_{L^2(\mu_t)}. \tag{5.36}$$

Set

$$z := (\mathrm{Id} + \beta\theta_N c(t)T_{\mu_t})\, a. \tag{5.37}$$

Then $\int z \,\mathrm{d}\mu_t = 0$ and

$$\|z\|^2_{L^2(\mu_t)} - \mathcal{Q}_{N,t}(w) = \beta\theta_N c(t)\langle a, T_{\mu_t}a\rangle_{L^2(\mu_t)} + \beta^2\theta_N^2 c(t)^2\|T_{\mu_t}a\|^2_{L^2(\mu_t)} \geq 0. \tag{5.38}$$

This is where positivity of the repulsive interaction is used, in contrast to the screened-inverse estimate for the initial marginal.

Since $B_t = u_t + \beta^{-1}\nabla\log\mu_t$, the homogeneous part of (5.18) is

$$\partial_t w + \mathrm{div}(wu_t) = \beta^{-1}\,\mathrm{div}(\mu_t\nabla z) - \frac{c(t)}{N}\,\mathrm{div}(w\nabla h_t). \tag{5.39}$$

We compute first for smooth data. The continuity equation for μ_t gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \frac{w^2}{\mu_t}\,\mathrm{d}x = 2\int \frac{w}{\mu_t}\big(\partial_t w + \mathrm{div}(wu_t)\big)\,\mathrm{d}x. \tag{5.40}$$

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Symmetry of g and integration by parts likewise give

$$\frac{\mathrm{d}}{\mathrm{d}t}\iint \mathsf{g}(x-y)w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y = 2\int (\mathsf{g}*w)(\partial_t w + \operatorname{div}(wu_t))\,\mathrm{d}x + \iint (u_t(x)-u_t(y))\cdot\nabla\mathsf{g}(x-y)w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y. \tag{5.41}$$

In the terms containing $\partial_t w + \operatorname{div}(wu_t)$, the function $a+\beta\theta_N c(t)\mathsf{g}*w$ differs from z only by a spatial constant; the signed density in that pairing has mass zero. Substitution of (5.39) therefore yields

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{Q}_{N,t}(w) = &-\frac{2}{\beta}\int|\nabla z|^2\,\mathrm{d}\mu_t \\ &+\beta\theta_N c(t)\iint (u_t(x)-u_t(y))\cdot\nabla\mathsf{g}(x-y)w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y \\ &+\beta\theta_N c'(t)\iint \mathsf{g}(x-y)w(x)w(y)\,\mathrm{d}x\,\mathrm{d}y + \frac{2c(t)}{N}\int a\nabla h_t\cdot\nabla z\,\mathrm{d}\mu_t.\end{aligned} \tag{5.42}$$

The term containing c' is nonpositive. For the transport term, lemma 2.3 implies

$$\sup_x \int |x-y|\,|\nabla\mathsf{g}(x-y)|\,\mathrm{d}\mu_t(y) \le C\left(M\int_{|z|\le 1}|z|^{-\mathsf{s}}\,\mathrm{d}z + 1\right) \le C. \tag{5.43}$$

Schur's test bounds the transport term in absolute value by $C\|\nabla u_t\|_{L^\infty}\|a\|^2_{L^2(\mu_t)}$. The last term satisfies

$$\frac{2c(t)}{N}\left|\int a\nabla h_t\cdot\nabla z\,\mathrm{d}\mu_t\right| \le \frac{1}{\beta}\|\nabla z\|^2_{L^2(\mu_t)} + \frac{D}{N^2}\|a\|^2_{L^2(\mu_t)}, \tag{5.44}$$

where we used (5.3). Since z is centered, Poincaré's inequality and (5.38) give

$$\|\nabla z\|^2_{L^2(\mu_t)} \ge C_P^{-1}\|z\|^2_{L^2(\mu_t)} \ge C_P^{-1}\mathcal{Q}_{N,t}(w). \tag{5.45}$$

Combining these inequalities proves

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{Q}_{N,t}(w) \le \left(-\rho + \frac{D}{N^2} + C\|\nabla u_t\|_{L^\infty}\right)\mathcal{Q}_{N,t}(w), \qquad \rho := (\beta C_P)^{-1} > 0. \tag{5.46}$$

At fixed cutoff, spatial localization justifies the calculation for smooth initial signed densities, as in lemma 6.3. Approximation of a centered $L^2(\mu_r)$ initial value by smooth compactly supported centered functions, followed by the finite-time linear construction in proposition 5.2, extends the integrated form of (5.46) to such initial values. The bounds in (5.36) are uniform, so no cutoff-dependent constant enters this inequality.

Choose N_0 such that $D/N^2 \le \rho/2$ for $N \ge N_0$. Integrating (5.46) gives, for a homogeneous solution,

$$\mathcal{Q}_{N,t}(w_t)^{1/2} \le \exp\left(-\frac{\rho}{4}(t-r) + \frac{C}{2}\int_r^t \|\nabla u_s\|_{L^\infty}\,\mathrm{d}s\right)\mathcal{Q}_{N,r}(w_r)^{1/2}, \qquad t \ge r. \tag{5.47}$$

By (5.32), we can choose a fixed length $L > 0$ such that

$$\exp\left(-\frac{\rho L}{4} + \frac{C}{2}\int_0^\infty \|\nabla u_t\|_{L^\infty}\,\mathrm{d}t\right) \le \frac{1}{2}. \tag{5.48}$$

On $[jL,(j+1)L]$, write

$$w_{N,t} = w^{\mathrm{hom}}_{N,j,t} + w^{\mathrm{src}}_{N,j,t}, \qquad w^{\mathrm{hom}}_{N,j,jL} = w_{N,jL}, \qquad w^{\mathrm{src}}_{N,j,jL} = 0. \tag{5.49}$$

Here, w^{hom} solves (5.39), whereas w^{src} solves (5.18) with the prescribed $\eta_{N,t}$ and zero initial value. By (5.47)–(5.48), and by (5.22) together with (5.36), respectively,

$$\mathcal{Q}_{N,(j+1)L}(w^{\mathrm{hom}}_{N,j,(j+1)L})^{1/2} \le \frac{1}{2}\mathcal{Q}_{N,jL}(w_{N,jL})^{1/2}, \tag{5.50}$$

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$$\sup_{jL\le t\le (j+1)L} \mathcal{Q}_{N,t}(w^{\mathrm{src}}_{N,j,t})^{1/2} \le \frac{\sqrt{C_B}C_L}{N}. \tag{5.51}$$

The triangle inequality for $\mathcal{Q}^{1/2}_{N,(j+1)L}$ therefore gives

$$Y_{j+1} \le \frac{1}{2}Y_j + \frac{C_L\sqrt{C_B}}{N}, \qquad Y_j := \mathcal{Q}_{N,jL}(w_{N,jL})^{1/2}. \tag{5.52}$$

The initial estimate and (5.36) imply $Y_0 \le \sqrt{C_B}C_0/N$, and hence

$$Y_j \le 2^{-j}Y_0 + \frac{C_L\sqrt{C_B}}{N}\sum_{i=0}^{j-1} 2^{-i} \le \frac{\sqrt{C_B}(C_0+2C_L)}{N}. \tag{5.53}$$

Applying (5.22) within each interval proves (5.34) for $N \ge N_0$. For $2 \le N < N_0$, the full moment and contraction of conditional expectation give directly

$$N\|a_{N,t}\|_{L^2(\mu_t)} \le N_0\sqrt{M_* - 1}, \qquad t \ge 0. \tag{5.54}$$

This proves the assertion for every $N \ge 2$. □

5.4. **Product-space projection.** We record as a lemma the orthogonal projection argument used in [DGR26b]. Besides giving the final all-marginal estimate, it justifies the two-particle bounds used earlier in this section.

Lemma 5.4 (Product-space projection)**.** *Let μ be a probability measure and let $f_N \in L^2(\mu^{\otimes N})$ be a symmetric probability density, with $N \ge 2$. Set $P_N := f_N\mu^{\otimes N}$ and $f_N^{(k)} := \frac{\mathrm{d}P_N^{(k)}}{\mathrm{d}\mu^{\otimes k}}$, so that*

$$f_N^{(k)}(x_1,\dots,x_k) = \int f_N(x_1,\dots,x_N)\,\mathrm{d}\mu^{\otimes(N-k)}(x_{k+1},\dots,x_N), \qquad 1 \le k \le N, \tag{5.55}$$

with $f_N^{(N)} = f_N$. Then, for every $1 \le k \le N$,

$$\|f_N^{(k)} - 1\|^2_{L^2(\mu^{\otimes k})} \le k\|f_N^{(1)} - 1\|^2_{L^2(\mu)} + \frac{k(k-1)}{N(N-1)}\|f_N - 1\|^2_{L^2(\mu^{\otimes N})}. \tag{5.56}$$

Moreover, for $\eta(x,y) := f_N^{(2)}(x,y) - f_N^{(1)}(x) - f_N^{(1)}(y) + 1$,

$$\binom{N}{2}\|\eta\|^2_{L^2(\mu^{\otimes 2})} \le \|f_N - 1\|^2_{L^2(\mu^{\otimes N})}. \tag{5.57}$$

Proof. We use the Hoeffding decomposition (see, e.g., [Lee90, Section 1.6]). Set $f_N^{(0)} := 1$. For $1 \le j \le N$, the symmetric Hoeffding kernel is

$$h_j(x_1,\dots,x_j) := \sum_{A\subset[j]} (-1)^{j-|A|} f_N^{(|A|)}(x_A). \tag{5.58}$$

Here, x_A is the tuple with indices in A, in increasing order. The kernels satisfy

$$\int h_j(x_1,\dots,x_j)\,\mathrm{d}\mu(x_i) = 0, \qquad 1 \le i \le j, \tag{5.59}$$

for almost every choice of the remaining coordinates, and

$$f_N - 1 = \sum_{j=1}^{N}\sum_{\substack{A\subset[N]\\|A|=j}} h_j(x_A). \tag{5.60}$$

For distinct nonempty index sets $A, B \subset [N]$, the orthogonality relation is

$$\int h_{|A|}(x_A)h_{|B|}(x_B)\,\mathrm{d}\mu^{\otimes N} = 0. \tag{5.61}$$

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Together with the marginal identities (5.55), these properties give

$$\|f_N^{(k)} - 1\|^2_{L^2(\mu^{\otimes k})} = \sum_{j=1}^{k} \binom{k}{j} \|h_j\|^2_{L^2(\mu^{\otimes j})}, \qquad 1 \le k \le N. \tag{5.62}$$

For $2 \le j \le k \le N$,

$$\frac{\binom{k}{j}}{\binom{N}{j}} = \frac{k(k-1)}{N(N-1)} \prod_{i=2}^{j-1} \frac{k-i}{N-i} \le \frac{k(k-1)}{N(N-1)}, \tag{5.63}$$

where the product is one when $j = 2$. Separating $j = 1$, for which $h_1 = f_N^{(1)} - 1$, and summing the remaining terms proves (5.56). When $k = 1$, the assertion is an equality. Finally, $h_2 = \eta$, so (5.57) follows by retaining the $j = 2$ term in (5.62). □

If the full second moment is at most M_* and the one-mark error is at most U/N, lemma 5.4 gives

$$\chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) \le \frac{kU^2}{N^2} + (M_* - 1)\frac{k(k-1)}{N(N-1)}. \tag{5.64}$$

The standard information inequalities (see, e.g., [NS20, (16), (139)–(141)]) give

$$\mathsf{H}(P \mid \nu) \le \log(1 + \chi^2(P \mid \nu)) \le \chi^2(P \mid \nu), \tag{5.65}$$

$$\mathrm{TV}(P, \nu) \le \frac{1}{2}\sqrt{\chi^2(P \mid \nu)}. \tag{5.66}$$

Together with (5.64), these inequalities give the claimed relative-entropy and total-variation rates.

6. Confined backgrounds and completion of the proofs

We first verify the one-body hypotheses for perturbations of a confined thermal equilibrium measure. We then justify the regularized calculations and pass the estimates of Sections 3 and 5 to the singular particle system. The background stability result is stated in Section 6.1 and proved in Section A; the remainder of this section completes the proofs of theorems 1.1 and 1.2 and corollary 1.3.

6.1. Verification of the confined backgrounds.

The perturbation in (1.29) is measured through derivatives of $\log(\mu_t/\pi)$. This choice controls the logarithmic Hessian of μ_t throughout $\mathbb{R}^{\mathsf{d}}$, including the region where π is small. Smallness of an unweighted Sobolev norm of $\mu_t - \pi$ would not give the required estimate there.

Recall the space $\mathcal{X}^m$ defined by (1.31). Its elements have representatives of at most linear growth. For each such representative there is a vector $a_z \in \mathbb{R}^{\mathsf{d}}$ such that

$$\lim_{R\to\infty} \sup_{|x|\ge R} |\nabla z(x) - a_z| = 0, \qquad \lim_{R\to\infty} \sup_{|x|\ge R} |\nabla^j z(x)| = 0, \quad (2 \le j \le m). \tag{6.1}$$

The highest derivatives are uniformly continuous. These properties follow by approximation with the affine-plus-compact smooth functions used to define $\mathcal{X}^m$. They will be used in Section A.3 to construct the linearized evolution on this space.

Proposition 6.1 (Logarithmic stability at every fixed inverse temperature)**.** *Fix $\mathsf{d}, \mathsf{s}, \beta$ and an integer $m \ge 3$ as in theorem 1.2, and let V satisfy (1.23)–(1.24). There exist $\delta_0, C, \omega > 0$ with the following properties.*

(i) If $z_0 \in \mathcal{X}^m$ and $\|z_0\|_{\mathcal{X}^m} \le \delta_0$, the initial density (1.29) has a global mean-field solution satisfying

$$\|\log(\mu_t/\pi)\|_{\mathcal{X}^m} \le C\|z_0\|_{\mathcal{X}^m} e^{-\omega t}, \tag{6.2}$$

$$\|u_t\|_{W^{1,\infty}} \le C\|z_0\|_{\mathcal{X}^m} e^{-\omega t}. \tag{6.3}$$

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There are constants $a, M, C_P, C_1 > 0$, independent of $t \geq 0$ and of z_0 in the stated ball, such that

$$\mu_t(x) \leq Me^{-a|x|^2}, \qquad (t \geq 0,\ x \in \mathbb{R}^{\mathsf{d}}), \tag{6.4}$$

$$\mathrm{Var}_{\mu_t}(\psi) \leq C_P \int |\nabla \psi|^2 \, \mathrm{d}\mu_t, \qquad (\psi \in H^1(\mu_t),\ t \geq 0), \tag{6.5}$$

$$\sup_{t \geq 0} \left(\|\nabla^2 \log \mu_t\|_{L^\infty} + \|\nabla B_t\|_{L^\infty} + \|\nabla^2 B_t\|_{L^\infty} \right) \leq C_1. \tag{6.6}$$

Here, B_t and u_t are defined in (1.13) and (1.14). For smooth initial data with all higher $\mathcal{X}^\ell$ norms finite, the solution remains smooth on every finite interval. The constants in the preceding estimates do not depend on those higher norms.

(ii) Let π^ε be the thermal equilibrium measure for the kernel g_ε of (2.12), and let μ_t^ε solve the corresponding mean-field equation from

$$\mu_0^\varepsilon = \frac{\pi^\varepsilon e^{z_0}}{\int \pi^\varepsilon e^{z_0}}. \tag{6.7}$$

Set $\pi^0 := \pi$, $\mu_t^0 := \mu_t$, and $u_t^0 := u_t$. There is $\varepsilon_* > 0$ such that the estimates in (i) hold with the same constants for $0 < \varepsilon \leq \varepsilon_*$. For every $T < \infty$,

$$\sup_{0 \leq t \leq T} \left(\left\| \log \frac{\mu_t^\varepsilon}{\pi^\varepsilon} - \log \frac{\mu_t}{\pi} \right\|_{\mathcal{X}^m} + \|\mu_t^\varepsilon - \mu_t\|_{L^1} + \|u_t^\varepsilon - u_t\|_{W^{1,\infty}} \right) \xrightarrow[\varepsilon \downarrow 0]{} 0. \tag{6.8}$$

For the smooth data in (i), the densities also converge locally smoothly in space, uniformly on finite time intervals.

(iii) Let $z_0 \in \bigcap_{\ell \geq m} \mathcal{X}^\ell$ be arbitrary and smooth. There exist $T_0 > 0$ and $\varepsilon_* > 0$ such that the singular equation and all regularized equations with $0 < \varepsilon \leq \varepsilon_*$ have solutions on $[0, T_0]$. The bounds (6.4)–(6.6) hold on this interval with constants independent of $t \in [0, T_0]$ and $\varepsilon \in [0, \varepsilon_*]$, and (6.8) holds for every $T \leq T_0$. The time T_0 and these constants may depend on $\|z_0\|_{\mathcal{X}^m}$. No exponential-decay assertion is made in this case.

We prove the proposition in Section A. Here, we check the integrability expenditure along the mean-field trajectory. Let $\|z_0\|_{\mathcal{X}^m} \leq \delta \leq \delta_0$, and choose $\varepsilon_* > 0$ as in part (ii). By (6.3),

$$\frac{\beta}{\kappa} \int_0^\infty \sup_{0 \leq \varepsilon \leq \varepsilon_*} \|\nabla u_t^\varepsilon\|_{L^\infty} \, \mathrm{d}t \leq \frac{\beta C \delta}{\kappa \omega}. \tag{6.9}$$

Here, κ is chosen for the Gaussian upper bound, uniform in $t \geq 0$ and $\varepsilon \in [0, \varepsilon_*]$, in (6.4). Since $\beta_0 > \beta/2$, the right side can be made smaller than the positive bound in (1.20) by decreasing δ. Thus, the same one-body argument verifies both the expenditure condition and the background hypotheses used in proposition 5.3.

6.2. **Approximation and completion of the confined proofs.** For theorem 1.1, the hypotheses concern a prescribed mean-field trajectory. We keep that trajectory fixed by compensating the force truncation with a one-body term that vanishes as the cutoff is removed. For theorem 1.2, the particle system and mean-field equation instead use the same cutoff, with the backgrounds constructed in proposition 6.1. After treating the fixed-trajectory construction, lemma 6.3 justifies nonlinear testing and passage to the singular system for both families, as well as for the self-similar family used in Section 7.

Lemma 6.2 (Regularization with a fixed mean-field trajectory)**.** *Assume the one-body hypotheses of theorem 1.1 on $[0, T]$. For the heat truncation (2.12), define*

$$V_t^\varepsilon = V + (\mathsf{g} - \mathsf{g}_\varepsilon) * \mu_t. \tag{6.10}$$

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Use the auxiliary particle equation

$$\mathrm{d}x_i^\varepsilon = -\left(\nabla V_t^\varepsilon(x_i^\varepsilon) + \frac{1}{N}\sum_{j\neq i}\nabla \mathsf{g}_\varepsilon(x_i^\varepsilon - x_j^\varepsilon)\right)\mathrm{d}t + \sqrt{2/\beta}\,\mathrm{d}B_i, \qquad P_{N,0}^\varepsilon = \mathbb{Q}_{N,\beta_0}^\varepsilon(\mu_0), \tag{6.11}$$

where the superscript on $\mathbb{Q}$ means that g is replaced by g_ε in (1.6)–(1.7). At fixed N,ε, this equation has a unique strong solution on $[0,T]$. Its exact mean-field trajectory is μ_t; writing u_t^ε for its continuity velocity, we have

$$-\nabla V_t^\varepsilon - \nabla \mathsf{g}_\varepsilon * \mu_t = B_t, \qquad u_t^\varepsilon = u_t. \tag{6.12}$$

Moreover,

$$\sup_{0\le t\le T}\|\nabla V_t^\varepsilon - \nabla V\|_{W^{2,\infty}} \le C\varepsilon^{(\mathsf{d}-\mathsf{s}-1)/2} \longrightarrow 0. \tag{6.13}$$

The moving-reference defect and the first-marginal equation are exactly (2.32) and (5.18), with $c=1$, kernel g_ε, and the unchanged background μ_t. The moving reference remains $\mathbb{Q}_{N,\beta}^\varepsilon(\mu_t)$, while the initial laws in (6.11) converge in total variation to (1.8).

Proof. **Density derivatives and the force correction.** Write $U_t = -\log\mu_t$ and let L be a common bound for $\|\nabla^2 U_t\|_\infty$. At each fixed time, $|\nabla U_t(x)| \le |\nabla U_t(0)| + L|x|$. The Gaussian upper bound therefore makes $\nabla\mu_t = -\mu_t\nabla U_t$ integrable. Spatial cutoffs give $\int \nabla\mu_t\,\mathrm{d}x = 0$, whence

$$|\nabla U_t(0)| = \left|\int (\nabla U_t(0) - \nabla U_t(x))\,\mathrm{d}\mu_t(x)\right| \le L\int |x|\,\mathrm{d}\mu_t(x) \le C. \tag{6.14}$$

The final constant is uniform in t. Together with

$$\nabla\mu_t = -\mu_t\nabla U_t, \qquad \nabla^2\mu_t = \mu_t(\nabla U_t\otimes\nabla U_t - \nabla^2 U_t), \tag{6.15}$$

this gives common polynomial–Gaussian bounds for the density and its first two derivatives, in particular a uniform $W^{2,\infty}$ bound.

Differentiating the heat representation gives the L^1-valued integral

$$\nabla\mathsf{g} - \nabla\mathsf{g}_\varepsilon = a_{\mathsf{d},\mathsf{s}}\int_0^\varepsilon r^{(\mathsf{d}-\mathsf{s})/2-1}\nabla p_r\,\mathrm{d}r. \tag{6.16}$$

The logarithmic constant subtraction disappears upon differentiation. Since $\|\nabla p_r\|_1 = C_{\mathsf{d}} r^{-1/2}$,

$$\|\nabla\mathsf{g} - \nabla\mathsf{g}_\varepsilon\|_{L^1} \le C\int_0^\varepsilon r^{(\mathsf{d}-\mathsf{s}-3)/2}\,\mathrm{d}r \le C\varepsilon^{(\mathsf{d}-\mathsf{s}-1)/2}. \tag{6.17}$$

The exponent is positive: for $\mathsf{d}=2$ we have $\mathsf{s}<1$, while for $\mathsf{d}>2$ we have $\mathsf{d}-\mathsf{s}-1 > \mathsf{d}/2-1 > 0$. Placing zero, one, or two derivatives on μ_t in the convolution proves (6.13). The potential in (6.10) is well defined by local integrability and the Gaussian upper bound.

At fixed cutoff, the force and all its positive-order derivatives are bounded. Equation (6.13) and the bounded Hessian of V make the drift in (6.11) globally Lipschitz, with at most linear growth, uniformly on $[0,T]$. The existence and uniqueness theorem for stochastic differential equations with globally Lipschitz coefficients gives the asserted strong solution on this interval (see, e.g., [Øks03, Section 5.2]). The cancellation (6.12) shows that μ_t solves its mean-field equation exactly. Thus, the time-dependent correction is confined to the auxiliary system and tends to zero by (6.13). The limiting particle equation is (1.3).

The reference defect and the marginal equation. The cancellation in (6.12) allows the computation leading to (2.32) to be used without changing μ_t or u_t. We record the identities needed

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for the approximation argument. Let $\Gamma^{\varepsilon}_{N,t} = \mathbb{Q}^{\varepsilon}_{N,\beta}(\mu_t)$ and use $E^{\varepsilon}_{N,t}$ and C^{ε}_{N} for (2.1) with the truncated kernel. Differentiating the reference density gives

$$\nabla_i \log \Gamma^{\varepsilon}_{N,t} = \nabla \log \mu_t(x_i) - \beta \left(\frac{1}{N} \sum_{j \neq i} \nabla \mathsf{g}_{\varepsilon}(x_i - x_j) - \nabla \mathsf{g}_{\varepsilon} * \mu_t(x_i) \right). \tag{6.18}$$

Consequently, the particle drift is $u_t(x_i) + \beta^{-1} \nabla_i \log \Gamma^{\varepsilon}_{N,t}$, and its generator satisfies

$$(L^{\varepsilon}_{N,t})^* \Gamma^{\varepsilon}_{N,t} = -\sum_i \operatorname{div}_i (u_t(x_i) \Gamma^{\varepsilon}_{N,t}). \tag{6.19}$$

The continuity equation $\partial_t \mu_t + \operatorname{div}(\mu_t u_t) = 0$ cancels the product-background derivative. In the remaining energy, material differentiation of the pair term gives $N^{-1} \sum_{i<j} (u_t(x_i) - u_t(x_j)) \cdot \nabla \mathsf{g}_{\varepsilon}(x_i - x_j)$; the one-body and background terms supply the two cross terms and the background double integral. Thus,

$$\left(\partial_t + \sum_i u_t(x_i) \cdot \nabla_i \right) E^{\varepsilon}_{N,t} = C^{\varepsilon}_{N}[u_t]. \tag{6.20}$$

Integrating the material derivative of the unnormalized Gibbs density, with spatial cutoffs and the Gaussian tails, gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \mathsf{K}^{\varepsilon}_{N,\beta}(\mu_t) = -\beta \mathbb{E}_{\Gamma^{\varepsilon}_{N,t}}[C^{\varepsilon}_{N}[u_t]]. \tag{6.21}$$

Combining these identities proves

$$\partial_t \Gamma^{\varepsilon}_{N,t} - (L^{\varepsilon}_{N,t})^* \Gamma^{\varepsilon}_{N,t} = -\beta \left(C^{\varepsilon}_{N}[u_t] - \mathbb{E}_{\Gamma^{\varepsilon}_{N,t}}[C^{\varepsilon}_{N}[u_t]] \right) \Gamma^{\varepsilon}_{N,t}. \tag{6.22}$$

For the first marginal, write $h^{\varepsilon}_t = \mathsf{g}_{\varepsilon} * \mu_t$ and use the decomposition (5.13) for the approximate law. Subtracting the exact mean-field equation from the first particle marginal equation gives

$$\begin{aligned} \partial_t w^{\varepsilon}_{N,t} = \mathcal{L}^*_t w^{\varepsilon}_{N,t} + \frac{N-1}{N} \operatorname{div}(\mu_t \nabla \mathsf{g}_{\varepsilon} * w^{\varepsilon}_{N,t}) - \frac{1}{N} \operatorname{div}(w^{\varepsilon}_{N,t} \nabla h^{\varepsilon}_t) \\ - \frac{1}{N} \operatorname{div}(\mu_t \nabla h^{\varepsilon}_t) + \frac{N-1}{N} S^{\varepsilon}_{\eta^{\varepsilon}_{N,t}}, \qquad \mathcal{L}^*_t = \beta^{-1} \Delta - \operatorname{div}(B_t \cdot). \end{aligned} \tag{6.23}$$

In particular, the backward propagator is unchanged. The two terms with coefficient $-1/N$ arise from $(N-1)/N - 1 = -1/N$, when the particle interaction is compared with the mean-field force; see (5.18). The fixed-cutoff testing of these identities is justified in lemma 6.3.

Convergence of the initial Gibbs preparations. Set

$$H_{\varepsilon}(x) := a_{\mathsf{d},\mathsf{s}} \int_0^{\varepsilon} r^{(\mathsf{d}-\mathsf{s})/2-1} p_r(x) \, \mathrm{d}r \geq 0. \tag{6.24}$$

Here, r is the heat parameter in (2.12), and $\|H_{\varepsilon}\|_1 = C_{\mathsf{d},\mathsf{s}} \varepsilon^{(\mathsf{d}-\mathsf{s})/2}$. For $\mathsf{s} > 0$, $\mathsf{g} - \mathsf{g}_{\varepsilon} = H_{\varepsilon}$. For $\mathsf{s} = 0$, the difference is $H_{\varepsilon} - H_{\varepsilon}(e)$, and $H_{\varepsilon}(e) \to 0$ by the factor $e^{-1/(4r)}$ in the heat kernel. It follows that $h^{\varepsilon}_{\mu_0} \to h_{\mu_0}$ uniformly, $I^{\varepsilon}_{\mu_0} \to I_{\mu_0}$, and the pair potential converges off the collision diagonal.

For $\mathsf{s} > 0$, the pair potential is nonnegative and the one-body convolution is bounded above uniformly in ε. The unnormalized Gibbs density is therefore bounded by $C_N \mu_0^{\otimes N}$. For the logarithm and $0 < \varepsilon \leq 1$,

$$\mathsf{g}_{\varepsilon}(x) \geq -C - \log(1 + |x|). \tag{6.25}$$

Indeed, $\mathsf{g} - H_{\varepsilon} \geq \mathsf{g} - H_1$; the right-hand side is smooth on bounded sets and has this lower bound at infinity, and the normalization constants $H_{\varepsilon}(e)$ are bounded. The positive part of $h^{\varepsilon}_{\mu_0}$ is bounded uniformly by local integrability of the logarithm and boundedness of μ_0. The energies $I^{\varepsilon}_{\mu_0}$ are

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bounded as well. Since $1+|x-y| \le (1+|x|)(1+|y|)$, a fixed-$N$ majorant for the unnormalized density is

$$C_N \prod_{i=1}^{N} (1+|x_i|)^{\beta_0(N-1)/N} e^{-a|x_i|^2}. \tag{6.26}$$

This is integrable. Dominated convergence and the positive lower bound (2.4) on the normalizers give total-variation convergence of the normalized initial laws. □

Lemma 6.3 (Nonlinear testing and passage to the actual singular law)**.** *Fix a finite interval $[0,T]$ on which the relevant background bounds hold. Consider one of the following heat-regularized particle systems:*

(i) the auxiliary equation (6.11), with $\mu_t^\varepsilon = \mu_t$ and $c \equiv 1$;

(ii) the system with the original confinement, $c \equiv 1$, and the cutoff backgrounds of proposition 6.1;

(iii) the self-similar system of Section 7, with its Gaussian backgrounds and decreasing Riesz coupling $c(t)$.

In (ii) and (iii), use the same heat cutoff in the particle equation, mean-field equation, and Gibbs laws. In all three cases, set

$$P_{N,0}^\varepsilon := \mathbb{Q}^\varepsilon_{N,\beta_0 c(0)}(\mu_0^\varepsilon), \qquad \Gamma_{N,t}^\varepsilon := \mathbb{Q}^\varepsilon_{N,\beta c(t)}(\mu_t^\varepsilon). \tag{6.27}$$

Then the following conclusions hold.

(a) At each fixed $N, \varepsilon > 0$, the identities (3.5), (3.7), (5.8), and (5.18) are justified on $[0,T]$. The resulting estimates of Sections 3 and 5 have constants independent of the cutoff whenever the one-body and static bounds entering those estimates have constants independent of the cutoff.

(b) For each fixed N, the particle path laws satisfy

$$\mathrm{Law}(X_N^\varepsilon) \rightharpoonup \mathrm{Law}(X_N) \quad \text{in } \mathcal{P}(C([0,T];\mathbb{R}^{\mathsf{d}N})), \qquad \varepsilon \downarrow 0, \tag{6.28}$$

where X_N is the corresponding singular particle system with initial law $\mathbb{Q}_{N,\beta_0 c(0)}(\mu_0)$. In case (iii), it is the exact self-similar transform of the unconfined particle system. For each $1 \le k \le N$ and $t \in [0,T]$,

$$\chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) \le \liminf_{\varepsilon \downarrow 0} \chi^2(P_{N,t}^{\varepsilon,(k)} \mid (\mu_t^\varepsilon)^{\otimes k}). \tag{6.29}$$

In particular, any bound on the right independent of the cutoff passes to the singular law, including the full-particle bound at $k = N$.

Proof. **Testing at fixed particle number and cutoff.** Fix N and $\varepsilon > 0$. The force and its positive-order derivatives are bounded. The particle drift is globally Lipschitz and has at most linear growth, including the time-dependent one-body correction by (6.13). Since the Hessian of $U_t^\varepsilon = -\log \mu_t^\varepsilon$ is bounded, its gradient has at most linear growth. The reference $\Gamma_{N,t}^\varepsilon$ therefore has Gaussian or polynomial–Gaussian tails, a score of at most linear growth, and bounded logarithmic Hessian, with constants that may depend on N, ε, T.

The coefficient in the relative-density equation (3.4) is

$$R_{N,t}^\varepsilon := \frac{\partial_t \Gamma_{N,t}^\varepsilon - (L_{N,t}^\varepsilon)^* \Gamma_{N,t}^\varepsilon}{\Gamma_{N,t}^\varepsilon}, \qquad \sup_{0 \le t \le T} \|R_{N,t}^\varepsilon\|_{W^{1,\infty}(\mathbb{R}^{\mathsf{d}N})} \le C_{N,\varepsilon,T}. \tag{6.30}$$

To prove the bound for $c \equiv 1$, use (6.22) and the fixed-cutoff bounds on $|z||\nabla \mathsf{g}_\varepsilon(z)|$, $\nabla \mathsf{g}_\varepsilon(z)$, and $|z||\nabla^2 \mathsf{g}_\varepsilon(z)|$, together with the bound on ∇u_t^ε. For decreasing Riesz coupling, (2.32) also contains $c' E_{N,t}^\varepsilon$. The potential g_ε and its first derivative are bounded when $\mathsf{s} > 0$, so this contribution and its first spatial derivative satisfy the same fixed-N, ε, T bound.

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For an explicit justification of the relative-density equation, set

$$b_0 = \beta_0 c(0), \qquad b_1 = \beta c(0), \tag{6.31}$$

and write $E^\varepsilon_{N,0} = N\mathsf{F}^\varepsilon_N(\cdot, \mu^\varepsilon_0)$. The initial density relative to the moving reference is

$$q^\varepsilon_{N,0} = \frac{\mathsf{K}^\varepsilon_{N,b_1}(\mu^\varepsilon_0)}{\mathsf{K}^\varepsilon_{N,b_0}(\mu^\varepsilon_0)} \exp((b_1 - b_0)E^\varepsilon_{N,0}). \tag{6.32}$$

For $\mathsf{s} > 0$, the cutoff potential and $E^\varepsilon_{N,0}$ are bounded. For $\mathsf{s} = 0$, the cutoff is smooth at the origin and grows at most logarithmically at infinity, while its force is bounded. Hence, for fixed N, ε, there are $C, m < \infty$ such that

$$C^{-1}(1 + |X_N|)^{-m} \le q^\varepsilon_{N,0}(X_N) \le C(1 + |X_N|)^m, \qquad |\nabla q^\varepsilon_{N,0}(X_N)| \le C(1 + |X_N|)^m. \tag{6.33}$$

Solve (3.4) with initial density (6.32) by the Feynman–Kac formula (see, e.g., [Øks03, Section 8.2]), evaluating the time-dependent coefficients in reverse time for this initial-value problem. The auxiliary diffusion drift $2\beta^{-1}\nabla \log \Gamma - b^N$ is globally Lipschitz and has at most linear growth. Its polynomial moment bounds and (6.30) propagate the polynomial upper bound in (6.33). Jensen's inequality applied to the reciprocal gives a polynomial lower bound, and differentiation of the spatial stochastic flow gives a polynomial bound for the first derivative of q. All these bounds are on a fixed finite interval and may depend on N, ε, T.

Combined with the Gaussian or polynomial–Gaussian reference tails, they justify the power tests, logarithms, and integrations by parts in (3.5) and (3.7). Local parabolic regularity gives the classical equation on compact subsets of $(0, T] \times \mathbb{R}^{\mathsf{d}N}$ (see, e.g., [Lun95, Chapter 5]). The product $q\Gamma$ solves the particle forward equation and has mass one, by spatial cutoff and the tail bounds. It starts from $P^\varepsilon_{N,0}$, so uniqueness identifies it with the density of the regularized particle system. The same domination justifies differentiation of the initial Gibbs marginal and the first-marginal equation. These fixed-N, ε estimates justify the calculations; the cutoff-independent bounds come from Sections 3 and 5.

Convergence of the initial laws. For the auxiliary system (6.11), total-variation convergence follows from lemma 6.2. For the other two families, the convergence and bounds for the initial mean-field densities give

$$\mu^\varepsilon_0 \longrightarrow \mu_0 \quad \text{locally uniformly}, \qquad 0 < \mu^\varepsilon_0(x) \le Me^{-a|x|^2}, \qquad 0 < \varepsilon \le \varepsilon_*. \tag{6.34}$$

The heat-kernel identity (6.24) gives, for each $R < \infty$,

$$\sup_{x \in B_R} |\mathsf{g}_\varepsilon * \mu^\varepsilon_0(x) - \mathsf{g} * \mu_0(x)| \le \|(\mathsf{g}_\varepsilon - \mathsf{g}) * \mu^\varepsilon_0\|_{L^\infty} + \sup_{x \in B_R} |\mathsf{g} * (\mu^\varepsilon_0 - \mu_0)(x)| \underset{\varepsilon \downarrow 0}{\longrightarrow} 0. \tag{6.35}$$

Indeed, the first term is at most $M\|H_\varepsilon\|_{L^1}$ for $\mathsf{s} > 0$, with the additional constant $H_\varepsilon(e)$ for $\mathsf{s} = 0$. For the second term, split the convolution into $|y| \le R'$ and $|y| > R'$. Local uniform convergence and local integrability of g control the first part; the Gaussian upper bound in (6.34) makes the second part uniformly small as $R' \to \infty$. The same bounds give $I^\varepsilon_{\mu^\varepsilon_0} \to I_{\mu_0}$.

Let $F^\varepsilon_{N,0}$ denote the unnormalized initial Gibbs density at coupling $b_0 = \beta_0 c(0)$. Off the collision diagonal,

$$F^\varepsilon_{N,0}(X_N) \longrightarrow F_{N,0}(X_N), \qquad 0 \le F^\varepsilon_{N,0}(X_N) \le C_N \prod_{i=1}^N (1 + |x_i|)^{b_0(N-1)/N} e^{-a|x_i|^2}. \tag{6.36}$$

For $\mathsf{s} > 0$, the polynomial factor can be omitted. The bound follows as in (6.26) and is integrable at fixed N. Dominated convergence and the lower partition bound therefore give

$$\mathrm{TV}(P^\varepsilon_{N,0}, P_{N,0}) \longrightarrow 0, \qquad \varepsilon \downarrow 0. \tag{6.37}$$

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Convergence of the particle paths. The target singular system has a global noncolliding strong solution by proposition B.1; for case (iii), use its exact self-similar transform. Couple the initial configurations so that $X^{\varepsilon}_{N,0} \to X_{N,0}$ in probability, and drive all equations by the same Brownian motions, independent of these initial configurations.

For $R > 1$, set

$$\mathcal{K}_R := \{X_N : |X_N| \le R,\ \min_{i<j} |x_i - x_j| \ge R^{-1}\}. \tag{6.38}$$

The particle drifts converge with their first derivatives uniformly on $[0,T] \times \mathcal{K}_{2R}$. For (6.11), this uses (6.13) for the one-body correction. Choose a smooth configuration-space cutoff equal to one on a neighborhood of $\mathcal{K}_R$ and supported in the interior of $\mathcal{K}_{2R}$. Write $b^{\varepsilon,R}$ and b^R for the extended drifts, and $X_N^{\varepsilon,R}$ and X_N^R for their solutions. Their Lipschitz constants are bounded by a constant L_R independent of small ε, and Gronwall gives

$$\sup_{t\le T} |X^{\varepsilon,R}_{N,t} - X^R_{N,t}| \le e^{L_R T} \left(|X^{\varepsilon}_{N,0} - X_{N,0}| + T \|b^{\varepsilon,R} - b^R\|_{L^\infty([0,T]\times\mathbb{R}^{\mathrm{d}N})} \right). \tag{6.39}$$

On the event $X_{N,t} \in \mathcal{K}_R$ for every $t \le T$, the limiting extended path equals the singular path. The positive distance from $\mathcal{K}_R$ to the boundary of the region where the cutoff equals one and (6.39) identify the approximate extended path with the original approximate path, with probability tending to one as $\varepsilon \downarrow 0$. Thus, for every $\eta > 0$,

$$\limsup_{\varepsilon\downarrow 0} \mathbb{P}\left(\sup_{t\le T} |X^{\varepsilon}_{N,t} - X_{N,t}| > \eta \right) \le \mathbb{P}\left(X_{N,t} \notin \mathcal{K}_R \text{ for some } t \le T \right) \xrightarrow[R\to\infty]{} 0. \tag{6.40}$$

The last limit follows from continuity, nonexplosion, and positive minimum pair separation on $[0,T]$. This proves (6.28) at each fixed N.

Passage of the second-moment bounds. Suppose that $P_j \rightharpoonup P$, $\nu_j \rightharpoonup \nu$, and $C := \liminf_j \|\mathrm{d}P_j/\mathrm{d}\nu_j - 1\|_{L^2(\nu_j)} < \infty$. Along a subsequence realizing this lower limit, Cauchy–Schwarz and weak convergence give, for every $\psi \in C_c(\mathbb{R}^{\mathrm{d}k})$,

$$\left| \int \psi \,\mathrm{d}(P - \nu) \right| = \lim_j \left| \int \psi \,\mathrm{d}(P_j - \nu_j) \right| \le C \left(\int \psi^2 \,\mathrm{d}\nu \right)^{1/2}. \tag{6.41}$$

The Riesz–Fréchet representation theorem and density of C_c in $L^2(\nu)$ give $\mathrm{d}P/\mathrm{d}\nu - 1 \in L^2(\nu)$ with norm at most C. Apply this with

$$P_j = P^{\varepsilon_j,(k)}_{N,t}, \quad \nu_j = (\mu^{\varepsilon_j}_t)^{\otimes k}, \quad P = P^{(k)}_{N,t}, \quad \nu = \mu_t^{\otimes k}. \tag{6.42}$$

Path-law convergence gives $P_j \rightharpoonup P$, while the corresponding background convergence gives $\nu_j \rightharpoonup \nu$. This proves (6.29). In particular, if an interval I carries the bound

$$\sup_{\varepsilon} \sup_{t\in I} \chi^2(P^{\varepsilon,(k)}_{N,t} \mid (\mu^{\varepsilon}_t)^{\otimes k}) \le C_{N,k}, \qquad \text{then} \qquad \sup_{t\in I} \chi^2(P^{(k)}_{N,t} \mid \mu_t^{\otimes k}) \le C_{N,k}. \tag{6.43}$$

For $I = [0,\infty)$, perform the path-law convergence on each finite interval and apply the bound at each fixed time before taking the supremum. No uniform-in-time path convergence is used. □

Proof of theorem 1.2. Proposition 6.1 supplies the one-body bounds uniformly in time and cutoff. Choose the perturbation radius so that (6.9) gives (1.20) with $T = \infty$, and then choose $p_0 > 2$ as in (3.18). By lemma 3.2, with $b_0 = \beta_0$ and $b_1 = \beta$, the initial relative L^{p_0} norm is bounded. Corollary 3.3 therefore gives the full second-moment bound on $[0,\infty)$.

Apply proposition 5.1 at $b = \beta_0$ and propagate its one-mark estimate by proposition 5.3. The product-space projection and information inequalities (5.64)–(5.66) give the claimed marginal rates, with constants independent of the cutoff. Finally, lemma 6.3 passes these bounds to the singular system. The dependence on the initial perturbation is only through the low norm stated in the theorem. □

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Proof of theorem 1.1 and corollary 1.3. Use the auxiliary regularization of lemma 6.2. Its mean-field trajectory and continuity velocity are exactly μ_t and u_t. The one-body constants, weighted backward propagator, and expenditure are therefore independent of the cutoff. The static bounds are uniform by proposition 2.1, and (6.22) supplies the same exponential estimate for the coefficient in the relative-density equation, with

$$\lambda_0(t) = \frac{\beta}{\kappa}\|\nabla u_t\|_{L^\infty}, \qquad \int_0^T \lambda_0(t)\,\mathrm{d}t = A_\mu(T). \tag{6.44}$$

Thus, the exponent equation is the same at every cutoff. Choose $p_0 > 2$ satisfying (3.18), as permitted by (1.20). Lemma 3.2, with $b_0 = \beta_0$ and $b_1 = \beta$, and corollary 3.3 give the full product-space second-moment bound.

For the first marginal, apply proposition 5.1 at $b = \beta_0$ and then proposition 5.2. The latter applies because (6.23) has the original backward propagator and the same two terms with coefficient $-1/N$. Equations (5.64)–(5.66) give the marginal rates with a constant independent of N, k, ε, and lemma 6.3 proves (1.21) after passage to the singular system.

For a confinement satisfying (1.23)–(1.24) and any smooth logarithmic perturbation in (1.30), the local part of proposition 6.1 provides the intrinsic one-body bounds on a positive interval. Since the right-hand side of (1.20) is positive, shortening the interval gives that condition, so theorem 1.1 proves the corollary. □

Proposition 6.4 (Stationary sharpness)**.** *Let $\beta > 0$ and let π be a thermal equilibrium measure in the class of theorem 1.2. There is $c > 0$, depending only on the fixed equilibrium π and the parameters $\mathsf{d}, \mathsf{s}, \beta$, such that, for every $N \geq 2$ and $2 \leq k \leq N$,*

$$\mathrm{TV}(\mathbb{Q}^{(k)}_{N,\beta}(\pi), \pi^{\otimes k}) \geq c\frac{k}{N}, \tag{6.45}$$

$$\mathsf{H}(\mathbb{Q}^{(k)}_{N,\beta}(\pi) \mid \pi^{\otimes k}) \geq 2c^2\frac{k^2}{N^2}, \tag{6.46}$$

$$\chi^2(\mathbb{Q}^{(k)}_{N,\beta}(\pi) \mid \pi^{\otimes k}) \geq 4c^2\frac{k^2}{N^2}. \tag{6.47}$$

Proof. For sufficiently large N, these lower bounds are a stationary specialization of [DGR26b, Theorem 2.1]. We give a proof using the upper bounds just established and the static fluctuation formula already used in this paper.

Set $\Gamma_N := \mathbb{Q}_{N,\beta}(\pi)$ and $f_{N,k} := \mathrm{d}\Gamma^{(k)}_N / \mathrm{d}\pi^{\otimes k}$. The stationary case of theorem 1.2 gives

$$\|f_{N,k} - 1\|_{L^2(\pi^{\otimes k})} \leq C_0\frac{k}{N}, \qquad 1 \leq k \leq N. \tag{6.48}$$

The nonnegative operator T_π is nonzero: on any fixed ball, h_π is bounded while $\mathsf{g}(x-y)$ diverges at the diagonal, and π has a positive lower bound there. Thus, its centered kernel cannot vanish almost everywhere. By density, choose a bounded real $\varphi \in L^2_0(\pi)$ with $T_\pi\varphi \neq 0$. The spectral theorem gives

$$v := \|\varphi\|^2_{L^2(\pi)}, \qquad v_\beta := \langle \varphi, (\mathrm{Id} + \beta T_\pi)^{-1}\varphi\rangle_{L^2(\pi)} < v. \tag{6.49}$$

The Gaussian upper bound on π verifies the logarithmic integrability hypotheses, when needed, of [DGR26a, Proposition 2.11]. In the notation

$$S_N := \frac{1}{\sqrt{N}}\sum_{i=1}^N \varphi(x_i), \tag{6.50}$$

that result gives $S_N \to \mathcal{N}(0, v_\beta)$ in law under Γ_N. Boundedness and centering of φ give a uniform eighth-moment bound for S_N under $\pi^{\otimes N}$: in its expansion, every index occurring only once integrates

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to zero, leaving at most four distinct indices and $O(N^4)$ terms. By (2.6),

$$\sup_{N\geq 2}\mathbb{E}_{\Gamma_N}[|S_N|^4]\leq \sup_{N\geq 2}\|G_{N,\beta}(\pi)\|_2\big(\mathbb{E}_{\pi^{\otimes N}}[|S_N|^8]\big)^{1/2}<\infty. \tag{6.51}$$

Hence, the second moments $\mathbb{E}_{\Gamma_N}[S_N^2]$ converge to v_β. Also, (6.48) at $k=1$ gives $\mathbb{E}_{\Gamma_N}[\varphi(x_1)^2]\to v$. Exchangeability yields

$$(N-1)\mathbb{E}_{\Gamma_N}[\varphi(x_1)\varphi(x_2)]=\mathbb{E}_{\Gamma_N}[S_N^2]-\mathbb{E}_{\Gamma_N}[\varphi(x_1)^2]\longrightarrow v_\beta-v<0. \tag{6.52}$$

For $2\leq k\leq N$, define the centered pair statistic

$$Z_k:=\frac{1}{k}\sum_{\substack{1\leq i,j\leq k\\ i\neq j}}\varphi(x_i)\varphi(x_j)=\left(\frac{1}{\sqrt{k}}\sum_{i=1}^k\varphi(x_i)\right)^2-\frac{1}{k}\sum_{i=1}^k\varphi(x_i)^2. \tag{6.53}$$

Its product expectation is zero, and the same eighth-moment calculation gives

$$K_\varphi:=\sup_{k\geq 2}\mathbb{E}_{\pi^{\otimes k}}[|Z_k|^4]<\infty. \tag{6.54}$$

For all sufficiently large N, (6.52) implies, uniformly for $2\leq k\leq N$,

$$\left|\int Z_k(f_{N,k}-1)\,\mathrm{d}\pi^{\otimes k}\right|=(k-1)\,|\mathbb{E}_{\Gamma_N}[\varphi(x_1)\varphi(x_2)]|\geq\frac{v-v_\beta}{4}\frac{k}{N}. \tag{6.55}$$

Let $Z_{k,R}:=\max\{-R,\min\{Z_k,R\}\}$. Since $\|Z_k-Z_{k,R}\|_2\leq K_\varphi^{1/2}/R$, (6.48) gives

$$\left|\int (Z_k-Z_{k,R})(f_{N,k}-1)\,\mathrm{d}\pi^{\otimes k}\right|\leq\frac{C_0K_\varphi^{1/2}}{R}\frac{k}{N}. \tag{6.56}$$

Choose $R\geq 8C_0K_\varphi^{1/2}/(v-v_\beta)$, independently of N,k. By total-variation duality,

$$2R\ \mathrm{TV}(\Gamma_N^{(k)},\pi^{\otimes k})\geq\left|\int Z_{k,R}(f_{N,k}-1)\,\mathrm{d}\pi^{\otimes k}\right|\geq\frac{v-v_\beta}{8}\frac{k}{N}. \tag{6.57}$$

This proves (6.45) for all sufficiently large N.

For each remaining $N\geq 2$, we claim that $\Gamma_N^{(2)}\neq\pi^{\otimes 2}$. Otherwise, its one- and two-body expectations would give $\mathbb{E}_{\Gamma_N}[N\mathsf{F}_N(X_N,\pi)]=-I_\pi$, and (1.7) would imply

$$\mathsf{H}(\Gamma_N\mid\pi^{\otimes N})=\beta I_\pi-\log\mathsf{K}_{N,\beta}(\pi)\leq 0. \tag{6.58}$$

Thus, $\Gamma_N=\pi^{\otimes N}$, which is impossible: $N\mathsf{F}_N$ is not constant, since it diverges at a pair collision with the other coordinates separated. All the expectations used here are finite by (2.4) and (2.6). By projection, $\mathrm{TV}(\Gamma_N^{(k)},\pi^{\otimes k})\geq\mathrm{TV}(\Gamma_N^{(2)},\pi^{\otimes 2})>0$ for every $k\geq 2$. Decreasing c over the finite set of remaining particle numbers proves (6.45) for every $N\geq 2$. Finally, Pinsker's inequality and Cauchy–Schwarz give $\mathsf{H}\geq 2\,\mathrm{TV}^2$ and $\chi^2\geq 4\,\mathrm{TV}^2$, respectively, and prove the remaining bounds. □

Remark 6.5 (Periodic geometry). Fix $\beta>0$ and $\beta_0>\beta/2$. On the unit-volume torus, take $V=0$ and the mean-zero spectral log/Riesz kernel. With initial law $\mathbb{Q}_{N,\beta_0}(\mu_0)$, the same terminal estimates hold for every smooth strictly positive mean-field measure on each interval satisfying

$$\frac{\beta}{\kappa}\int_0^T\|\nabla u_t\|_{L^\infty}\,\mathrm{d}t<\log\left(2\min\left\{1,\frac{\beta_0}{\beta}\right\}\right), \tag{6.59}$$

and hence on an initial interval for arbitrary such data. They hold uniformly for all time when $\|\mu_0-1\|_{H^\ell}$ is sufficiently small, for any fixed integer $\ell>\mathsf{d}/2+2$.

The static inputs follow from [DGR26a, Remark 2.1]. On each finite interval, smoothness and strict positivity give upper and positive lower density bounds, hence a one-body Poincaré constant. The

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weighted elliptic and parabolic estimates apply on the compact manifold. To obtain a background satisfying the expenditure condition for all time, write $w = \mu - 1$. Its equation is

$$\partial_t w = (\beta^{-1}\Delta + \Delta \mathsf{g} *)w + \operatorname{div}(w \nabla \mathsf{g} * w). \tag{6.60}$$

The nonzero Fourier multipliers of its linear semigroup are $\exp[-4\pi^2|n|^2(\beta^{-1} + \widehat{\mathsf{g}}(n))t]$. They have a positive diffusive gap and gain one derivative with an integrable $t^{-1/2}$ singularity. The Sobolev algebra estimate and the torus L^1 bound on the force give a quadratic bound $C\|w\|_{H^\ell}^2$ for the flux. The resulting contraction in the exponentially weighted trajectory norm yields, for sufficiently small $\|w_0\|_{H^\ell}$,

$$\sup_{t \geq 0} e^{\omega t}\|w_t\|_{H^\ell} \leq C\|w_0\|_{H^\ell}. \tag{6.61}$$

In particular, $1/2 \leq \mu_t \leq 3/2$ after decreasing the initial radius. Since $u_t = -\nabla \mathsf{g} * w_t - \beta^{-1}\nabla \log(1 + w_t)$, Sobolev embedding and the force bound give

$$\frac{\beta}{\kappa}\int_0^\infty \|\nabla u_t\|_{L^\infty}\,\mathrm{d}t \leq \frac{\beta C}{\kappa}\int_0^\infty \|w_t\|_{H^\ell}\,\mathrm{d}t \leq \frac{\beta C}{\kappa\omega}\|\mu_0 - 1\|_{H^\ell}. \tag{6.62}$$

Choose κ for the common density bound. The right side is smaller than the bound in (6.59) when the initial radius is sufficiently small. The Fourier and force estimates are uniform for the periodic heat cutoffs, so the same choice works before passing to the singular kernel.

For the periodic version of lemma 4.1, use

$$\|\nabla\psi(\cdot + z) - \nabla\psi\|_{L^2(\mathbb{T}^{\mathsf{d}})} \leq 2^{1-\alpha}|z|^\alpha \|\nabla\psi\|_{L^2(\mathbb{T}^{\mathsf{d}})}^{1-\alpha}\|\nabla^2\psi\|_{L^2(\mathbb{T}^{\mathsf{d}})}^{\alpha}, \qquad 0 < \alpha < 1, \tag{6.63}$$

where $z \in [-1/2, 1/2]^{\mathsf{d}}$ and $\psi \in H^2(\mathbb{T}^{\mathsf{d}})$. This follows by interpolating the bounds $2\|\nabla\psi\|_2$ and $|z|\|\nabla^2\psi\|_2$ for the left side. For $\max\{0, \mathsf{s} + 1 - \mathsf{d}/2\} < \alpha < 1$, the periodic heat cutoffs satisfy

$$\sup_{0 \leq \varepsilon \leq 1}\int_{\mathbb{T}^{\mathsf{d}}} \operatorname{dist}(z, 0)^{2\alpha}|\nabla \mathsf{g}_\varepsilon(z)|^2\,\mathrm{d}z < \infty. \tag{6.64}$$

The upper and positive lower bounds on μ_t allow comparison of the weighted and unweighted norms. Applying Cauchy–Schwarz in the two-point pairing with (6.63) and (6.64) gives the required periodic mixed gradient–Hessian estimate. Thus, the arguments from the full moment through the first marginal and product-space projection apply on the torus. For arbitrary smooth positive initial data, the local contraction argument gives the same background bounds on an initial interval; shortening that interval gives (6.59).

7. Unconfined consequences in self-similar variables

We apply the self-similar transformation of Serfaty and the third author [RS24, Section 2] to the unconfined system, obtained from (1.3) by setting $V = 0$. The transformed equation has confinement $|y|^2/4$. For the logarithmic interaction, theorem 1.2 applies directly. For positive Riesz exponents, the interaction coefficient decreases in time; lemma 7.3 supplies the Gaussian background estimates needed to apply corollary 3.3 and proposition 5.3.

An initial length scale $L > 0$ will determine the strength of the transformed Riesz interaction. Set

$$\tau := \log(1 + t/L^2), \qquad y_i(\tau) := \frac{x_i(t)}{Le^{\tau/2}}. \tag{7.1}$$

Write $D_\tau(y) := Le^{\tau/2}y$, so that $t = L^2(e^\tau - 1)$ and $\mathrm{d}t/\mathrm{d}\tau = L^2e^\tau$. Define the transformed measures by

$$\bar{\mu}_\tau(y) := (Le^{\tau/2})^{\mathsf{d}}\mu_{L^2(e^\tau-1)}(Le^{\tau/2}y), \qquad \bar{P}_{N,\tau} := (D_\tau^{-1})_{\#}^{\otimes N}P_{N,L^2(e^\tau-1)}. \tag{7.2}$$

Brownian scaling and the identity $\nabla \mathsf{g}(ay) = a^{-\mathsf{s}-1}\nabla \mathsf{g}(y)$ give

$$\mathrm{d}y_i = \left[-\frac{1}{2}y_i - \frac{c(\tau)}{N}\sum_{j \neq i}\nabla \mathsf{g}(y_i - y_j)\right]\mathrm{d}\tau + \sqrt{2/\beta}\,\mathrm{d}\widetilde{B}_i, \tag{7.3}$$

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where the $\widetilde{B}_i$ are independent Brownian motions and

$$c(\tau) := \begin{cases} 1, & \mathsf{s} = 0, \\ L^{-\mathsf{s}} e^{-\mathsf{s}\tau/2}, & \mathsf{s} > 0. \end{cases} \tag{7.4}$$

The force coefficient is $(Le^{\tau/2})^{-\mathsf{s}}$, while $\mathrm{d}t/\mathrm{d}\tau = (Le^{\tau/2})^2$ leaves the noise coefficient unchanged. This is the calculation of [RS24, Lemma 2.1], with the initial dilation L included. The transformed mean-field equation, its drift, and its continuity velocity are respectively

$$\partial_\tau \bar{\mu}_\tau = \beta^{-1}\Delta\bar{\mu}_\tau + \operatorname{div}\left(\bar{\mu}_\tau\left(\frac{y}{2} + c(\tau)\nabla \mathsf{g} * \bar{\mu}_\tau\right)\right), \tag{7.5}$$

$$\bar{B}_\tau := -\frac{y}{2} - c(\tau)\nabla \mathsf{g} * \bar{\mu}_\tau, \qquad \bar{u}_\tau := \bar{B}_\tau - \beta^{-1}\nabla \log \bar{\mu}_\tau. \tag{7.6}$$

The bars distinguish these objects from the measures and fields in physical coordinates.

Theorem 7.1 (Explicit unconfined classes)**.** *Fix $\mathsf{d} \geq 2$, $\beta > 0$, an initial inverse temperature $\beta_0 > \beta/2$, and an integer $m \geq 3$. In both cases below, take the initial particle law to be $P_{N,0} = \mathbb{Q}_{N,\beta_0}(\mu_0)$.*

(i) Suppose $\mathsf{s} = 0$, and let π denote the density of the thermal equilibrium measure (1.28) for $V(y) = |y|^2/4$. There are $\delta > 0$ and $C < \infty$, depending only on $\mathsf{d}, \beta, \beta_0, m$, such that for every $L > 0$ and every z_0 of the form (1.30) with $\|z_0\|_{\mathcal{X}^m} \leq \delta$, the initial density

$$\mu_0(x) = L^{-\mathsf{d}} \frac{\pi(x/L)e^{z_0(x/L)}}{\int \pi e^{z_0}} \tag{7.7}$$

satisfies both (1.33) and (1.34) for the unconfined dynamics. The constant C is independent of N, k, t, L and of the higher derivatives and support of the perturbation.

(ii) Suppose $0 < \mathsf{s} < \mathsf{d}/2$, and set

$$\gamma(y) := \left(\frac{\beta}{4\pi}\right)^{\mathsf{d}/2} e^{-\beta|y|^2/4}. \tag{7.8}$$

There are $L_* < \infty$, $\delta > 0$, and $C < \infty$, depending only on $\mathsf{d}, \mathsf{s}, \beta, \beta_0, m$, such that for every $L \geq L_*$ and $z_0 \in \mathcal{X}^m$ with $\|z_0\|_{\mathcal{X}^m} \leq \delta$, the initial density

$$\mu_0(x) = L^{-\mathsf{d}} \frac{\gamma(x/L)e^{z_0(x/L)}}{\int \gamma e^{z_0}} \tag{7.9}$$

satisfies both (1.33) and (1.34). The constant C is independent of N, k, t, L within this class.

The passage from smooth to finite-order initial data in part (ii) is justified at the end of the proof, after (7.44).

Remark 7.2 (The one-dimensional cases). The change of variables (7.1) is valid in every dimension. The restriction $\mathsf{d} \geq 2$ in theorem 7.1 and lemma 7.3 comes from the locally integrable force estimates in lemma 2.3: in the stated range, $\mathsf{s} + 1 < \mathsf{d}$. These estimates enter both the background construction below and the first-marginal argument. For $\mathsf{d} = 1$ and $\mathsf{s} > 0$, we expect that replacing the absolute-force estimates by more refined singular-integral estimates should allow one to treat the one-dimensional Riesz cases in the present Hilbert–Schmidt range $0 < \mathsf{s} < 1/2$.

The particle-law passage also uses the collision-free solution from proposition B.1. For $\mathsf{d} = 1$ and $\mathsf{s} = 0$, the collision-free construction is not available at fixed β uniformly in N. We expect that an analogue should be accessible by working directly with an appropriate one-dimensional logarithmic particle dynamics through collisions, but we do not pursue this additional issue here.

For the Riesz case, we construct the background as a perturbation of the Gaussian (7.8), which is invariant for the noninteracting transformed equation. Large L makes the initial coupling small, and (7.4) makes the resulting forcing decay. The following lemma verifies the background hypotheses uniformly in the heat cutoff.

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Lemma 7.3 (A decaying weak interaction around the Gaussian)**.** *Fix $\mathsf{d} \geq 2$, $0 < \mathsf{s} < \mathsf{d}/2$, $\beta > 0$, and an integer $m \geq 3$. There exist $\delta_* > 0$, $C < \infty$, and $0 < \omega < \min\{1/2, \mathsf{s}/2\}$, depending only on these parameters, with the following property. Let*

$$c(\tau) = c_0 e^{-\mathsf{s}\tau/2}, \qquad c_0 \geq 0, \qquad z_0 \in \mathcal{X}^m, \qquad \|z_0\|_{\mathcal{X}^m} + c_0 \leq \delta_*. \tag{7.10}$$

For $0 \leq \varepsilon \leq 1$, with $\mathsf{g}_0 := \mathsf{g}$, equation (7.5) with g replaced by g_ε and initial density $\gamma e^{z_0}/\int \gamma e^{z_0}$ has a global positive solution $\bar{\mu}^\varepsilon_\tau$. Its continuity velocity $\bar{u}^\varepsilon_\tau$, defined by (7.6) with the same replacement, satisfies

$$\|\log(\bar{\mu}^\varepsilon_\tau/\gamma)\|_{\mathcal{X}^m} + \|\bar{u}^\varepsilon_\tau\|_{W^{1,\infty}} \leq C(\|z_0\|_{\mathcal{X}^m} + c_0)e^{-\omega\tau}, \qquad \tau \geq 0. \tag{7.11}$$

For some $a > 0$, the background bounds are

$$\bar{\mu}^\varepsilon_\tau(y) \leq Ce^{-a|y|^2}, \tag{7.12}$$

$$\sup_{\substack{\tau \geq 0 \\ 0 \leq \varepsilon \leq 1}} \left(\|\nabla^2 \log \bar{\mu}^\varepsilon_\tau\|_{L^\infty} + \|\nabla \bar{B}^\varepsilon_\tau\|_{L^\infty} + \|\nabla^2 \bar{B}^\varepsilon_\tau\|_{L^\infty} \right) \leq C, \tag{7.13}$$

$$\mathrm{Var}_{\bar{\mu}^\varepsilon_\tau}(\varphi) \leq \frac{4}{\beta} \int |\nabla \varphi|^2 \, \mathrm{d}\bar{\mu}^\varepsilon_\tau, \qquad \varphi \in C^\infty_c(\mathbb{R}^{\mathsf{d}}). \tag{7.14}$$

The constants are independent of $\tau, \varepsilon, c_0, z_0$ under (7.10). For every $T < \infty$,

$$\sup_{0 \leq \tau \leq T} \left(\|\log(\bar{\mu}^\varepsilon_\tau/\bar{\mu}^0_\tau)\|_{\mathcal{X}^m} + \|\bar{u}^\varepsilon_\tau - \bar{u}^0_\tau\|_{W^{1,\infty}} \right) \underset{\varepsilon \downarrow 0}{\longrightarrow} 0. \tag{7.15}$$

If z_0 has finite higher norms, they remain finite on compact time intervals; the solutions are smooth for smooth initial data.

Proof. **The logarithmic equation.** Fix $0 \leq \varepsilon \leq 1$, and write $\mu_z := \gamma e^z / \int \gamma e^z$. We solve for z modulo constants in the space (1.31). The logarithmic equation is

$$\partial_\tau z = (\beta^{-1}\Delta - \tfrac{1}{2} y \cdot \nabla) z + \beta^{-1} |\nabla z|^2 + c(\tau) \mathcal{F}_\varepsilon(z), \tag{7.16}$$

where

$$\mathcal{F}_\varepsilon(z) := \mathrm{div}(\nabla \mathsf{g}_\varepsilon * \mu_z) - \frac{\beta}{2} y \cdot (\nabla \mathsf{g}_\varepsilon * \mu_z) + (\nabla \mathsf{g}_\varepsilon * \mu_z) \cdot \nabla z. \tag{7.17}$$

The Gaussian versions of (A.30) and (A.32), together with (A.23), give on a fixed small ball

$$\sup_{0 \leq \varepsilon \leq 1} \|\mathcal{F}_\varepsilon(0)\|_{\mathcal{X}^{m-1}} \leq C, \qquad \sup_{0 \leq \varepsilon \leq 1} \|\mathcal{F}_\varepsilon(z) - \mathcal{F}_\varepsilon(w)\|_{\mathcal{X}^{m-1}} \leq C\|z - w\|_{\mathcal{X}^m}. \tag{7.18}$$

For the divergence and the term containing y, the identities used in (7.18) are

$$\mathrm{div}(\nabla \mathsf{g}_\varepsilon * \mu_z) = \sum_{i=1}^{\mathsf{d}} (\partial_i \mathsf{g}_\varepsilon) * (\partial_i \mu_z), \tag{7.19}$$

$$y \cdot (\nabla \mathsf{g}_\varepsilon * \mu_z)(y) = \sum_{i=1}^{\mathsf{d}} \left[((x_i \partial_i \mathsf{g}_\varepsilon) * \mu_z)(y) + (\partial_i \mathsf{g}_\varepsilon * (x_i \mu_z))(y) \right]. \tag{7.20}$$

The first identity transfers the divergence to the density, so derivatives through order $m-1$ require derivatives of μ_z only through order m. The second identity places the factor y either on the kernel or on the density, as in (A.23). Thus, lemma 2.3 gives the one-derivative loss in (7.18). Membership in the completion $\mathcal{X}^{m-1}$ also requires the decay

$$\lim_{R \to \infty} \sup_{\substack{0 \leq \varepsilon \leq 1 \\ \|z\|_{\mathcal{X}^m} \leq \delta_*}} \sup_{|y| \geq R} \sum_{j=1}^{m-1} |\nabla^j \mathcal{F}_\varepsilon(z)(y)| = 0. \tag{7.21}$$

Indeed, derivatives of μ_z through order m, and their first spatial moments, have uniform Gaussian upper bounds on this ball. The kernels $\nabla \mathsf{g}_\varepsilon(x)$ and $x_i \partial_i \mathsf{g}_\varepsilon(x)$ are locally integrable and tend to zero

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as $|x| \to \infty$, uniformly in ε, since $\mathsf{s} > 0$. Splitting each integral in the convolution variable x into $|x| \le |y|/2$ and its complement proves (7.21).

Normalization is checked by the identities

$$\mu_z \mathcal{F}_\varepsilon(z) = \operatorname{div}(\mu_z \nabla \mathsf{g}_\varepsilon * \mu_z), \tag{7.22}$$

$$\mu_z \left[(\beta^{-1}\Delta - \tfrac{1}{2} y \cdot \nabla) z + \beta^{-1} |\nabla z|^2 \right] = \beta^{-1} \Delta \mu_z + \tfrac{1}{2} \operatorname{div}(y \mu_z). \tag{7.23}$$

Both right-hand sides have integral zero. Choosing the representative centered with respect to γ and differentiating $\log \int e^z \, \mathrm{d}\gamma$ therefore cancels the centering constant, exactly as in (A.71)–(A.72). Hence a solution of (7.16) in the quotient gives the normalized solution of (7.5).

Decay and cutoff convergence. Choose $0 < \omega < \min\{1/2, \mathsf{s}/2\}$, and equip trajectories with the norm

$$\|z\|_{\omega,m} := \sup_{\tau \ge 0} e^{\omega \tau} \|z_\tau\|_{\mathcal{X}^m}. \tag{7.24}$$

In this proof, write P_τ for the Ornstein–Uhlenbeck semigroup (A.43) with $A = \mathrm{Id}/2$, whose generator is $\beta^{-1}\Delta - \frac{1}{2} y \cdot \nabla$. The Duhamel map is

$$(\mathcal{T}_\varepsilon z)_\tau := P_\tau z_0 + \int_0^\tau P_{\tau - r} \left[\beta^{-1} |\nabla z_r|^2 + c_0 e^{-\mathsf{s} r/2} \mathcal{F}_\varepsilon(z_r) \right] \mathrm{d}r. \tag{7.25}$$

The Ornstein–Uhlenbeck estimates (A.45) and (7.18) imply, for $\|z\|_{\omega,m}, \|w\|_{\omega,m} \le R$,

$$\|\mathcal{T}_\varepsilon z\|_{\omega,m} \le C \|z_0\|_{\mathcal{X}^m} + C(c_0 + R^2 + c_0 R), \tag{7.26}$$

$$\|\mathcal{T}_\varepsilon z - \mathcal{T}_\varepsilon w\|_{\omega,m} \le C(R + c_0) \|z - w\|_{\omega,m}. \tag{7.27}$$

The inhomogeneous term $c(\tau)\mathcal{F}_\varepsilon(0)$ is controlled by

$$\sup_{\tau \ge 0} e^{\omega \tau} \int_0^\tau (1 + (\tau - r)^{-1/2}) e^{-(\tau - r)/2 - \mathsf{s} r/2} \, \mathrm{d}r \le \int_0^\infty (1 + r^{-1/2}) e^{-(1/2 - \omega) r} \, \mathrm{d}r < \infty. \tag{7.28}$$

Here, $\omega < \mathsf{s}/2$ bounds the remaining factor $e^{-(\mathsf{s}/2 - \omega) r}$ by one. The quadratic and Lipschitz terms in (7.26)–(7.27) have the same integrable time kernel. The constant is independent of the heat cutoff. Choose $R = 2C(\|z_0\|_{\mathcal{X}^m} + c_0)$ and then δ_* small enough that (7.26) maps this ball into itself and the coefficient in (7.27) is at most $1/2$. The fixed point proves global existence and the logarithmic-density bound in (7.11).

For every integer $\ell > m$, the differentiated product and convolution estimates used in (A.76) give

$$\|\beta^{-1} |\nabla z|^2 + c(\tau) \mathcal{F}_\varepsilon(z)\|_{\mathcal{X}^{\ell - 1}} \le C_\ell \left[(\|z\|_{\mathcal{X}^m} + c(\tau)) \|z\|_{\mathcal{X}^\ell} + c(\tau) \right]. \tag{7.29}$$

The short-interval argument (A.81), now with the displayed inhomogeneous term, propagates finite higher norms on every finite interval. Their size does not enter the low-norm contraction.

The force-difference estimates (A.35)–(A.36) and (A.23) give

$$\sup_{\|z\|_{\mathcal{X}^m} \le R} \|\mathcal{F}_\varepsilon(z) - \mathcal{F}_0(z)\|_{\mathcal{X}^{m-1}} \le C_R \left(\varepsilon^{(\mathsf{d} - \mathsf{s} - 1)/2} + \varepsilon^{(\mathsf{d} - \mathsf{s})/2} \right) \xrightarrow[\varepsilon \downarrow 0]{} 0. \tag{7.30}$$

Let z^ε and z^0 be the trajectories constructed with g_ε and g, respectively, from the same initial datum z_0. Thus,

$$z^\varepsilon = \mathcal{T}_\varepsilon z^\varepsilon, \qquad z^0 = \mathcal{T}_0 z^0, \qquad z^\varepsilon|_{\tau = 0} = z^0|_{\tau = 0} = z_0. \tag{7.31}$$

Subtracting these identities and using the contraction coefficient $1/2$ in (7.27) gives

$$\|z^\varepsilon - z^0\|_{\omega,m} \le 2 \|\mathcal{T}_\varepsilon z^0 - \mathcal{T}_0 z^0\|_{\omega,m} \le C c_0 \left(\varepsilon^{(\mathsf{d} - \mathsf{s} - 1)/2} + \varepsilon^{(\mathsf{d} - \mathsf{s})/2} \right) \xrightarrow[\varepsilon \downarrow 0]{} 0. \tag{7.32}$$

The last inequality follows from (7.30) and the integrable kernel in (7.28). This proves convergence in (7.15) for the logarithmic densities. Approximation in $\mathcal{X}^m$ gives the same construction for finite-order initial data; for smooth data, the higher-norm comparison gives smooth solutions.

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The background bounds. Choose the representative satisfying $\int z_\tau^\varepsilon \,\mathrm{d}\gamma = 0$. By (A.16), its absolute value is bounded by $C\|z_\tau^\varepsilon\|_{\mathcal{X}^m}(1+|y|)$. Jensen's inequality bounds the normalizing integral from below by one. Completing the square therefore gives (7.12); differentiating the normalized density gives common Gaussian bounds for its derivatives through order m.

The force estimate (A.11), with derivatives placed on the density, now bounds $\nabla \mathsf{g}_\varepsilon * \bar{\mu}_\tau^\varepsilon$ and its first two derivatives uniformly in τ, ε. The continuity velocity is

$$\bar{u}_\tau^\varepsilon = -\beta^{-1}\nabla z_\tau^\varepsilon - c(\tau)\nabla \mathsf{g}_\varepsilon * \bar{\mu}_\tau^\varepsilon. \tag{7.33}$$

The logarithmic-density bound already proved and $c(\tau) = c_0 e^{-\mathsf{s}\tau/2}$ give the velocity estimate in (7.11). For the convergence of the velocities, subtract (7.33) at cutoffs ε and zero. The force-difference bound (A.35) and the Lipschitz estimate for the normalized density give, for every $T < \infty$,

$$\sup_{\tau\in[0,T]} \|\bar{u}_\tau^\varepsilon - \bar{u}_\tau^0\|_{W^{1,\infty}} \le C\left((1+c_0)\sup_{\tau\in[0,T]}\|z_\tau^\varepsilon - z_\tau^0\|_{\mathcal{X}^m} + c_0\varepsilon^{(\mathsf{d}-\mathsf{s}-1)/2}\right) \xrightarrow[\varepsilon\downarrow 0]{} 0. \tag{7.34}$$

The limit follows from (7.32). This completes (7.15). Moreover, decreasing δ_* if necessary,

$$\nabla^2(-\log\bar{\mu}_\tau^\varepsilon) = \frac{\beta}{2}\mathrm{Id} - \nabla^2 z_\tau^\varepsilon \ge \frac{\beta}{4}\mathrm{Id}, \qquad \tau \ge 0, \quad 0 \le \varepsilon \le 1. \tag{7.35}$$

The curvature criterion [BGL14, Section 4.8] gives the Poincaré inequality (7.14). Finally, (7.6), (7.35), and the convolution bounds give (7.13). □

Proof of theorem 7.1. **The initial Gibbs measures.** At $\tau = 0$, write $X_N = LY_N$ and $\mu_0 = (D_0)_\#\bar{\mu}_0$. Expanding the three terms in (1.6) gives the modulated-energy transformation identity (cf. [RS24, Lemma 2.3])

$$N\mathsf{F}_N(LY_N, \mu_0) = \begin{cases} N\mathsf{F}_N(Y_N, \bar{\mu}_0) + \frac{1}{2}\log L, & \mathsf{s} = 0, \\ L^{-\mathsf{s}} N\mathsf{F}_N(Y_N, \bar{\mu}_0), & \mathsf{s} > 0. \end{cases} \tag{7.36}$$

Thus,

$$\mathsf{K}_{N,\beta_0}(\mu_0) = \begin{cases} L^{-\beta_0/2}\mathsf{K}_{N,\beta_0}(\bar{\mu}_0), & \mathsf{s} = 0, \\ \mathsf{K}_{N,\beta_0 L^{-\mathsf{s}}}(\bar{\mu}_0), & \mathsf{s} > 0. \end{cases} \tag{7.37}$$

The deterministic logarithmic factor cancels between the Gibbs weight and its normalizing constant. In both cases,

$$\bar{P}_{N,0} = \mathbb{Q}_{N,\beta_0 c(0)}(\bar{\mu}_0). \tag{7.38}$$

For $\mathsf{s} = 0$, the transformed dynamics have coupling one and confinement $|y|^2/4$, so theorem 1.2 gives both marginal bounds for every $\tau \ge 0$. Neither the transformed equation nor its initial density depends on L.

The Riesz interaction. Set $c_0 = L^{-\mathsf{s}}$. Choose $L_* \ge 1$ and $\delta > 0$ initially so that $\delta + L_*^{-\mathsf{s}} \le \delta_*$ in lemma 7.3. Its background bounds and proposition 2.1 give static constants and a commutator coefficient $\kappa > 0$ uniform over $\tau \ge 0$, $0 \le \varepsilon \le 1$, $0 \le c_0 \le L_*^{-\mathsf{s}}$, and $\|z_0\|_{\mathcal{X}^m} \le \delta$. Fix $b > 0$ as in (2.45), and write $\dot{c} := \mathrm{d}c/\mathrm{d}\tau$. Since

$$\dot{c}(\tau) = -\frac{\mathsf{s}}{2}c(\tau), \qquad \int_0^\infty |\dot{c}(\tau)|\,\mathrm{d}\tau = c_0, \tag{7.39}$$

the coefficient in (2.45) has the common integrable bound

$$\int_0^\infty \left(\frac{\beta}{\kappa}\sup_{0\le\varepsilon\le 1}\|\nabla\bar{u}_\tau^\varepsilon\|_{L^\infty} + \frac{\beta}{b}|\dot{c}(\tau)|\right)\mathrm{d}\tau \le \frac{\beta C}{\kappa\omega}(\delta + L_*^{-\mathsf{s}}) + \frac{\beta}{b}L_*^{-\mathsf{s}}. \tag{7.40}$$

Here, we used (7.11). The second term comes from the derivative of the coupling in the moving-reference defect (2.32).

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Choose a finite $p_0 > 2$ such that

$$1 - \frac{1}{p_0} < \min\left\{1, \frac{\beta_0}{\beta}\right\}. \tag{7.41}$$

This is possible because $\beta_0 > \beta/2$. Increase L_* and decrease δ so that the right-hand side of (7.40) is strictly less than

$$\log\left(2\left(1 - \frac{1}{p_0}\right)\right). \tag{7.42}$$

These choices depend only on the fixed parameters in the theorem. The initial comparison in lemma 3.2, with $b_0 = \beta_0 c_0$ and $b_1 = \beta c_0$, uses the nonnegative coupling

$$p_0 b_0 - (p_0 - 1)b_1 = c_0(p_0\beta_0 - (p_0 - 1)\beta) \geq 0. \tag{7.43}$$

The comparison is uniform for $0 \leq c_0 \leq L_*^{-\mathsf{s}}$, since all three coupling parameters in lemma 3.2 lie in a fixed bounded nonnegative interval. The strict margin chosen in (7.40)–(7.42) is common to the same initial-data class. Consequently, corollary 3.3 gives a full second moment uniform in N, transformed time, the heat cutoff, and $L \geq L_*$.

Apply proposition 5.1 at the initial coupling $\beta_0 c_0$. Also, (7.11) gives

$$\sup_{0\leq\varepsilon\leq 1} \int_0^\infty \|\nabla \bar{u}_\tau^\varepsilon\|_{L^\infty} \, \mathrm{d}\tau \leq \frac{C}{\omega}(\|z_0\|_{\mathcal{X}^m} + c_0). \tag{7.44}$$

Together with the Poincaré and derivative bounds of lemma 7.3, these estimates verify the hypotheses of proposition 5.3. The term containing $\dot{c}$ in (5.42) is nonpositive, so the homogeneous contraction estimate applies without an additional error. Lemma 5.4 then gives the marginal bounds.

For each fixed N, L, and finite transformed-time interval, case (iii) of lemma 6.3 passes these bounds to the singular particle system. That system is the image under (7.1) of the unconfined strong solution in proposition B.1. The localization is performed at fixed N, L; the marginal estimates have constants independent of both parameters. For finite-order initial data, approximate z_0 in $\mathcal{X}^m$ by the smooth defining class and use the density, initial-measure, and path-localization arguments of lemma 6.3. The constants depend only on the common low-norm bound.

Return to physical coordinates. For $t = L^2(e^\tau - 1)$, (7.2) gives

$$\frac{\mathrm{d}\bar{P}_{N,\tau}^{(k)}}{\mathrm{d}\bar{\mu}_\tau^{\otimes k}}(Y_k) = \frac{\mathrm{d}P_{N,t}^{(k)}}{\mathrm{d}\mu_t^{\otimes k}}(D_\tau^{\otimes k} Y_k). \tag{7.45}$$

Changing variables in the defining integrals therefore yields

$$\chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) = \chi^2(\bar{P}_{N,\tau}^{(k)} \mid \bar{\mu}_\tau^{\otimes k}), \tag{7.46}$$

$$\mathsf{H}(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) = \mathsf{H}(\bar{P}_{N,\tau}^{(k)} \mid \bar{\mu}_\tau^{\otimes k}), \tag{7.47}$$

$$\mathrm{TV}(P_{N,t}^{(k)}, \mu_t^{\otimes k}) = \mathrm{TV}(\bar{P}_{N,\tau}^{(k)}, \bar{\mu}_\tau^{\otimes k}). \tag{7.48}$$

For relative entropy, this is [RS24, Lemma 2.2]; the same Radon–Nikodym identity gives the other two equalities. For each fixed $L > 0$, the time change $t = L^2(e^\tau - 1)$ is a bijection of $[0, \infty)$ onto itself. Hence (7.46) gives

$$\sup_{t\geq 0} \chi^2(P_{N,t}^{(k)} \mid \mu_t^{\otimes k}) = \sup_{\tau\geq 0} \chi^2(\bar{P}_{N,\tau}^{(k)} \mid \bar{\mu}_\tau^{\otimes k}) \leq C\frac{k^2}{N^2}. \tag{7.49}$$

The entropy and total-variation identities give the corresponding time-uniform bounds. Thus, both assertions hold with the same constants as in the transformed estimates, independently of L in the respective classes. □

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8. Failure of unrestricted full-particle chi-square control

We prove theorem 1.7. The construction has two parts. First, a mean-field density depending on one coordinate allows an explicit comparison between the product variance of a terminal test and the Brownian contribution to its fluctuation variance; see proposition 8.4. Second, an entropy estimate controls the nonlinear remainder in the empirical fluctuation equation, and proposition 8.8 converts the variance comparison into divergence of the full-particle chi-square. The singular particle law and the identities used in this argument are justified in Section 8.7.

For the calculations in this section, it is convenient to use the angular variables $x = 2\pi z$ and $\tau = 4t$ for the periodic particle and mean-field equations specified in Section 1.5. We reuse g, μ, P_N, F_N, $\mathsf{K}_{N,b}$, and $\mathbb{Q}_{N,b}$ for the transformed kernel, measures, energy, partition function, and Gibbs law. These local conventions end with this section. The partition and static fluctuation estimates used below are those of [DGR26a].

8.1. **An exact invariant stripe reduction.** Write $\mathbb{T} := \mathbb{R}/(2\pi\mathbb{Z})$, and denote normalized Haar measure by m, with the dimension determined by the domain of integration. In these variables,

$$\widehat{\mathsf{g}}(k) := \frac{\mathbf{1}_{\{k\neq 0\}}}{2|k|^4}, \qquad D := \frac{\pi^2}{\beta}, \qquad \mathrm{d}m(x) := (2\pi)^{-4}\,\mathrm{d}x \quad \text{on } \mathbb{T}^4. \tag{8.1}$$

The kernel is π^2 times the pullback of the kernel in (1.41). The particle equation and generator are

$$\mathrm{d}x_i(\tau) = -\frac{1}{N}\sum_{j\neq i}\nabla\mathsf{g}(x_i(\tau)-x_j(\tau))\,\mathrm{d}\tau + \sqrt{2D}\;\mathrm{d}B_i(\tau), \tag{8.2}$$

$$L_N := D\sum_{i=1}^N \Delta_i - \frac{1}{N}\sum_{i\neq j}\nabla\mathsf{g}(x_i-x_j)\cdot\nabla_i. \tag{8.3}$$

We use (1.5), (1.6) and (1.7) without introducing a second set of definitions. The coupling in the initial Gibbs law becomes D^{-1}: multiplication of the modulated energy by π^2 is canceled by $D^{-1} = \beta/\pi^2$. Thus the transformed initial law is $\mathbb{Q}_{N,D^{-1}}(\mu_0)$.

On the circle, $\mathrm{d}m(x) := \mathrm{d}x/(2\pi)$. We call a density of the form $\rho(x) = \rho_1(x_1)$ a stripe density: it is constant along the three-dimensional tori $\{x_1 = \text{constant}\}$. Its convolution with g also depends only on x_1. Integrating g in the other three coordinates keeps precisely its Fourier modes $(n,0,0,0)$, giving the circle kernel

$$W(x) = \sum_{n=1}^\infty \frac{\cos(nx)}{n^4}. \tag{8.4}$$

The stripe equation is exactly

$$\partial_\tau\rho = D\rho'' + \partial_x(\rho W'*\rho), \qquad b_\tau = -W'*\rho_\tau, \tag{8.5}$$

where convolution and densities are relative to m. For terminal data depending only on x_1, the solution of the backward linearized adjoint equation also depends only on x_1. The reduced operator is

$$\mathcal{L}_\tau f(x) = Df''(x) - \int W'(x-y)(f'(x)-f'(y))\rho_\tau(y)m(\,\mathrm{d}y). \tag{8.6}$$

The same coordinate integration gives (8.6) from the full linearized adjoint. Fubini applies because g and $\nabla\mathsf{g}$ are locally integrable. We use the same symbol ρ for the circle density and its extension $x\mapsto\rho(x_1)$ to $\mathbb{T}^4$; the domain of integration specifies the dimension of m.

On $[0,2\pi]$,

$$W(x) = \frac{\pi^4}{90} - \frac{\pi^2x^2}{12} + \frac{\pi x^3}{12} - \frac{x^4}{48}. \tag{8.7}$$

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Thus, $W \in C^{2,1}$ on the circle, W is even, and

$$W'(0) = W'(\pi) = 0, \qquad W''(0) = -a, \quad W''(\pi) = b, \qquad a = \frac{\pi^2}{6}, \quad b = \frac{\pi^2}{12}. \tag{8.8}$$

The negative curvature at zero expands an isolated packet; the positive curvature at the antipode can confine a packet through the mass in the other packet.

8.2. **Two Gaussian packets with unequal masses.** Let k_δ be the periodized centered Gaussian of variance δ^2, written as a density relative to m:

$$k_\delta(x) = \frac{\sqrt{2\pi}}{\delta} \sum_{j\in\mathbb{Z}} \exp\left(-\frac{(x+2\pi j)^2}{2\delta^2}\right). \tag{8.9}$$

It is smooth, even, strictly positive, and has integral one. Set

$$p = \frac{1}{6}, \qquad q = \frac{5}{6}, \qquad \rho_0^\delta(x) = pk_\delta(x) + qk_\delta(x-\pi), \qquad D_\delta = \kappa\delta^2. \tag{8.10}$$

The constant κ is given in (8.12) below. Write ρ_τ^δ for its solution of (8.5). For every fixed $\delta > 0$, this is a smooth strictly positive trajectory on each compact time interval. Indeed, the drift is uniformly Lipschitz with constant $\|W''\|_\infty$, independent of the density. The nonlinear diffusion is globally well posed by coupling, positivity follows from the maximum principle, and positive diffusion and convolution bootstrap the smooth initial density. All smooth norms needed below are finite for fixed δ; they need not be uniform as $\delta \downarrow 0$.

The initial law is invariant under $x \mapsto -x$. Uniqueness preserves this symmetry, and periodicity also makes it even about π. Therefore,

$$b_\tau^\delta(0) = b_\tau^\delta(\pi) = 0. \tag{8.11}$$

The two components start near 0 and π. The drift generated by $p\delta_0 + q\delta_\pi$ has derivatives $-\kappa$ at 0 and κ_b at π, where

$$\kappa := qb - pa = \frac{\pi^2}{24} > 0, \qquad \kappa_b := qa - pb = \frac{\pi^2}{8} > 0. \tag{8.12}$$

Condition on which component of the initial mixture was chosen. Let Y_a^δ be the real-valued displacement from 0 for the component of mass p, and let Y_b^δ be the displacement from π for the component of mass q. Lift the periodic drift to $\mathbb{R}$ and solve

$$\mathrm{d}Y_a^\delta(\tau) = b_\tau^\delta(Y_a^\delta(\tau))\,\mathrm{d}\tau + \sqrt{2\kappa}\,\delta\,\mathrm{d}B_a(\tau), \tag{8.13}$$

$$\mathrm{d}Y_b^\delta(\tau) = b_\tau^\delta(\pi + Y_b^\delta(\tau))\,\mathrm{d}\tau + \sqrt{2\kappa}\,\delta\,\mathrm{d}B_b(\tau), \tag{8.14}$$

with centered Gaussian initial displacements of variance δ^2, independent of the Brownian motions. Modulo 2π, their laws, with weights p and q, recover $\rho_\tau^\delta m$.

Lemma 8.1 (Narrow-packet mean-field limit)**.** *For every fixed $S < \infty$, as $\delta \downarrow 0$, the measures $\rho_\tau^\delta m$ converge uniformly for $0 \le \tau \le S$ in W_1 to*

$$\rho_* = p\delta_0 + q\delta_\pi. \tag{8.15}$$

The functions b^δ and $(b^\delta)'$ converge uniformly on $[0,S]\times\mathbb{T}$ to $b_* := -W' * \rho_*$ and b_*'. With a common initial Gaussian and Brownian motion, Y_a^δ/δ converges in $L^2(\Omega; C([0,S]))$ to the solution of

$$\mathrm{d}Z_\tau = -\kappa Z_\tau\,\mathrm{d}\tau + \sqrt{2\kappa}\,\mathrm{d}B_\tau, \qquad Z_0 \sim \mathcal{N}(0,1). \tag{8.16}$$

In particular,

$$\mathbb{E}[Z_\tau^2] = 1. \tag{8.17}$$

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Proof. By (8.11),

$$|b^\delta_\tau(y)| + |b^\delta_\tau(\pi + y)| \le 2\|W''\|_{L^\infty}|y|, \qquad y \in \mathbb{R}. \tag{8.18}$$

Apply Doob's maximal inequality (see, e.g., [RY99, Chapter II, Section 1]) and Gronwall to (8.13) and (8.14). For every finite $r \ge 2$, this gives

$$\mathbb{E}\left[\sup_{\tau\le S}|Y^\delta_a(\tau)|^r + \sup_{\tau\le S}|Y^\delta_b(\tau)|^r\right] \le C_{S,r}\delta^r. \tag{8.19}$$

Here, $C_{S,r}$ is independent of $0 < \delta \le 1$. Consequently,

$$\sup_{\tau\le S}\mathsf{W}_1(\rho^\delta_\tau m, \rho_*) \le C_S\delta. \tag{8.20}$$

Because W' and W'' are Lipschitz, this implies uniform convergence of b^δ and $(b^\delta)'$, with error $O_S(\delta)$.

At zero, $b'_*(0) = pa - qb = -\kappa$. We have that

$$\left|\delta^{-1}b^\delta_\tau(\delta z) + \kappa z\right| \le C_S\delta|z| + C\delta|z|^2, \qquad \forall\, 0 \le \tau \le S,\ z \in \mathbb{R}. \tag{8.21}$$

For large z, the same estimate follows from the Lipschitz bound after increasing C. Construct Y^δ_a and Z with $Y^\delta_a(0) = \delta Z_0$ and $B_a = B$ in (8.13) and (8.16). The Brownian terms then cancel in the equation for $\delta^{-1}Y^\delta_a - Z$. Using (8.19) with $r = 4$ in the resulting integral equation yields

$$\mathbb{E}\left[\sup_{0\le\tau\le S}|\delta^{-1}Y^\delta_a(\tau) - Z_\tau|^2\right] \le C_S\delta^2. \tag{8.22}$$

The Ornstein–Uhlenbeck variance is

$$\mathbb{E}[Z^2_\tau] = e^{-2\kappa\tau} + 2\kappa\int_0^\tau e^{-2\kappa(\tau-r)}\,\mathrm{d}r = 1, \tag{8.23}$$

which proves (8.17). □

Choose an even $\chi \in C^\infty_c((-\pi/4, \pi/4))$ equal to one on $(-\pi/8, \pi/8)$, and extend $\psi(x) := x\chi(x)$ periodically. Then ψ is smooth and odd, equals x near zero, and vanishes near π; in particular,

$$\psi'(0) = 1, \qquad \psi'(\pi) = 0, \qquad \rho^\delta_\tau(\psi) = 0. \tag{8.24}$$

The next variance is the variance of $N^{-1/2}\sum_i \psi(x_i)$ for independent x_i with law $\rho^\delta_S m$. It is therefore the comparison variance for the product reference in proposition 8.8.

Corollary 8.2 (Product variance of the terminal test)**.** *For every fixed $S < \infty$, as $\delta \downarrow 0$,*

$$\mathrm{Var}_{\rho^\delta_S m}(\psi) = p\delta^2 + o(\delta^2). \tag{8.25}$$

Proof. For the component of mass p centered at zero, $\psi(Y^\delta_a) = Y^\delta_a$ until the displacement leaves a fixed neighborhood. The fourth-moment estimate in (8.19) makes the error in the second moment $O_S(\delta^4)$. The same bound makes the contribution of the component centered at π $O_S(\delta^4)$ because ψ vanishes near π. Symmetry makes the mean zero. The L^2 convergence in lemma 8.1 and $\mathbb{E}[Z^2_S] = 1$ therefore give (8.25). □

8.3. **The backward test and the Brownian covariance.** Let f^δ_r solve

$$\partial_r f^\delta_r + \mathcal{L}^\delta_r f^\delta_r = 0, \qquad f^\delta_S = \psi. \tag{8.26}$$

For each fixed $\delta > 0$, the solution is classical and smooth. Set $g^\delta_r := (f^\delta_r)'$. Differentiating (8.6) gives

$$-\partial_r g^\delta_r = \kappa\delta^2(g^\delta_r)'' + b^\delta_r(g^\delta_r)' + (b^\delta_r)'g^\delta_r + \int W''(x-y)g^\delta_r(y)\rho^\delta_r(y)m(\,\mathrm{d}y). \tag{8.27}$$

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Lemma 8.3 (Convergence of the adjoint gradients)**.** *The functions g^δ converge uniformly on $[0,S]\times\mathbb{T}$ to the bounded Lipschitz mild solution of*

$$-\partial_r g_r^*(x) = b_*(x)\partial_x g_r^*(x) + b_*'(x) g_r^*(x) + \int_{\mathbb{T}} W''(x-y) g_r^*(y)\,\mathrm{d}\rho_*(y), \qquad g_S^* = \psi'. \tag{8.28}$$

Here, the limit is taken as $\delta \downarrow 0$, so the diffusion coefficient $D_\delta = \kappa\delta^2$ vanishes. At 0 and π,

$$g_r^*(0) = p + qe^{-b(S-r)}, \qquad g_r^*(\pi) = p(1-e^{-b(S-r)}). \tag{8.29}$$

Proof. Reverse time in (8.27). The Markov representation for the transport-diffusion equation gives contraction on bounded functions; see, e.g., [Øks03, Sections 7.1 and 8.1]. Synchronous coupling gives the Lipschitz bound $e^{L(t-s)}$, where $L = \|W''\|_\infty$. The other terms in (8.27) are bounded operators on bounded functions. Multiplication by $(b^\delta)'$ has a Lipschitz bound independent of δ, because W'' is Lipschitz; the integral operator maps bounded functions to functions with Lipschitz norm bounded by $\|W'''\|_{L^\infty}\|g\|_\infty$.

Duhamel's formula and Gronwall therefore give

$$\sup_{0<\delta\le 1}\sup_{0\le r\le S}\left(\|g_r^\delta\|_{L^\infty} + \|\partial_x g_r^\delta\|_{L^\infty}\right) \le C_S. \tag{8.30}$$

The same argument constructs g^* along the deterministic flow of b_*. The transport-diffusion semigroup converges on Lipschitz functions to the deterministic semigroup, with error $O_S(\delta)$: couple the stochastic and deterministic characteristics, use the uniform $O(\delta)$ drift difference from (8.20), and then Gronwall. For a uniformly Lipschitz g, the difference of the integral operators is also $O_S(\delta)$, since $y \mapsto W''(x-y)g(y)$ has a Lipschitz bound uniform in x. Multiplication coefficients converge uniformly. Subtracting the two mild equations and applying Gronwall gives

$$\sup_{0\le r\le S}\|g_r^\delta - g_r^*\|_{L^\infty} \le C_S\delta. \tag{8.31}$$

Since $b_*(0) = b_*(\pi) = 0$, the solutions of $\dot{x} = b_*(x)$ starting at either point remain there. The transport term in (8.28) therefore contributes nothing when the mild equation is evaluated at 0 or π, and the resulting two equations form a closed system. With $A_r = g_r^*(0)$ and $B_r = g_r^*(\pi)$,

$$-\partial_r\begin{pmatrix}A_r\\B_r\end{pmatrix} = b\begin{pmatrix}-q & q\\ p & -p\end{pmatrix}\begin{pmatrix}A_r\\B_r\end{pmatrix}, \qquad (A_S,B_S) = (1,0). \tag{8.32}$$

The terms involving $W''(0)$ cancel: at zero, for example, $(-\kappa)A_r - paA_r + qbB_r = -qbA_r + qbB_r$. The vector (p,p) is constant under this system and the other mode decays with rate b. Solving gives (8.29). □

The limiting values in (8.29) determine the Brownian variance in proposition 8.8. For the family just constructed, this variance is

$$\mathcal{V}_\delta = 2\kappa\delta^2\int_0^S\int |g_r^\delta(x)|^2\rho_r^\delta(x)m(\,\mathrm{d}x)\,\mathrm{d}r. \tag{8.33}$$

Proposition 8.4 (Variance comparison)**.** *For the fixed angular horizon $S = 20$,*

$$\lim_{\delta\downarrow 0}\frac{\mathcal{V}_\delta}{\delta^2} = 2\kappa p^2 S + \frac{\kappa pq}{b}(1-e^{-2bS}), \tag{8.34}$$

$$\lim_{\delta\downarrow 0}\frac{\mathcal{V}_\delta}{\mathrm{Var}_{\rho_S^\delta m}(\psi)} = 2\kappa pS + \frac{\kappa q}{b}(1-e^{-2bS}) = \frac{5\pi^2}{18} + \frac{5}{12}(1-e^{-10\pi^2/3}) > 2. \tag{8.35}$$

Consequently, $\mathcal{V}_\delta > 2\,\mathrm{Var}_{\rho_S^\delta m}(\psi)$ for all sufficiently small $\delta > 0$.

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Proof. By lemmas 8.1 and 8.3, the integrand in (8.33), after division by $2\kappa\delta^2$, converges uniformly in r to

$$p\big(p+qe^{-b(S-r)}\big)^2+q\big(p(1-e^{-b(S-r)})\big)^2=p^2+pqe^{-2b(S-r)}. \tag{8.36}$$

Integrating this identity over $0\le r\le S$ proves (8.34). Dividing by the variance asymptotic in (8.25) then gives (8.35). For $S=20$, its first term is $5\pi^2/18>5/2$, so the limiting ratio is strictly larger than two. □

The choice $p=1/6$ and $q=5/6$ gives $\kappa>0$. With $D_\delta=\kappa\delta^2$, the rescaled displacement of the packet centered at zero converges to a nondegenerate stationary Ornstein–Uhlenbeck process. Consequently, its contribution to the product-reference variance remains $p\delta^2+o(\delta^2)$, while the common-translation Brownian variance accumulates linearly in S. Strict positivity of κ is needed for this simultaneous presence of positive noise and stationary packet variance, not merely for localization at scale δ on a fixed time interval.

8.4. **Exponential moments of bounded centered statistics.** We next establish the exponential moments used in Section 8.5. They control the reference defect (8.51) and the nonlinear empirical remainder (8.71). We use the bounded-kernel case of the degenerate U-statistic estimates recalled in [DGR26a, Propositions 3.1 and 3.3]; the corresponding exponential estimate is also due to Jabin–Wang [JW18, Theorem 4].

Lemma 8.5 (Uniform small exponential moment)**.** *Let H be a bounded symmetric kernel and μ a probability measure such that*

$$\int H(x,y)\mu(\mathrm{d}y)=\int H(x,y)\mu(\mathrm{d}x)=0\quad \text{for } \mu\text{-a.e. remaining variable.} \tag{8.37}$$

For independent $x_1,\dots,x_N$ with common law μ, set

$$U_N:=\frac{1}{N}\sum_{i\neq j}H(x_i,x_j). \tag{8.38}$$

There are universal constants $a,C>0$ such that, if $\|H\|_\infty>0$,

$$\sup_N \mathbb{E}_{\mu^{\otimes N}}\Big[e^{a|U_N|/\|H\|_\infty}\Big]\le C. \tag{8.39}$$

Proof. This is the bounded-kernel consequence of the cited U-statistic bounds. In the density setting of [JW18, Theorem 4], apply that theorem to both signs of a sufficiently small multiple of $H/\|H\|_\infty$, and use $e^{|x|}\le e^x+e^{-x}$. The theorem includes the diagonal sum; removing it changes the exponent by at most a universal constant. The decoupling formulation in [DGR26a, Propositions 3.1 and 3.3] gives the same conclusion for an arbitrary probability measure μ. □

For a bounded symmetric c with its diagonal set to zero, define

$$Q_{N,\mu}[c]:=N\iint c(x,y)(\mu_N-\mu)(\mathrm{d}x)(\mu_N-\mu)(\mathrm{d}y). \tag{8.40}$$

If $\widetilde{c}$ is the Hoeffding centering of c, then

$$Q_{N,\mu}[c]=\frac{1}{N}\sum_{i\neq j}\widetilde{c}(x_i,x_j)-\frac{2}{N}\sum_i c_\mu(x_i)+\bar{c}_\mu, \tag{8.41}$$

where $c_\mu(x):=\int c(x,y)\mu(\mathrm{d}y)$, $\bar{c}_\mu:=\iint c\,\mathrm{d}\mu^{\otimes 2}$, and $\widetilde{c}(x,y):=c(x,y)-c_\mu(x)-c_\mu(y)+\bar{c}_\mu$. Since

$$\left|-\frac{2}{N}\sum_i c_\mu(x_i)+\bar{c}_\mu\right|\le 3\|c\|_\infty, \tag{8.42}$$

lemma 8.5 gives a small exponential moment of $|Q_{N,\mu}[c]|$, uniformly in N.

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8.5. **Global entropy control relative to the moving Gibbs law.** In this and the next two subsections, t denotes angular time. Fix $D > 0$, $S < \infty$, and a smooth probability density ρ_t on $\mathbb{T}^4$, and set $\mathrm{d}\mu_t := \rho_t \, \mathrm{d}m$. Assume that

$$\partial_t \rho_t = D\Delta \rho_t + \operatorname{div}(\rho_t \nabla(\mathsf{g} * \mu_t)), \qquad \inf_{[0,S]\times\mathbb{T}^4} \rho > 0. \tag{8.43}$$

Set $\mu_t(v) := \int_{\mathbb{T}^4} v \, \mathrm{d}\mu_t$, and $\Gamma_{N,t} := \mathbb{Q}_{N,D^{-1}}(\mu_t)$. The particle law $P_{N,t}$ solves (8.2) with initial law $\Gamma_{N,0}$, independent of the Brownian motions; its construction is given in Section 8.7. Write

$$G_{N,t} := \frac{\mathrm{d}\Gamma_{N,t}}{\mathrm{d}\mu_t^{\otimes N}}. \tag{8.44}$$

By (1.7), $G_{N,t} = \mathsf{K}_{N,D^{-1}}(\mu_t)^{-1} e^{-D^{-1}N\mathsf{F}_N(\cdot,\mu_t)}$, and hence

$$\|G_{N,t}\|^2_{L^2(\mu_t^{\otimes N})} = \frac{\mathsf{K}_{N,2D^{-1}}(\mu_t)}{\mathsf{K}_{N,D^{-1}}(\mu_t)^2}. \tag{8.45}$$

The static partition theorem in [DGR26a, Theorem 2.2(i), Remark 2.1], applied at D^{-1} and $2D^{-1}$, bounds the numerator. Jensen's inequality and the definition in (1.6) give $\mathsf{K}_{N,D^{-1}}(\mu) \geq e^{D^{-1}I_\mu}$, uniformly in N. Hence

$$\sup_N \sup_{t\leq S} \|G_{N,t}\|_{L^2(\mu_t^{\otimes N})} < \infty. \tag{8.46}$$

To check that the partition theorem applies uniformly for $0 \leq t \leq S$, boundedness of ρ on $[0,S]\times\mathbb{T}^4$ and $\mathsf{g} \in L^1(m) \cap L^2(m)$ give

$$\sup_{0\leq t\leq S} \left(\|\rho_t\|_{L^\infty} + \|\mathsf{g} * \mu_t\|_{L^\infty} + |I_{\mu_t}| + \|\widetilde{\mathsf{g}}_{\mu_t}\|_{L^2(\mu_t^{\otimes 2})} \right) < \infty, \tag{8.47}$$

where $\widetilde{\mathsf{g}}_{\mu_t}$ is the kernel centered in each variable under μ_t. The bounded convolution also bounds the one-body logarithmic exponential moments. These are the scalar controls in [DGR26a, Appendix A]. For the trajectory $\rho_t^\delta(x_1)$ constructed in Section 8.2, these bounds may depend on the fixed δ.

For a Lipschitz vector field $v : \mathbb{T}^4 \to \mathbb{R}^4$, define

$$c_v(x,y) := \mathbf{1}_{\{x\neq y\}} (v(x) - v(y)) \cdot \nabla \mathsf{g}(x-y), \tag{8.48}$$

Then

$$\|c_v\|_{L^\infty} \leq C \|\nabla v\|_{L^\infty}. \tag{8.49}$$

The mean-field continuity velocity is

$$u_t := -\nabla(\mathsf{g} * \mu_t) - D\nabla \log \rho_t, \qquad \partial_t \rho_t + \operatorname{div}(\rho_t u_t) = 0. \tag{8.50}$$

Lemma 8.6 (Exact reference defect and entropy estimate)**.** *The moving Gibbs defect is*

$$\partial_t \Gamma_{N,t} - L_N^* \Gamma_{N,t} = R_{N,t}\Gamma_{N,t}, \qquad R_{N,t} = -\frac{1}{2D}\left(Q_{N,\mu_t}[c_{u_t}] - \mathbb{E}_{\Gamma_{N,t}}[Q_{N,\mu_t}[c_{u_t}]] \right). \tag{8.51}$$

For the actual particle law $P_{N,t}$ started from $\Gamma_{N,0}$,

$$\sup_N \sup_{t\leq S} \mathsf{H}(P_{N,t} \mid \Gamma_{N,t}) < \infty, \qquad \sup_N \sup_{t\leq S} \mathsf{H}(P_{N,t} \mid \mu_t^{\otimes N}) < \infty. \tag{8.52}$$

Proof. Off the diagonal, the Gibbs score satisfies

$$D\nabla_i \log \Gamma_{N,t} = b_i^N - u_t(x_i). \tag{8.53}$$

Consequently, $L_N^*\Gamma_{N,t} = -\sum_i \operatorname{div}_i(u_t(x_i)\Gamma_{N,t})$. The material derivative of (1.6) and the derivative of its partition function are

$$\left(\partial_t + \sum_i u_t(x_i)\cdot\nabla_i \right) N\mathsf{F}_N(\cdot,\mu_t) = \frac{1}{2} Q_{N,\mu_t}[c_{u_t}], \tag{8.54}$$

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$$\partial_t \log \mathsf{K}_{N,D^{-1}}(\mu_t) = -\frac{1}{2D}\mathbb{E}_{\Gamma_{N,t}}[Q_{N,\mu_t}[c_{u_t}]]. \tag{8.55}$$

These are the angular versions of the computation in Section 2; substitution in the normalized density gives (8.51).

The kernel c_{u_t} is uniformly bounded on the fixed time interval. By lemma 8.5 and Equations (8.41) and (8.46) and Cauchy–Schwarz, there are $a > 0$ and $K < \infty$, independent of N and t, such that

$$\mathbb{E}_{\Gamma_{N,t}}[e^{a|R_{N,t}|}] \le K. \tag{8.56}$$

Indeed, if $a_0 > 0$ is small enough for the product exponential moment, then

$$\mathbb{E}_{\Gamma_{N,t}}[e^{a_0|Q_{N,\mu_t}[c_{u_t}]|}] \le \|G_{N,t}\|_{L^2(\mu_t^{\otimes N})}\left(\mathbb{E}_{\mu_t^{\otimes N}}[e^{2a_0|Q_{N,\mu_t}[c_{u_t}]|}]\right)^{1/2} \le C. \tag{8.57}$$

Jensen then bounds $|\mathbb{E}_{\Gamma_{N,t}}[Q_{N,\mu_t}[c_{u_t}]]|$ by $a_0^{-1}\log C$. Choosing a in terms of a_0 and D proves (8.56). Set $h_N(t) := \mathsf{H}(P_{N,t} \mid \Gamma_{N,t})$. The exact entropy identity is

$$h_N'(t) = -D\int \left|\nabla \log \frac{\mathrm{d}P_{N,t}}{\mathrm{d}\Gamma_{N,t}}\right|^2 \mathrm{d}P_{N,t} - \mathbb{E}_{P_{N,t}}[R_{N,t}]. \tag{8.58}$$

Its singular-law justification is given in Section 8.7. The Donsker–Varadhan lemma relative to $\Gamma_{N,t}$ and (8.56) imply

$$h_N'(t) \le a^{-1}(h_N(t) + \log K), \qquad h_N(0) = 0. \tag{8.59}$$

Gronwall proves the first bound in (8.52) for every finite S. Finally,

$$\mathsf{H}(P_{N,t} \mid \mu_t^{\otimes N}) = h_N(t) + \mathbb{E}_{P_{N,t}}[\log G_{N,t}] \le 2h_N(t) + \log\int G_{N,t}^2 \,\mathrm{d}\mu_t^{\otimes N}. \tag{8.60}$$

Here, the Donsker–Varadhan lemma is applied relative to $\Gamma_{N,t}$ with observable $\log G_{N,t}$, and $\int G_{N,t}\,\mathrm{d}\Gamma_{N,t} = \int G_{N,t}^2\,\mathrm{d}\mu_t^{\otimes N}$. This proves the second bound. □

Lemma 8.7 (Two consequences for the actual laws)**.** *If $(c_t)_{0\le t\le S}$ is a family of symmetric commutator kernels with $\sup_{t\le S}\|c_t\|_{L^\infty} < \infty$, then*

$$\sup_N \sup_{t\le S} \mathbb{E}_{P_{N,t}}[|Q_{N,\mu_t}[c_t]|] < \infty. \tag{8.61}$$

If $(v_t)_{0\le t\le S}$ is a family of real-valued functions with $\sup_{t\le S}\|v_t\|_{L^\infty} < \infty$, then

$$\sup_N \sup_{t\le S} \mathbb{E}_{P_{N,t}}\left[\left|N^{-1/2}\sum_i (v_t(x_i) - \mu_t(v_t))\right|^2\right] < \infty. \tag{8.62}$$

Proof. For the first assertion, apply (8.57) with c_t in place of c_{u_t} and then the Donsker–Varadhan lemma with (8.52). For the second assertion, Hoeffding's lemma gives

$$\mathbb{E}_{\mu_t^{\otimes N}}\left[e^{\theta N^{-1/2}\sum_i (v_t(x_i)-\mu_t(v_t))}\right] \le e^{\theta^2 \operatorname{osc}(v_t)^2/8}, \qquad \theta \in \mathbb{R}; \tag{8.63}$$

see also [DGR26a, Section 8.2]. Integrating (8.63) with $\theta = \sqrt{2a}\,Z$, where Z is an independent standard Gaussian, yields

$$\mathbb{E}_{\mu_t^{\otimes N}}\left[e^{a|N^{-1/2}\sum_i (v_t(x_i)-\mu_t(v_t))|^2}\right] \le \left(1 - \frac{a}{2}\operatorname{osc}(v_t)^2\right)^{-1/2}. \tag{8.64}$$

Choose $a > 0$ so that $a\sup_{t\le S}\operatorname{osc}(v_t)^2 < 1$. Cauchy–Schwarz transfers the moment with parameter $a/2$ to $\Gamma_{N,t}$, and the Donsker–Varadhan lemma transfers its first moment to $P_{N,t}$. This proves (8.62). □

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8.6. **Brownian noise as a terminal chi-square obstruction.** We now use lemma 8.7 to control the remainder and predictable quadratic variation in the empirical fluctuation equation (8.73). The variance A in (8.68) is the Brownian variance computed in (8.33) for the stripe construction. It suffices that the initial statistics have a tight family of laws.

Proposition 8.8 (Noise-variance obstruction)**.** *Let ρ_t satisfy (8.43), and let $P_{N,0} = \Gamma_{N,0}$. For $\psi \in C^\infty(\mathbb{T}^4;\mathbb{R})$, let f^r solve*

$$\partial_r f^r + \mathcal{L}_r f^r = 0, \qquad f^S = \psi, \tag{8.65}$$

$$\mathcal{L}_r f(x) := D\Delta f(x) - \int \nabla \mathsf{g}(x-y)\cdot(\nabla f(x) - \nabla f(y))\,\mathrm{d}\mu_r(y). \tag{8.66}$$

Then f is the unique classical solution, with

$$\sup_{0\le r\le S}(\|f^r\|_{L^\infty} + \|\nabla f^r\|_{L^\infty} + \|\nabla^2 f^r\|_{L^\infty}) < \infty. \tag{8.67}$$

Set

$$V := \mathrm{Var}_{\mu_S}(\psi) > 0, \qquad A := 2D\int_0^S \mu_r(|\nabla f^r|^2)\,\mathrm{d}r. \tag{8.68}$$

If $A > 2V$, then

$$\lim_{N\to\infty} \chi^2(P_{N,S} \mid \mu_S^{\otimes N}) = +\infty. \tag{8.69}$$

Proof. **The backward solution and empirical identity.** The local diffusion with drift $-\nabla\mathsf{g} * \mu_r$ has smooth coefficients. The remaining term $\nabla\mathsf{g} * (\rho_r \nabla f)$ maps C^m to C^{m-1}, with norm bounded by $C_{m,\rho}\|\nabla\mathsf{g}\|_{L^1}$. The usual one-derivative parabolic smoothing estimate therefore makes its Duhamel kernel integrable. Successive short-interval contraction and differentiation give existence, uniqueness, and (8.67); see, e.g., [Lun95, Chapter 5] for the local parabolic estimates. In the stripe case, this is the solution already constructed in (8.26).

Set $\mu_{N,r} := N^{-1}\sum_i \delta_{x_i(r)}$ and $\xi_{N,r} := \sqrt{N}(\mu_{N,r} - \mu_r)$. The finite-particle adjoint identity of [DGR26a, Proposition 8.2], in the angular normalization, is

$$\xi_{N,S}(\psi) = Y_N + M_N(S) + R_N, \qquad Y_N := \xi_{N,0}(f^0), \tag{8.70}$$

$$R_N := -\frac{1}{2\sqrt{N}}\int_0^S Q_{N,\mu_r}[c_{\nabla f^r}]\,\mathrm{d}r, \tag{8.71}$$

$$M_N(t) := \sqrt{\frac{2D}{N}}\sum_i \int_0^t \nabla f^r(x_i(r))\cdot \mathrm{d}B_i(r), \qquad 0\le t\le S. \tag{8.72}$$

The process M_N is a continuous square-integrable martingale. Here, $c_{\nabla f}$ is defined by (8.48), with diagonal zero. The coefficient can also be checked directly: substituting $\mu_{N,r} = \mu_r + N^{-1/2}\xi_{N,r}$ in the symmetrized empirical drift gives

$$\mathrm{d}\xi_{N,r}(f^r) = \xi_{N,r}(\partial_r f^r + \mathcal{L}_r f^r)\,\mathrm{d}r - \frac{Q_{N,\mu_r}[c_{\nabla f^r}]}{2\sqrt{N}}\,\mathrm{d}r + \mathrm{d}M_N(r). \tag{8.73}$$

The first term vanishes by (8.65). All pair sums retain the normalization $1/N$.

The Hessian bound makes $c_{\nabla f^r}$ uniformly bounded, so (8.61) gives $\mathbb{E}[|R_N|] = O(N^{-1/2})$. The predictable quadratic variation of M_N is

$$\langle M_N\rangle_S = 2D\int_0^S \mu_{N,r}(|\nabla f^r|^2)\,\mathrm{d}r. \tag{8.74}$$

Consequently,

$$\langle M_N\rangle_S - A = \frac{2D}{\sqrt{N}}\int_0^S \xi_{N,r}(|\nabla f^r|^2)\,\mathrm{d}r. \tag{8.75}$$

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Thus, (8.62) and Cauchy–Schwarz give $\mathbb{E}[|\langle M_N\rangle_S - A|] \le C_{D,S,f} N^{-1/2} \to 0$. We also have the deterministic bound $\langle M_N\rangle_S \le 2DS \sup_{0\le r\le S} \|\nabla f^r\|_{L^\infty}^2$.

The initial statistic and the independent Gaussian limit. The initial variables Y_N are tight. Indeed, Cauchy–Schwarz, (8.46), and the bounded fourth moment of the independent centered sum give

$$\mathbb{E}[Y_N^2] \le \|G_{N,0}\|_{L^2(\mu_0^{\otimes N})} \left(\mathbb{E}_{\mu_0^{\otimes N}}[|\xi_{N,0}(f^0)|^4]\right)^{1/2} \le C_{\rho_0,D,f^0}. \tag{8.76}$$

Along any subsequence one can therefore choose a further subsequence on which Y_N converges in law, say to a probability law ν.

To prove that the limiting Gaussian variable from $M_N(S)$ is independent of the limit of Y_N, set, for $\theta \in \mathbb{R}$,

$$Z_N(r) := \exp\left(i\theta M_N(r) + \frac{\theta^2}{2}\langle M_N\rangle_r \right). \tag{8.77}$$

Its modulus is bounded by the deterministic bound on $\langle M_N\rangle_S$, so Z_N is a true martingale and $\mathbb{E}[Z_N(S) \mid \mathcal{F}_0] = 1$. Since Y_N is $\mathcal{F}_0$-measurable, write $e^{i\theta M_N(S)} = Z_N(S) e^{-\theta^2\langle M_N\rangle_S/2}$ and use the conditional expectation above. For every $\alpha, \theta \in \mathbb{R}$, the mean-value estimate for the last exponential gives

$$\left|\mathbb{E}[e^{i\alpha Y_N + i\theta M_N(S)}] - e^{-\theta^2 A/2}\mathbb{E}[e^{i\alpha Y_N}]\right| \le C_\theta \mathbb{E}[|\langle M_N\rangle_S - A|] \longrightarrow 0. \tag{8.78}$$

Thus, the joint characteristic functions converge to those of an independent pair with laws ν and $\mathcal{N}(0,A)$. Hence, along the further subsequence,

$$\xi_{N,S}(\psi) \;\longrightarrow\; \nu * \mathcal{N}(0,A) \quad \text{in law.} \tag{8.79}$$

Under the product law $\mu_S^{\otimes N}$, the same function $X_N \mapsto N^{-1/2}\sum_i(\psi(x_i) - \mu_S(\psi))$ converges in law to $\mathcal{N}(0,V)$ by the usual central limit theorem.

The limiting chi-square divergence. For any probability measure ν, the convolution $\nu * \mathcal{N}(0,A)$ in (8.79) has infinite chi-square relative to $\mathcal{N}(0,V)$ when $A > 2V$. To see this without an assumption on ν, choose $R < \infty$ with $\nu([-R,R]) > 0$. The Lebesgue density p of $\nu * \mathcal{N}(0,A)$ satisfies

$$p(x) \ge \frac{\nu([-R,R])}{\sqrt{2\pi A}} \exp\left(-\frac{(|x|+R)^2}{2A} \right). \tag{8.80}$$

If γ_V is the density of $\mathcal{N}(0,V)$, then

$$\frac{p(x)^2}{\gamma_V(x)} \ge \frac{\nu([-R,R])^2\sqrt{2\pi V}}{2\pi A} \exp\left[\left(\frac{1}{2V} - \frac{1}{A}\right)x^2 - \frac{2R}{A}|x| - \frac{R^2}{A} \right]. \tag{8.81}$$

The positive quadratic coefficient makes this lower bound nonintegrable.

Suppose, contrary to (8.69), that there are $N_j \uparrow \infty$ such that

$$\sup_j \chi^2(P_{N_j,S} \mid \mu_S^{\otimes N_j}) < \infty. \tag{8.82}$$

Apply data processing to the map $X_N \mapsto N^{-1/2}\sum_i(\psi(x_i) - \mu_S(\psi))$ under these two laws; see [NS20, Section 2]. Along the further subsequence selected above, its laws converge to $\nu * \mathcal{N}(0,A)$ and $\mathcal{N}(0,V)$, respectively. Joint lower semicontinuity under weak convergence would then give

$$\chi^2(\nu * \mathcal{N}(0,A) \mid \mathcal{N}(0,V)) \le \liminf_j \chi^2(P_{N_j,S} \mid \mu_S^{\otimes N_j}) < \infty, \tag{8.83}$$

contrary to (8.81). Here, lower semicontinuity follows, for example, from [vEH14, Theorem 19 and (6)], applied to the Rényi divergence of order two. Every bounded subsequence is excluded, proving (8.69). □

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The static central limit theorem of [DGR26a, Proposition 2.11] additionally identifies ν as a centered Gaussian law. We use this identification only for the central limit statement in theorem 1.7; the chi-square divergence proved above requires only tightness of Y_N.

8.7. **Actual singular laws and the entropy identity.** We justify the entropy and empirical identities for the singular diffusion (8.2). The collision-free construction in Section B has the same local form on the torus: the periodic kernel is $-\pi^2 \log|x|$ plus a smooth function near zero, and the internal contribution of the smooth part to the cluster-dispersion equation is bounded by a constant times the dispersion, as is the exterior-field remainder. Here, we instead use the square-integrable force to give a short approximation argument that also identifies the reversible stationary law needed for entropy differentiation. Throughout this subsection, N and $D > 0$ are fixed.

Let

$$\mathrm{d}\Pi_N := Z_N^{-1} \exp\left(-\frac{1}{DN} \sum_{i<j} \mathsf{g}(x_i - x_j) \right) \mathrm{d}m^{\otimes N}. \tag{8.84}$$

Stationary domination and convergence of the particle laws. Let $\mathsf{g}_\varepsilon := e^{\varepsilon\Delta}\mathsf{g}$, and let Π_N^ε and $\Gamma_{N,0}^\varepsilon$ be the corresponding stationary and initial laws. The lower bound on g and the bound on $\|\nabla \mathsf{g}\|_{L^2(\mathbb{T}^4)}$ are preserved by heat smoothing. Dominated convergence of the stationary densities, including their normalizing constants, therefore gives

$$\sup_{0<\varepsilon\le 1} \left(\left\| \frac{\mathrm{d}\Pi_N^\varepsilon}{\mathrm{d}m^{\otimes N}} \right\|_{L^\infty} + \int |b_N^\varepsilon|^2 \, \mathrm{d}\Pi_N^\varepsilon \right) < \infty. \tag{8.85}$$

Moreover,

$$\frac{\mathrm{d}\Gamma_{N,0}^\varepsilon}{\mathrm{d}\Pi_N^\varepsilon}(X) = C_{N,\varepsilon} \prod_{i=1}^N \rho_0(x_i) \exp\Big(D^{-1} (\mathsf{g}_\varepsilon * \mu_0)(x_i) \Big). \tag{8.86}$$

Since ρ_0 is bounded above and below and $\sup_\varepsilon \|\mathsf{g}_\varepsilon * \mu_0\|_\infty < \infty$, normalization yields

$$0 < c_N \le \frac{\mathrm{d}\Gamma_{N,0}^\varepsilon}{\mathrm{d}\Pi_N^\varepsilon} \le C_N < \infty. \tag{8.87}$$

This domination also holds on path space. Indeed, for every nonnegative measurable path functional Φ, disintegrating with respect to the initial configuration gives

$$\mathbb{E}_{\Gamma_{N,0}^\varepsilon}[\Phi(X)] = \int \mathbb{E}_x[\Phi(X)] \frac{\mathrm{d}\Gamma_{N,0}^\varepsilon}{\mathrm{d}\Pi_N^\varepsilon}(x) \, \mathrm{d}\Pi_N^\varepsilon(x) \le C_N \mathbb{E}_{\Pi_N^\varepsilon}[\Phi(X)]. \tag{8.88}$$

Here, $\mathbb{E}_x$ refers to the same cutoff diffusion with deterministic initial configuration x. Stationarity, (8.85), and (8.88) imply

$$\mathbb{E}\left[\int_s^t |b_N^\varepsilon(X_r)|^2 \, \mathrm{d}r \right] \le C_N |t-s|, \qquad \mathbb{E}\left[\left| \int_s^t b_N^\varepsilon(X_r) \, \mathrm{d}r \right|^2 \right] \le C_N |t-s|^2. \tag{8.89}$$

Kolmogorov's tightness criterion (see, e.g., [RY99, Chapter I, Theorem 2.1]) applies to the drift integrals, and the Brownian terms are tight in continuous paths.

Fix two particle indices $i \neq j$, and choose a smooth function $0 \le F_r \le 1$ of their relative position that equals one when $\mathrm{dist}(x_i, x_j) \le r$ and vanishes when this distance is at least $2r$. For sufficiently small $r > 0$, the four-dimensional relative coordinate gives

$$\|F_r\|_{L^1(m^{\otimes N})} + \|F_r\|_{L^2(m^{\otimes N})}^2 \le C_N r^4, \quad \|\Delta F_r\|_{L^1(m^{\otimes N})} \le C_N r^2, \quad \|\nabla F_r\|_{L^2(m^{\otimes N})} \le C_N r. \tag{8.90}$$

For the stationary cutoff diffusion, Itô's formula gives

$$F_r(X_t) = F_r(X_0) + \int_0^t (D\Delta F_r + b_N^\varepsilon \cdot \nabla F_r)(X_u) \, \mathrm{d}u + \sqrt{2D} \int_0^t \nabla F_r(X_u) \cdot \mathrm{d}B_u. \tag{8.91}$$

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Taking the supremum and expectation, and applying Doob's inequality and the Itô isometry to the last term (see, e.g., [RY99, Chapters II and IV]), yields

$$\begin{aligned}\mathbb{E}_{\Pi_N^\varepsilon}\left[\sup_{t\le S} F_r(X_t)\right] &\le \|F_r\|_{L^1(\Pi_N^\varepsilon)} + DS\|\Delta F_r\|_{L^1(\Pi_N^\varepsilon)} \\ &\quad + S\|b_N^\varepsilon\|_{L^2(\Pi_N^\varepsilon)}\|\nabla F_r\|_{L^2(\Pi_N^\varepsilon)} + \sqrt{8DS}\|\nabla F_r\|_{L^2(\Pi_N^\varepsilon)} \\ &\le C_{N,D,S}(r^4 + r^2 + r) \le C_{N,D,S} r. \end{aligned} \tag{8.92}$$

By (8.88), the same estimate holds up to a constant for the initial law $\Gamma_{N,0}^\varepsilon$. Summing over pairs gives

$$\mathbb{P}\left(\min_{0\le t\le S}\min_{i<j}\operatorname{dist}(x_i^\varepsilon(t), x_j^\varepsilon(t)) < r\right) \le C_{N,D,S} r. \tag{8.93}$$

The event in (8.93) is open in the space of continuous paths. Portmanteau therefore gives the same bound for any path limit, and letting $r \downarrow 0$ excludes collisions. On each collision-free compact configuration set, the drifts converge in every smooth norm. The stopped martingale problem consequently passes to the limit. Pathwise uniqueness away from collisions identifies the limit with the singular strong solution and removes the subsequence. These arguments apply both to stationary and to Gibbs initial laws. Stationarity and reversibility pass to the limit, and (8.88) identifies the latter path law as a bounded change of its initial distribution.

Entropy differentiation. Write $P_{N,t} = p_t \Pi_N$ and $\Gamma_{N,t} = e^{A_t}\Pi_N$. The function A_t is smooth with bounded derivatives on each fixed time interval, because the singular pair factors cancel in this ratio. The bounds $0 < c_N \le p_0 \le C_N$ pass to p_t under the reversible Markov semigroup.

We first justify integration by parts across the collision set. Choose χ_r vanishing near all pair collisions and converging to one elsewhere, with $\int |\nabla \chi_r|^2 \,\mathrm{d}\Pi_N \to 0$, as supplied by (8.90). For a smooth test φ,

$$\left|\int \varphi \nabla A_t \cdot \nabla \chi_r \,\mathrm{d}\Pi_N\right| \le \|\varphi\|_{L^\infty}\|\nabla A_t\|_{L^\infty}\left(\int |\nabla \chi_r|^2 \,\mathrm{d}\Pi_N\right)^{1/2} \longrightarrow 0. \tag{8.94}$$

Furthermore, $L_N A_t = D\Delta A_t + b_N \cdot \nabla A_t \in L^2(\Pi_N)$ by (8.85). Thus A_t belongs to the generator domain, and the reversible gradient form is the closure of $D\int |\nabla\varphi|^2 \,\mathrm{d}\Pi_N$ without a collision-boundary term. The entropy chain rule for this form, applied to the bounded positive semigroup density p_t, gives the relative de Bruijn identity (cf. [BGL14, Chapter 5])

$$\frac{\mathrm{d}}{\mathrm{d}t}\int p_t \log p_t \,\mathrm{d}\Pi_N = -D\int |\nabla \log p_t|^2 p_t \,\mathrm{d}\Pi_N. \tag{8.95}$$

The identity holds for a.e. t and in integrated form starting at zero; form approximation and the bounds on p_t justify the logarithmic test. Differentiating the smooth cross term now gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{H}(P_{N,t} \mid \Gamma_{N,t}) = -D\int |\nabla \log p_t|^2 p_t \,\mathrm{d}\Pi_N - \int p_t(\partial_t A_t + L_N A_t)\,\mathrm{d}\Pi_N. \tag{8.96}$$

Since $L_N e^{A_t} = e^{A_t}(L_N A_t + D|\nabla A_t|^2)$, the reference defect is

$$R_{N,t} = \partial_t A_t - L_N A_t - D|\nabla A_t|^2. \tag{8.97}$$

Reversibility gives $\int p_t L_N A_t \,\mathrm{d}\Pi_N = -D\int \nabla p_t \cdot \nabla A_t \,\mathrm{d}\Pi_N$. Substitution into (8.96) therefore yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{H}(P_{N,t} \mid \Gamma_{N,t}) = -D\int |\nabla \log p_t - \nabla A_t|^2 p_t \,\mathrm{d}\Pi_N - \int R_{N,t} p_t \,\mathrm{d}\Pi_N, \tag{8.98}$$

which is (8.58). The empirical Itô identities follow by stopping away from the collision set and then removing the stopping, using compact-time positive pair separation and the bounded symmetrized commutator in (8.73).

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Finiteness of chi-square at fixed particle number. The law $P_{N,t}$ is bounded above by a fixed-N multiple of Π_N. The latter has a bounded density relative to $m^{\otimes N}$, and ρ_t is bounded below on $[0,S]\times\mathbb{T}^4$. Consequently, $\chi^2(P_{N,t}\mid \mu_t^{\otimes N})$ is finite for every fixed N and $t\le S$.

8.8. **Completion of the theorem and the fluctuation covariance.**

Proof of theorem 1.7. Choose the family in (8.10) and the terminal test in (8.24). By proposition 8.4, there is $\delta_0>0$ such that $\mathcal{V}_\delta > 2\,\mathrm{Var}_{\rho_S^\delta m}(\psi)$ for all $0<\delta<\delta_0$. Set $\beta_*=24/\delta_0^2$. Given any $\beta>\beta_*$, fix $\delta=\sqrt{24/\beta}<\delta_0$.

The angular four-dimensional trajectory $\rho_\tau^\delta(x_1)$ is smooth and strictly positive on $[0,20]$. The actual full singular particle laws exist by Section 8.7. Static moments, the global moving-reference entropy estimate, and the backward-test estimates satisfy every hypothesis of proposition 8.8. Writing $\mathrm{d}\mu_{20}^\delta(x):=\rho_{20}^\delta(x_1)\,\mathrm{d}m(x)$ on $\mathbb{T}^4$, we obtain

$$\lim_{N\to\infty}\chi^2(P_{N,20}\mid(\mu_{20}^\delta)^{\otimes N})=+\infty. \tag{8.99}$$

Under (8.1), this is exactly Equation (1.42) at physical time 5 with $\beta=\pi^2/(\kappa\delta^2)=24/\delta^2$ and

$$\mu_0(z)=\rho_0^\delta(2\pi z_1). \tag{8.100}$$

The initial uniform chi-square bound is (8.46). The full entropy bound is (8.52); finiteness at each fixed N was established in Section 8.7. Finally, $k=N$ in Equation (1.45) would give a bounded sequence in Equation (1.42), a contradiction.

Moreover, the static central limit theorem of [DGR26a, Proposition 2.11] identifies the initial covariance as a positive screened covariance. Hence the terminal covariance on the test ψ satisfies

$$\Sigma_\delta(\psi,\psi)=\Sigma_{\delta,0}(f_0^\delta,f_0^\delta)+\mathcal{V}_\delta\ge\mathcal{V}_\delta>2\,\mathrm{Var}_{\rho_S^\delta m}(\psi). \tag{8.101}$$

The initial Gaussian fluctuation and the asymptotically independent Brownian Gaussian fluctuation therefore give the central limit theorem (1.44), and (8.101) gives its strict variance inequality. □

Appendix A. Thermal equilibrium measures and logarithmic mean-field stability

We prove proposition 6.1 in four steps. The thermal equilibrium estimates in Section A.1 give the density bounds and one-body Poincaré inequality. Section A.2 establishes the convolution estimates needed to construct the linearized evolution in Section A.3. Finally, Section A.4 uses the one-derivative smoothing estimate to solve the nonlinear logarithmic equation.

The kernel and cutoff are normalized as in (1.1) and (2.12). A superscript ε denotes replacement of g by g_ε, with $\varepsilon=0$ corresponding to the singular kernel. Bounds stated uniformly in the cutoff hold for $\varepsilon\in[0,\varepsilon_*]$, for a sufficiently small fixed $\varepsilon_*>0$. The time intervals and any uniformity in time are specified separately.

A.1. **Thermal equilibrium measures.** For the Coulomb interaction, existence, uniqueness, and the Euler–Lagrange characterization of the thermal equilibrium measure are proved in [AS20, Lemmas 2.1–2.2]. We give the argument for the present log/Riesz range, including the derivative estimates and convergence under heat truncation. Convexity of $-\log\pi$ outside a ball will give the one-body Poincaré inequality.

Lemma A.1 (Thermal regularity and a one-body spectral gap)**.** *For every confinement satisfying* (1.23)–(1.24)*, every $\beta>0$, and kernel* (1.1)*, the thermal equation* (1.28) *has a unique probability density π. It is smooth and positive. Set $U_\pi:=-\log\pi$. There are constants $a,M,C_P>0$ such that*

$$\pi(x)\le Me^{-a|x|^2},\qquad \|\nabla^j\pi\|_{L^1}+\sup_x e^{a|x|^2/2}|\nabla^j\pi(x)|<\infty,\quad (j\ge0), \tag{A.1}$$

$$\nabla U_\pi=\beta\nabla V+\beta\nabla\mathsf{g}*\pi,\qquad \|\nabla^2U_\pi\|_\infty<\infty,\qquad \nabla^2U_\pi\ge\frac{\beta\lambda_A}{2}\mathrm{Id}\quad \textit{outside a ball}, \tag{A.2}$$

and the probability measure $\pi(x)\,\mathrm{d}x$ satisfies (4.15). The corresponding bounds hold uniformly for π^ε. For every integer $j \geq 0$ and $0 < a' < a/2$,

$$\|\nabla^j(\pi^\varepsilon - \pi)\|_{L^1} + \sup_x e^{a'|x|^2}|\nabla^j(\pi^\varepsilon - \pi)(x)| \underset{\varepsilon\downarrow 0}{\longrightarrow} 0. \tag{A.3}$$

Proof. **Existence and uniqueness.** Consider the free energy

$$\mathcal{F}(\rho) = \beta^{-1}\int \rho\log\rho\,\mathrm{d}x + \int V(x)\rho(x)\,\mathrm{d}x + \frac{1}{2}\iint \mathsf{g}(x-y)\rho(x)\rho(y)\,\mathrm{d}x\,\mathrm{d}y. \tag{A.4}$$

A Gaussian trial density has finite free energy. For $\mathsf{s} > 0$, the interaction is nonnegative. For $\mathsf{s} = 0$, use

$$-\log|x-y| \geq -\log(1+|x|) - \log(1+|y|), \qquad \log(1+r) \leq \epsilon r^2 + C_\epsilon. \tag{A.5}$$

Combining part of the quadratic confinement with entropy relative to a broader Gaussian gives coercivity in both cases. Consequently, a minimizing sequence has uniformly bounded second moment and relative entropy with respect to such a Gaussian. Tightness and lower semicontinuity give a minimizer. To see the latter point directly for the logarithm, its negative tail is uniformly integrable by the second-moment bound, whereas its positive singular part is lower semicontinuous after truncation. The entropy plus a quadratic term is a Gaussian relative entropy up to a constant and is lower semicontinuous. The same argument applies to the cutoffs uniformly.

The entropy term excludes a set of positive Lebesgue measure on which a minimizer vanishes. Indeed, adding a small bounded density on such a set, while subtracting the same mass on a set where the minimizer is bounded above and below, produces a negative entropy variation of order $\epsilon|\log\epsilon|$. Restricting first to sets where the potentials are bounded makes the other variations of order ϵ. Exhaustion gives the claimed positivity almost everywhere. Bounded variations where the density and its reciprocal are bounded then yield (1.28), again by exhaustion.

The heat representation shows that the interaction quadratic form is nonnegative on zero-mass densities. Strict convexity of entropy therefore gives uniqueness of the minimizer. The same convexity shows that any solution of the Euler–Lagrange relation is the minimizer.

Uniform density and derivative bounds. Use the normalizations $b = 0$ and $V(0) = 0$ following (1.24), and write

$$\widetilde{V}(x) = V(x) - \frac{1}{2}x\cdot Ax. \tag{A.6}$$

Given $\delta > 0$, choose R_δ so that $|\nabla\widetilde{V}(x)| \leq \delta$ for $|x| \geq R_\delta$. Integrate $\nabla\widetilde{V}$ along the segment from 0 to x, using its global bound inside B_{R_δ} and the bound δ outside. This gives

$$|\widetilde{V}(x)| \leq \delta|x| + C_\delta. \tag{A.7}$$

After decreasing δ and completing the square,

$$V(x) \geq \frac{\lambda_A}{4}|x|^2 - C. \tag{A.8}$$

Let γ_A be the Gaussian proportional to $e^{-\beta x\cdot Ax/2}$. The function $\widetilde{V}$ is integrable under γ_A, and in both interaction cases $\sup_y(\mathsf{g}_\varepsilon * \gamma_A)(y) \leq C$ uniformly. Jensen's inequality under γ_A gives a lower bound on the thermal normalizer:

$$Z_{\pi^\varepsilon} = Z_{\gamma_A}\int e^{-\beta(\widetilde{V}+h^\varepsilon_{\pi^\varepsilon})}\,\mathrm{d}\gamma_A \geq Z_{\gamma_A}\exp\left[-\beta\int\widetilde{V}\,\mathrm{d}\gamma_A - \beta\int(\mathsf{g}_\varepsilon*\gamma_A)\,\mathrm{d}\pi^\varepsilon\right] \geq c_{V,A,\beta} > 0. \tag{A.9}$$

For a Riesz kernel, $h^\varepsilon_{\pi^\varepsilon} \geq 0$, and (A.8) immediately gives a Gaussian upper bound. For the logarithm, the uniform second moment and (A.5) give

$$h^\varepsilon_{\pi^\varepsilon}(x) \geq -\log(1+|x|) - C, \qquad \pi^\varepsilon(x) \leq C(1+|x|)^\beta e^{-c|x|^2}. \tag{A.10}$$

The second inequality follows from (A.8) and (A.9). It implies (A.1) for $j = 0$ with a weaker Gaussian exponent. The singular minimizer satisfies the same bound.

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The force estimate in lemma 2.3 gives

$$\|\nabla \mathsf{g}_\varepsilon * f\|_{L^\infty} \le \|\nabla \mathsf{g}_\varepsilon\|_{L^1(B_1)}\|f\|_{L^\infty} + \|\nabla \mathsf{g}_\varepsilon\|_{L^\infty(B_1^c)}\|f\|_{L^1} \le C(\|f\|_{L^1} + \|f\|_{L^\infty}). \tag{A.11}$$

The convolution $\nabla \mathsf{g} * \pi$ is continuous: its local part is controlled by translation continuity in L^1, and its far-field part by dominated convergence. Differentiating the Euler–Lagrange equation gives

$$\nabla \pi = -\beta(\nabla V + \nabla \mathsf{g} * \pi)\pi, \qquad \nabla^j(\nabla \mathsf{g} * \pi) = \nabla \mathsf{g} * \nabla^j \pi. \tag{A.12}$$

The first identity gives a polynomial times Gaussian bound for $\nabla\pi$. The second identity and (A.11) then permit induction in the derivative order. This proves smoothness and (A.1), also for the cutoff equilibrium densities. Each polynomial factor is absorbed by the weaker Gaussian weight in that display.

The derivatives of $\nabla \widetilde{V}$, and the force and its derivatives convolved with a Gaussian-decaying density, tend to zero at infinity. For the derivatives this follows by splitting $|y| \le |x|/2$ and its complement, differentiating the smooth far-field kernel on the first region, and using the Gaussian derivative bounds on the second. The same splitting is uniform in the cutoff. Since $\nabla^2 V = A + \nabla^2 \widetilde{V}$ and $\nabla^2 \widetilde{V} \to 0$, this proves (A.2) after enlarging the exterior ball.

The one-body Poincaré inequality. By (A.2), U_π has bounded Hessian and is uniformly convex outside a ball, with constants independent of ε for U_{π^ε}. A bounded smooth radial modification χ may therefore be chosen, independently of ε, so that $\nabla^2(U_{\pi^\varepsilon} + \chi) \ge \rho_1 \mathrm{Id}$ for some $\rho_1 > 0$. Indeed, choose $\chi'(r) = Kr$ on the fixed interior ball, then decrease χ' to zero on a sufficiently wide annulus with its negative radial second derivative smaller than half the exterior convexity constant. Keep $\chi' \ge 0$ on that annulus. Applying the curvature criterion to the measure proportional to $e^{-U_{\pi^\varepsilon} - \chi}$, and then the bounded-perturbation principle (see, e.g., [BGL14, Sections 4.2 and 4.8]), gives

$$\mathrm{Var}_{\pi^\varepsilon}(\psi) \le e^{\operatorname{osc}\chi}\rho_1^{-1}\int |\nabla\psi|^2 \,\mathrm{d}\pi^\varepsilon, \qquad (\psi \in H^1(\pi^\varepsilon),\ 0 \le \varepsilon \le \varepsilon_*). \tag{A.13}$$

Thus, the Poincaré constant is independent of the heat cutoff.

Convergence of the thermal equilibrium measures. Tightness and the preceding bounds give subsequential convergence of the cutoff minimizers π^ε as $\varepsilon \downarrow 0$. Lower semicontinuity and testing against the singular minimizer π identify every limit with π; alternatively pass the Euler–Lagrange equation by the same near-/far-field splitting and use uniqueness. The force satisfies

$$\|\nabla \mathsf{g}_\varepsilon - \nabla \mathsf{g}\|_{L^1(\mathbb{R}^{\mathsf{d}})} \longrightarrow 0, \qquad \sup_{r \le |x| \le R} |\nabla^j(\nabla \mathsf{g}_\varepsilon - \nabla \mathsf{g})(x)| \longrightarrow 0 \tag{A.14}$$

as $\varepsilon \downarrow 0$, for every $0 < r < R < \infty$ and $j \ge 0$; moreover, $\sup_{0 \le \varepsilon \le \varepsilon_*} \|\nabla \mathsf{g}_\varepsilon\|_{L^\infty(B_1^c)} < \infty$, by lemma 2.3. The differentiated Euler–Lagrange equations and the uniform Gaussian bounds give local convergence of every derivative by induction. To obtain (A.3), first use local derivative convergence on B_R. On B_R^c, the Gaussian derivative bound, uniform in ε, gives, for $a' < a/2$,

$$\sup_{0 \le \varepsilon \le \varepsilon_*} \sup_{|x| \ge R} e^{a'|x|^2}|\nabla^j(\pi^\varepsilon - \pi)(x)| \le C_j e^{-(a/2 - a')R^2}. \tag{A.15}$$

The corresponding L^1 tail is uniformly small as well. Combining these tail bounds with convergence on B_R, then letting $\varepsilon \downarrow 0$ and $R \to \infty$, proves (A.3).

□

A.2. **Elementary bounds in the logarithmic derivative spaces.** Recall that constants are identified with zero in $\mathcal{X}^j$. When a pointwise representative is needed we impose $\pi(z) = 0$. Then

$$|z(x)| \le \|\nabla z\|_\infty \left(|x| + \int |y| \,\mathrm{d}\pi(y)\right), \qquad \|z\|_{L^2(\pi)} \le \sqrt{C_P}\|z\|_{\mathcal{X}^j}. \tag{A.16}$$

In particular, bounded sets in $\mathcal{X}^j$ have a common linear-growth bound.

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For an integrable function f with Gaussian-decaying derivatives, the interaction term in the logarithmic equation is

$$\mathcal{R}_\pi f := \pi^{-1}\operatorname{div}(\pi(\nabla \mathsf{g} * f)) = \operatorname{div}(\nabla \mathsf{g} * f) - \nabla U_\pi \cdot (\nabla \mathsf{g} * f). \tag{A.17}$$

The next lemma separates the linear perturbation of the Ornstein–Uhlenbeck generator from the quadratic remainder. Both map $\mathcal{X}^m$ into $\mathcal{X}^{m-1}$; the loss of one derivative is compensated by the smoothing estimate in Section A.3.

Lemma A.2 (Convolution and nonlinear estimates)**.** *For $m \geq 3$, the operator*

$$Dz := -(\nabla \widetilde{V} + \nabla \mathsf{g} * \pi) \cdot \nabla z + \mathcal{R}_\pi(\pi(z - \pi(z))) \tag{A.18}$$

maps $\mathcal{X}^m$ boundedly into $\mathcal{X}^{m-1}$. For the representative Dz defined by (A.18),

$$\lim_{R\to\infty} \sup_{\|z\|_{\mathcal{X}^m}\leq 1} \sup_{|x|\geq R} \sum_{j=0}^{m-1} |\nabla^j Dz(x)| = 0. \tag{A.19}$$

Let P_t^A be the Ornstein–Uhlenbeck semigroup given by (A.43). For every $0 < a < b < \infty$, the set

$$\{P_t^A Dz : \ t \in [a,b], \ \|z\|_{\mathcal{X}^m} \leq 1\} \tag{A.20}$$

has compact closure in $\mathcal{X}^m$. In particular, $P_t^A D : \mathcal{X}^m \to \mathcal{X}^m$ is compact for every $t > 0$.

Let $\mu_z = \pi e^z / \int \pi e^z$, take $\pi(z) = 0$, and set

$$\mathcal{N}(z) := \beta^{-1}|\nabla z|^2 + (\nabla \mathsf{g} * (\mu_z - \pi)) \cdot \nabla z + \mathcal{R}_\pi(\mu_z - \pi - \pi z). \tag{A.21}$$

For every sufficiently small fixed ball in $\mathcal{X}^m$,

$$\|\mathcal{N}(z)\|_{\mathcal{X}^{m-1}} \leq C\|z\|_{\mathcal{X}^m}^2, \qquad \|\mathcal{N}(z) - \mathcal{N}(w)\|_{\mathcal{X}^{m-1}} \leq C(\|z\|_{\mathcal{X}^m} + \|w\|_{\mathcal{X}^m})\|z - w\|_{\mathcal{X}^m}. \tag{A.22}$$

The corresponding cutoff maps converge to these maps in operator norm for D and locally uniformly, with their Lipschitz bounds, for $\mathcal{N}$.

Proof. **Convolution bounds and the derivative count.** We first estimate $\mathcal{R}_\pi f$ and the terms defining D. If z is centered, (A.16) and lemma A.1 give common Gaussian bounds with polynomial prefactors for all derivatives of πz through order m. Every convolution with the force is bounded by splitting its local L^1 and bounded far-field parts. The coefficient $\nabla U_\pi = \beta A x + \beta \nabla \widetilde{V} + \beta \nabla \mathsf{g} * \pi$ requires one additional observation. The coefficient $\nabla \widetilde{V}$ and its derivatives are bounded and tend to zero at infinity. For the linear part of ∇U_π, use

$$x_i(\partial_j \mathsf{g} * f)(x) = ((z_i \partial_j \mathsf{g}) * f)(x) + (\partial_j \mathsf{g} * (y_i f))(x). \tag{A.23}$$

Both kernels on the right are locally integrable and bounded on B_1^c. The second term is estimated by (A.11) with $y_i f$ in place of f. In the divergence term of $\mathcal{R}_\pi$, write

$$\operatorname{div}(\nabla \mathsf{g} * f) = \sum_{i=1}^{\mathsf{d}} (\partial_i \mathsf{g}) * (\partial_i f). \tag{A.24}$$

Apply (A.23) after transferring further derivatives to f. Derivatives of $\mathcal{R}_\pi f$ through order $m-1$ therefore use derivatives of f through order m, and no higher derivative of the singular kernel is needed. More explicitly,

$$\sum_{j=0}^{m-1} \|\nabla^j \mathcal{R}_\pi f\|_{L^\infty} \leq C \sum_{j=0}^{m} \left(\|(1+|x|)\nabla^j f\|_{L^1} + \|(1+|x|)\nabla^j f\|_{L^\infty} \right). \tag{A.25}$$

For $f = \pi(z - \pi(z))$, (A.16) and Leibniz' rule give

$$\sum_{j=0}^{m} \left(\|(1+|x|)\nabla^j [\pi(z-\pi(z))]\|_{L^1} + \|(1+|x|)\nabla^j [\pi(z-\pi(z))]\|_{L^\infty} \right) \leq C\|z\|_{\mathcal{X}^m}. \tag{A.26}$$

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These estimates and the bounded derivatives of the coefficient of ∇z in (A.18) prove the boundedness of D.

Decay at infinity. For positive Riesz exponents, the first term on the right of (A.23) tends to zero, because $z_i\partial_j \mathsf{g}(z) = O(|z|^{-\mathsf{s}})$; the remaining terms also decay. For the logarithm, the same conclusion uses $\int f = 0$. The kernel $z \mapsto z_i\partial_j \mathsf{g}(z)$ is bounded on $\mathbb{R}^{\mathsf{d}} \setminus \{0\}$, and its gradient is bounded by $C|z|^{-1}$. Its value at the origin does not affect the convolution. Thus, subtracting its value at x gives

$$|((z_i\partial_j \mathsf{g}) * f)(x)| \le \frac{C}{|x|}\int |y|\,|f(y)|\,\mathrm{d}y + C\int_{|y|>|x|/2} |f(y)|\,\mathrm{d}y \xrightarrow[|x|\to\infty]{} 0. \tag{A.27}$$

The moments and tails on the right are common on the bounded input sets under consideration. Derivatives of f have zero integral as well. This proves uniform decay for the centered $f = \pi(z-\pi(z))$ and for every derivative needed above. The coefficients $\nabla\widetilde{V}$ and $\nabla \mathsf{g} * \pi$, together with all derivatives required here, tend to zero at infinity. Their products with ∇z therefore tend to zero at infinity, uniformly on the bounded input sets. This proves (A.19). It also shows that these images lie in the stated completion, by smooth spatial cutoff.

Compactness after Ornstein–Uhlenbeck smoothing. Fix $0 < a < b < \infty$. For $t \in [a,b]$, the Gaussian formula (A.43) for P_t^A gives two additional bounded derivatives of the image after convolution. For $t \in [a,b]$ and $\|z\|_{\mathcal{X}^m} \le 1$, the derivatives of $P_t^A Dz$ vanish uniformly at infinity, since the singular values of e^{-At} stay away from zero and the Gaussian convolution kernels have uniform integrable tails. Hence the image is equicontinuous through order m, uniformly vanishes at infinity, and is bounded. Arzelà–Ascoli on a large ball, followed by the uniform tail bound, proves compactness in the full bounded-derivative norm and hence in $\mathcal{X}^m$. This argument is uniform for t in a compact positive interval.

The nonlinear estimates. Take $\pi(z) = 0$. Jensen gives $\int e^z\,\mathrm{d}\pi \ge 1$, and (A.16) controls $e^{|z(x)|}$ by an exponential of $C\|z\|_{\mathcal{X}^m}(1+|x|)$. Taylor's formula gives

$$e^z - 1 - z = z^2\int_0^1 (1-r)e^{rz}\,\mathrm{d}r, \qquad \int e^z\,\mathrm{d}\pi - 1 = \int (e^z - 1 - z)\,\mathrm{d}\pi. \tag{A.28}$$

In particular, the exact quadratic remainder of the normalized density is

$$\mu_z - \pi - \pi z = \frac{\pi}{\int e^z\,\mathrm{d}\pi}\left[e^z - 1 - z - (1+z)\int (e^z - 1 - z)\,\mathrm{d}\pi\right]. \tag{A.29}$$

Differentiate this identity through order m. Each term in its right-hand side contains at least two factors of z or its derivatives; undifferentiated factors of z have at most linear growth. The Gaussian derivative bounds for π absorb these polynomial factors and $e^{C\|z\|(1+|x|)}$ on a fixed $\mathcal{X}^m$ ball. The analogous first-order identity gives, for every integer $q \ge 0$,

$$\sum_{j=0}^{m}\Big(\|(1+|x|)^q\nabla^j(\mu_z - \pi)\|_{L^1} + \|(1+|x|)^q\nabla^j(\mu_z - \pi)\|_{L^\infty}\Big) \le C_q\|z\|_{\mathcal{X}^m}, \tag{A.30}$$

$$\sum_{j=0}^{m}\Big(\|(1+|x|)^q\nabla^j(\mu_z - \pi - \pi z)\|_{L^1} + \|(1+|x|)^q\nabla^j(\mu_z - \pi - \pi z)\|_{L^\infty}\Big) \le C_q\|z\|_{\mathcal{X}^m}^2. \tag{A.31}$$

For the difference estimates, differentiate the normalized density along the segment joining w to z:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\mu_{w+\theta(z-w)} = \mu_{w+\theta(z-w)}\left[z - w - \int (z-w)\,\mathrm{d}\mu_{w+\theta(z-w)}\right]. \tag{A.32}$$

The same weighted bounds, integrated in θ, yield

$$\sum_{j=0}^{m}\Big(\|(1+|x|)\nabla^j[\mu_z - \mu_w - \pi(z-w)]\|_{L^1} + \|(1+|x|)\nabla^j[\mu_z - \mu_w - \pi(z-w)]\|_{L^\infty}\Big)$$

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$$\leq C(\|z\|_{\mathcal{X}^m} + \|w\|_{\mathcal{X}^m})\|z - w\|_{\mathcal{X}^m}. \tag{A.33}$$

The same sum without the linear subtraction is bounded by $C\|z - w\|_{\mathcal{X}^m}$. Apply (A.25) to the quadratic remainder and its difference. The two remaining terms in (A.21) satisfy

$$\||\nabla z|^2\|_{\mathcal{X}^{m-1}} + \|[\nabla \mathsf{g} * (\mu_z - \pi)] \cdot \nabla z\|_{\mathcal{X}^{m-1}} \leq C\|z\|^2_{\mathcal{X}^m}, \tag{A.34}$$

by Leibniz' rule, (A.11), and (A.30); their differences satisfy the corresponding bilinear bound. These are precisely (A.22). The density remainders have integral zero, so the decay argument above applies to their convolution terms. The remaining term $|\nabla z|^2$ tends to the constant $|a_z|^2$ at infinity, and its positive-order derivatives vanish there, as required by the quotient space.

Cutoff convergence. The preceding estimates hold on every fixed bounded $\mathcal{X}^m$ ball, with constants depending on its radius. The heat representation gives the following bounds on the force difference $\nabla \mathsf{g}_\varepsilon - \nabla \mathsf{g}$:

$$\|\nabla \mathsf{g}_\varepsilon - \nabla \mathsf{g}\|_{L^1} \leq C\varepsilon^{(\mathsf{d}-\mathsf{s}-1)/2}, \tag{A.35}$$

$$\||x|(\nabla \mathsf{g}_\varepsilon - \nabla \mathsf{g})\|_{L^1} \leq C\varepsilon^{(\mathsf{d}-\mathsf{s})/2}. \tag{A.36}$$

Indeed, these follow by integrating respectively $\|\nabla p_r\|_{L^1} = Cr^{-1/2}$ and $\||x|\nabla p_r\|_{L^1} = C$ over $0 < r < \varepsilon$ with the weight in (2.12). Both powers are positive in the stated range.

The operators D^ε and D use the centered densities $\pi^\varepsilon(z - \pi^\varepsilon(z))$ and $\pi(z - \pi(z))$, respectively. Choose the representative satisfying $\pi(z) = 0$. By (A.16),

$$\sup_{\substack{\|z\|_{\mathcal{X}^m} \leq 1 \\ \pi(z)=0}} |\pi^\varepsilon(z)| \leq C\|(1 + |x|)(\pi^\varepsilon - \pi)\|_{L^1} \underset{\varepsilon \downarrow 0}{\longrightarrow} 0. \tag{A.37}$$

Here, (A.3) implies convergence of the weighted norm on the right. For the density difference, the exact identity is

$$\pi^\varepsilon(z - \pi^\varepsilon(z)) - \pi z = (\pi^\varepsilon - \pi)z - \pi^\varepsilon \pi^\varepsilon(z), \qquad \pi(z) = 0. \tag{A.38}$$

Leibniz' rule, $|z(x)| \leq C(1 + |x|)$, and the unit bound on the positive-order derivatives of z give, for $p \in \{1, \infty\}$,

$$\begin{aligned} &\sup_{\substack{\|z\|_{\mathcal{X}^m} \leq 1 \\ \pi(z)=0}} \sum_{j=0}^m \left\|(1 + |x|)\nabla^j[\pi^\varepsilon(z - \pi^\varepsilon(z)) - \pi z]\right\|_{L^p} \\ &\leq C \sum_{j=0}^m \left\|(1 + |x|)^2 \nabla^j(\pi^\varepsilon - \pi)\right\|_{L^p} + C\|(1 + |x|)(\pi^\varepsilon - \pi)\|_{L^1} \underset{\varepsilon \downarrow 0}{\longrightarrow} 0. \end{aligned} \tag{A.39}$$

The term containing $\pi^\varepsilon(z)$ is bounded using (A.37) and the cutoff-uniform Gaussian derivative bounds for π^ε in lemma A.1. The remaining terms tend to zero by (A.3). Combining this with (A.35)–(A.36) in (A.23) proves

$$\|D^\varepsilon - D\|_{\mathcal{X}^m \to \mathcal{X}^{m-1}} \underset{\varepsilon \downarrow 0}{\longrightarrow} 0. \tag{A.40}$$

For the normalized densities, use (A.29) and (A.32) with π^ε in place of π; subtracting a constant from z does not change either normalized density. The same estimates give, for each fixed $R > 0$,

$$\sup_{\|z\|_{\mathcal{X}^m} \leq R} \|\mathcal{N}^\varepsilon(z) - \mathcal{N}(z)\|_{\mathcal{X}^{m-1}} \underset{\varepsilon \downarrow 0}{\longrightarrow} 0. \tag{A.41}$$

The Lipschitz bounds on that ball are independent of $\varepsilon \in [0, \varepsilon_*]$. □

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A.3. **The linearized semigroup.** We first construct the linearized evolution in the logarithmic derivative norm, then prove its exponential decay. The one-body Poincaré inequality gives decay in weighted L^2. To obtain decay in $\mathcal{X}^m$, we also use compactness of the difference between the linearized and Ornstein–Uhlenbeck semigroups.

The linearized operator is

$$\mathcal{L}_\pi = -\beta^{-1}\mathcal{A}_\pi(\mathrm{Id} + \beta T_\pi), \tag{A.42}$$

where $\mathcal{A}_\pi$ is the weighted Laplacian in lemma 4.2 and T_π is the centered interaction operator in lemma 2.4.

The anisotropic Ornstein–Uhlenbeck semigroup with generator $\beta^{-1}\Delta - Ax\cdot\nabla$ acts on the quotient spaces by

$$P_t^A z(x) = \mathbb{E}\left[z\left(e^{-At}x + \Sigma_t^{1/2}Z\right)\right], \qquad \Sigma_t = \beta^{-1}A^{-1}(\mathrm{Id} - e^{-2At}), \qquad Z \sim \mathcal{N}(0, \mathrm{Id}). \tag{A.43}$$

Here, $\lambda_A > 0$ denotes the smallest eigenvalue of A. For a smooth representative of an element of $\mathcal{X}^j$,

$$\nabla^j P_t^A z = (e^{-At})^{\otimes j} P_t^A(\nabla^j z). \tag{A.44}$$

Strong continuity holds first on affine-plus-compact smooth functions. The boundedness supplied by (A.44) and density then extend it to $\mathcal{X}^j$. Differentiating the Gaussian formula (A.43) also gives

$$\|P_t^A\|_{\mathcal{X}^j\to\mathcal{X}^j} \le C_A e^{-\lambda_A t}, \qquad \|P_t^A\|_{\mathcal{X}^{j-1}\to\mathcal{X}^j} \le C_{A,j}(1 + t^{-1/2})e^{-\lambda_A t}, \quad j \ge 2,\ t > 0. \tag{A.45}$$

Throughout this subsection, constants may depend on the fixed parameters β, A, j.

Lemma A.3 (A stable linearized semigroup in logarithmic derivative norms)**.** *The operator $\mathcal{L}_\pi$ in (A.42) generates on $\mathcal{X}^m$ a strongly continuous semigroup $S(t)$. There exist $C > 0$ and an exponential decay rate $\omega_0 > 0$ such that*

$$\|S(t)\|_{\mathcal{X}^m\to\mathcal{X}^m} \le Ce^{-\omega_0 t}, \qquad \|S(t)\|_{\mathcal{X}^{m-1}\to\mathcal{X}^m} \le C(1 + t^{-1/2})e^{-\omega_0 t}. \tag{A.46}$$

The constants can be chosen uniformly for sufficiently small heat cutoffs.

Proof. **Construction and local smoothing.** On affine-plus-compact smooth representatives, the thermal equation (1.28) gives

$$\mathcal{L}_\pi = (\beta^{-1}\Delta - Ax\cdot\nabla) + D, \tag{A.47}$$

with D defined by (A.18). We construct S through the Volterra equation

$$S(t)z = P_t^A z + \int_0^t P_{t-r}^A D S(r) z \, \mathrm{d}r. \tag{A.48}$$

By lemma A.2 and (A.45), the integral operator on $C([0,T];\mathcal{X}^m)$ has norm at most $C(T + \sqrt{T})$. A contraction on a short interval, followed by continuation, gives a unique solution and bounds on every finite interval. Uniqueness gives the semigroup property, and the integral equation gives strong continuity at zero.

For data in $\mathcal{X}^{m-1}$, use continuity in that space together with

$$\sup_{0<t\le T} t^{1/2}\|S(t)z\|_{\mathcal{X}^m} < \infty. \tag{A.49}$$

The only additional convolution is

$$t^{1/2}\int_0^t (t-r)^{-1/2}r^{-1/2}\,\mathrm{d}r = \pi t^{1/2}. \tag{A.50}$$

Thus, the same contraction applies when T is small and gives the local one-derivative smoothing estimate. The integral term is continuous at zero in $\mathcal{X}^{m-1}$ because its norm there is bounded by a constant times $\int_0^t r^{-1/2}\,\mathrm{d}r$.

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Compactness of the semigroup difference. Fix $t > 0$ and $0 < \eta < t$. For the part of (A.48) where $t - r < \eta$, the local bounds give

$$\left\| \int_{t-\eta}^{t} P^A_{t-r} DS(r)\, \mathrm{d}r \right\|_{\mathcal{X}^m \to \mathcal{X}^m} \leq C_t(\eta + \sqrt{\eta}). \tag{A.51}$$

For $0 \leq r \leq t - \eta$, set $M_t := \sup_{0 \leq r \leq t} \|S(r)\|_{\mathcal{X}^m \to \mathcal{X}^m}$. By (A.20),

$$\mathcal{C}_{t,\eta} := \overline{\{P^A_s Dz : \ \eta \leq s \leq t, \ \|z\|_{\mathcal{X}^m} \leq M_t\}} \quad \text{is compact in } \mathcal{X}^m. \tag{A.52}$$

For every z with $\|z\|_{\mathcal{X}^m} \leq 1$,

$$\frac{1}{t-\eta} \int_0^{t-\eta} P^A_{t-r} DS(r) z \, \mathrm{d}r \in \overline{\mathrm{conv}}(\mathcal{C}_{t,\eta}). \tag{A.53}$$

The closed convex hull of a compact subset of a Banach space is compact, so this part of the integral is a compact operator. Taking $\eta \downarrow 0$ in (A.51), and using that compact operators are norm closed, gives

$$S(t) - P^A_t : \mathcal{X}^m \longrightarrow \mathcal{X}^m \quad \text{compact for every } t > 0. \tag{A.54}$$

Identification with the Hilbert-space semigroup. On $L^2_0(\pi)$, set $B := \mathrm{Id} + \beta T_\pi$. This operator is bounded and self-adjoint, and $B \geq \mathrm{Id}$ by positivity of the centered interaction. By the representation theorem for densely defined closed nonnegative forms (see, e.g., [Kat95, Chapter VI, Section 2]), there is a unique nonnegative self-adjoint operator H associated with the quadratic form

$$a \longmapsto \|\nabla(B^{1/2}a)\|^2_{L^2(\pi)}, \qquad \{a \in L^2_0(\pi) : B^{1/2}a \in H^1(\pi)\}. \tag{A.55}$$

The form is closed and densely defined because $B^{1/2}$ is bounded and invertible. The Poincaré inequality gives

$$\|\nabla(B^{1/2}a)\|^2_{L^2(\pi)} \geq C_P^{-1} \|B^{1/2}a\|^2_{L^2(\pi)} \geq C_P^{-1} \|a\|^2_{L^2(\pi)}. \tag{A.56}$$

Consequently, the semigroup realizing $-\beta^{-1}\mathcal{A}_\pi B$ is

$$S_2(t) = B^{-1/2} e^{-tH/\beta} B^{1/2}, \qquad \|S_2(t)\|_{L^2_0(\pi) \to L^2_0(\pi)} \leq \sqrt{\|B\|}\, e^{-t/(\beta C_P)}. \tag{A.57}$$

We check that this is the same evolution as (A.48) after choosing centered representatives. Schur's test, applied as in (5.3), gives

$$\|\nabla T_\pi a\|_{L^2(\pi)} = \|\nabla \mathsf{g} * (\pi a)\|_{L^2(\pi)} \leq C \|a\|_{L^2(\pi)}. \tag{A.58}$$

Thus, B maps $H^1(\pi)$ continuously to itself. Its inverse does also: for $a = B^{-1}f$, the identity $a = f - \beta T_\pi a$ and (A.58) control ∇a by $\|f\|_{H^1(\pi)}$. The Hilbert-space evolution satisfies

$$\langle \partial_t a, \psi \rangle_{H^1(\pi)^*, H^1(\pi)} = -\beta^{-1} \int \nabla(Ba) \cdot \nabla \psi \, \mathrm{d}\pi \qquad (\psi \in H^1(\pi)). \tag{A.59}$$

For a difference of two weak solutions in $L^2_{\mathrm{loc}}(H^1(\pi))$ with time derivative in $L^2_{\mathrm{loc}}(H^1(\pi)^*)$, testing by Ba yields

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle a, Ba \rangle_{L^2(\pi)} = -2\beta^{-1} \|\nabla(Ba)\|^2_{L^2(\pi)}. \tag{A.60}$$

Time mollification justifies the identity, since B is self-adjoint and preserves the form domain. It proves uniqueness for zero initial difference.

For affine-plus-compact smooth initial data, the centered representative of (A.48) satisfies (A.59). Writing $a_t := S(t)z - \pi(S(t)z)$ for this representative, (A.16) and (A.58) give, for every $T < \infty$,

$$a \in C([0,T]; L^2_0(\pi)) \cap L^2(0,T; H^1(\pi)), \qquad \partial_t a \in L^2(0,T; H^1(\pi)^*). \tag{A.61}$$

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Thus, the centered solution lies in the class for which weak uniqueness was just proved. The spatial cutoff and operator-core argument of lemma 4.2 justify the weak equation and the energy test. Uniqueness, and then density in $\mathcal{X}^m$, give

$$S(t)z - \pi(S(t)z) = S_2(t)(z - \pi(z)). \tag{A.62}$$

The expression on the left is independent of the representative chosen for the quotient element $S(t)z$.

Exponential decay in $\mathcal{X}^m$. Choose t_0 so that $\|P^A_{t_0}\|_{\mathcal{X}^m\to\mathcal{X}^m} < 1/4$. By (A.54), $S(t_0)$ is a compact perturbation of $P^A_{t_0}$. For $|\zeta| > 1/4$,

$$\zeta\mathrm{Id} - S(t_0) = (\zeta\mathrm{Id} - P^A_{t_0})\left[\mathrm{Id} - (\zeta\mathrm{Id} - P^A_{t_0})^{-1}(S(t_0) - P^A_{t_0})\right]. \tag{A.63}$$

Here, the spectral argument is on the complexification $\mathcal{X}^m_{\mathbb{C}}$ of the real Banach space $\mathcal{X}^m$, with $P^A_{t_0}$ and $S(t_0)$ extended complex linearly. The Fredholm alternative shows that every spectral value $\zeta \in \sigma(S(t_0)|_{\mathcal{X}^m_{\mathbb{C}}})$ with $|\zeta| > 1/4$ is an eigenvalue of $S(t_0)$ on this space; see, e.g., [EN00, Chapters IV–V].

Let $z \neq 0$ be a corresponding eigenvector in $\mathcal{X}^m_{\mathbb{C}}$. Its centered representative $a = z - \pi(z)$ is nonzero in $L^2_0(\pi;\mathbb{C})$. By (A.62) and the semigroup property,

$$|\zeta|^n\|a\|_{L^2(\pi)} = \|S_2(nt_0)a\|_{L^2(\pi)} \le \sqrt{\|B\|}\, e^{-nt_0/(\beta C_P)}\|a\|_{L^2(\pi)}. \tag{A.64}$$

Taking nth roots gives

$$|\zeta| \le e^{-t_0/(\beta C_P)} < 1. \tag{A.65}$$

All other spectral values lie in the disk of radius $1/4$. Hence $S(t_0)$ has spectral radius strictly below one. The spectral-radius formula gives $\|S(nt_0)\| \le Cq^n$ for some $q < 1$. Boundedness for $0 \le t \le t_0$ then proves the first estimate in (A.46). For $t \ge 1$, the semigroup property and the local smoothing estimate give

$$\|S(t)\|_{\mathcal{X}^{m-1}\to\mathcal{X}^m} \le \|S(t-1)\|_{\mathcal{X}^m\to\mathcal{X}^m}\|S(1)\|_{\mathcal{X}^{m-1}\to\mathcal{X}^m} \le Ce^{-\omega_0 t}. \tag{A.66}$$

Together with the local estimate for $0 < t \le 1$, this proves the second bound in (A.46).

Estimates uniform in the heat cutoff. The convergence in (A.40), the bounds on S^ε on $[0,T]$, independent of ε, and (A.48) imply, for every $T < \infty$,

$$\sup_{0\le t\le T}\|S^\varepsilon(t) - S(t)\|_{\mathcal{X}^m\to\mathcal{X}^m} \le C_T\|D^\varepsilon - D\|_{\mathcal{X}^m\to\mathcal{X}^{m-1}} \xrightarrow[\varepsilon\downarrow 0]{} 0. \tag{A.67}$$

Choose t_1 with $\|S(t_1)\| < 1/4$. For sufficiently small cutoffs,

$$\|S^\varepsilon(t_1)\|_{\mathcal{X}^m\to\mathcal{X}^m} < 1/2. \tag{A.68}$$

Iteration and the bounds on $[0,t_1]$, uniform for $0 \le \varepsilon \le \varepsilon_*$, give an exponential decay rate independent of ε. The local smoothing constants are also independent of ε. All these operators act on the same quotient space $\mathcal{X}^m$; the centering used in (A.62) does not change that space when π is replaced by π^ε. □

A.4. **The nonlinear equation and its stability.**

Proof of proposition 6.1. **The normalized logarithmic equation.** For $\mu_z = \pi e^z / \int \pi e^z$, expansion of (1.4) gives, modulo spatial constants,

$$\partial_t z = \mathcal{L}_\pi z + \mathcal{N}(z). \tag{A.69}$$

To verify that normalization does not alter the density equation, first observe that the definitions imply

$$\mathcal{L}_\pi z + \mathcal{N}(z) = \mu_z^{-1}\left[\beta^{-1}\Delta\mu_z + \mathrm{div}(\mu_z(\nabla V + \nabla \mathsf{g} * \mu_z))\right], \qquad \int (\mathcal{L}_\pi z + \mathcal{N}(z))\,\mathrm{d}\mu_z = 0. \tag{A.70}$$

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The integral identity follows by integration by parts using the Gaussian decay. Choose the representative with $\pi(z) = 0$. Its evolution and the derivative of its normalizer are

$$\partial_t z = \mathcal{L}_\pi z + \mathcal{N}(z) - \pi(\mathcal{L}_\pi z + \mathcal{N}(z)), \tag{A.71}$$

$$\partial_t \log \int e^z \, \mathrm{d}\pi = \int \partial_t z \, \mathrm{d}\mu_z = -\pi(\mathcal{L}_\pi z + \mathcal{N}(z)). \tag{A.72}$$

Subtracting the second equality from the first yields the original equation for $\partial_t \log \mu_z$. Thus, solving the quotient equation (A.69) produces the normalized mean-field solution.

Small-data stability and local existence. The integral equation in $\mathcal{X}^m$ is

$$z_t = S(t)z_0 + \int_0^t S(t-r)\mathcal{N}(z_r)\, \mathrm{d}r. \tag{A.73}$$

Fix $0 < \omega < \omega_0$. By lemmas A.2 and A.3,

$$\sup_{t\geq 0} e^{\omega t} \left\| \int_0^t S(t-r)\mathcal{N}(z_r)\, \mathrm{d}r \right\|_{\mathcal{X}^m} \leq C \left(\sup_{r\geq 0} e^{\omega r} \|z_r\|_{\mathcal{X}^m} \right)^2 \int_0^\infty (1 + r^{-1/2}) e^{-(\omega_0 - \omega)r} \, \mathrm{d}r. \tag{A.74}$$

The integral satisfies

$$\int_0^\infty (1 + r^{-1/2}) e^{-(\omega_0-\omega)r} \, \mathrm{d}r < \infty. \tag{A.75}$$

Applying the difference estimate in (A.22) in the same way shows that (A.73) is a contraction on a ball of radius $2C\|z_0\|_{\mathcal{X}^m}$ in the exponentially weighted trajectory norm, when the initial norm is sufficiently small. This proves global existence, uniqueness in that class, and (6.2). On any fixed bounded initial ball, the local Lipschitz estimate instead gives a contraction on a positive interval because $\int_0^T (1 + r^{-1/2})\, \mathrm{d}r = T + 2\sqrt{T}$ tends to zero. This proves local existence for the data in part (iii).

Higher regularity. For each integer $\ell > m$, the same product and convolution estimates give, on a bounded $\mathcal{X}^m$ ball,

$$\|\mathcal{N}(z)\|_{\mathcal{X}^{\ell-1}} \leq C_\ell \|z\|_{\mathcal{X}^m} \|z\|_{\mathcal{X}^\ell}. \tag{A.76}$$

Indeed, for $1 \leq j \leq \ell$, bounded-derivative interpolation gives

$$\|\nabla^j z\|_{L^\infty} \leq C_\ell \|\nabla z\|_{L^\infty}^{1-(j-1)/(\ell-1)} \|\nabla^\ell z\|_{L^\infty}^{(j-1)/(\ell-1)}. \tag{A.77}$$

Apply this inequality to the products in Leibniz' rule and to the differentiated Taylor remainder (A.29). On a fixed $\mathcal{X}^m$ ball, one obtains

$$\||\nabla z|^2\|_{\mathcal{X}^{\ell-1}} \leq C_\ell \|z\|_{\mathcal{X}^m} \|z\|_{\mathcal{X}^\ell}, \tag{A.78}$$

$$\|[\nabla \mathsf{g} * (\mu_z - \pi)] \cdot \nabla z\|_{\mathcal{X}^{\ell-1}} \leq C_\ell \|z\|_{\mathcal{X}^m} \|z\|_{\mathcal{X}^\ell}, \tag{A.79}$$

$$\sum_{j=0}^{\ell} \left(\|(1+|x|)\nabla^j(\mu_z - \pi - \pi z)\|_{L^1} + \|(1+|x|)\nabla^j(\mu_z - \pi - \pi z)\|_{L^\infty} \right) \leq C_\ell \|z\|_{\mathcal{X}^m} \|z\|_{\mathcal{X}^\ell}. \tag{A.80}$$

For the last estimate, every differentiated Taylor term contains at least two factors of z or its derivatives; (A.77) places at most one factor in the highest norm, and (A.16) and the Gaussian bounds control undifferentiated factors. Applying (A.25) with $m = \ell$ proves the same bound for $\mathcal{R}_\pi(\mu_z - \pi - \pi z)$. The three terms in (A.21) therefore give (A.76).

The one-derivative smoothing estimate for S at order ℓ follows from (A.48)–(A.50) with m replaced by ℓ. On an interval of length h, the high-norm contribution of the integral in (A.73) is at most

$$C_\ell(h + \sqrt{h}) \sup_{t\leq r\leq t+h} \|z_r\|_{\mathcal{X}^m} \sup_{t\leq r\leq t+h} \|z_r\|_{\mathcal{X}^\ell}. \tag{A.81}$$

Choose h so that the coefficient of the last supremum is below $1/2$, and continue on successive intervals. Every finite higher norm remains finite on compact time intervals. The differentiated

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equation then gives time regularity. Hence the smooth initial data in the proposition remain smooth, while the low-norm constants in (6.2) are independent of their higher derivatives.

Convergence of the cutoff solutions. By lemmas A.2 and A.3, the solutions lie in a common bounded $\mathcal{X}^m$ ball on each fixed finite interval of existence. To compare them, use the same Ornstein–Uhlenbeck semigroup for every cutoff. The decomposition (A.47) gives

$$z_t^\varepsilon = P_t^A z_0 + \int_0^t P_{t-r}^A (D^\varepsilon z_r^\varepsilon + \mathcal{N}^\varepsilon(z_r^\varepsilon))\,\mathrm{d}r. \tag{A.82}$$

This identity also holds for $\varepsilon = 0$. Fix a finite interval $[0,T]$ and a radius R containing both trajectories. On $[0,h]$, with $0 < h \le \min\{1,T\}$, subtract the two identities and use (A.45) and the uniform local Lipschitz bounds. We obtain

$$\begin{aligned}\sup_{0\le t\le h} \|z_t^\varepsilon - z_t\|_{\mathcal{X}^m} &\le C_R(h+\sqrt{h}) \sup_{0\le t\le h} \|z_t^\varepsilon - z_t\|_{\mathcal{X}^m} \\ &\quad + C_R(h+\sqrt{h}) \left(\|D^\varepsilon - D\|_{\mathcal{X}^m \to \mathcal{X}^{m-1}} + \sup_{\|w\|_{\mathcal{X}^m}\le R} \|\mathcal{N}^\varepsilon(w) - \mathcal{N}(w)\|_{\mathcal{X}^{m-1}} \right).\end{aligned} \tag{A.83}$$

Choose h, independently of ε, so that $C_R(h+\sqrt{h}) < 1/2$ and absorb the first term. The remaining terms tend to zero by (A.40) and (A.41). On the next interval of length at most h, the same estimate has the additional initial term $C\|z_h^\varepsilon - z_h\|_{\mathcal{X}^m}$, which already tends to zero. Iterating over finitely many intervals proves

$$\sup_{0\le t\le T} \|z_t^\varepsilon - z_t\|_{\mathcal{X}^m} \underset{\varepsilon\downarrow 0}{\longrightarrow} 0. \tag{A.84}$$

This argument applies both to the small global solutions and to the larger local solutions. The higher-regularity bounds above are uniform in the cutoff, so repeating the comparison at each finite higher order gives convergence in that order for smooth data. Representatives may be centered with respect to the fixed measure π, since the equation and norm are defined modulo constants.

Density estimates and convergence. For the representative $\pi(z_t) = 0$, Jensen's inequality and (A.16) give

$$\int e^{z_t}\,\mathrm{d}\pi \ge 1, \qquad \mu_t(x) \le M \exp\left[-a|x|^2 + \|\nabla z_t\|_{L^\infty} \left(|x| + \int |y|\,\mathrm{d}\pi(y) \right) \right]. \tag{A.85}$$

If $\sup_{t\in I}\|z_t\|_{\mathcal{X}^m} < \infty$, with $I = [0,\infty)$ for small data and $I = [0,T_0]$ for the local solution, (A.85) gives a Gaussian upper bound with constants independent of $t \in I$. The same constants may be chosen for μ_t^ε when the $\mathcal{X}^m$ bound is uniform in $\varepsilon \in [0,\varepsilon_*]$. Differentiating the normalized exponential gives the corresponding density-derivative bounds. In particular, (A.3) and (A.84) imply

$$\sup_{0\le t\le T} \|\mu_t^\varepsilon - \mu_t\|_{L^1(\mathbb{R}^d)} \underset{\varepsilon\downarrow 0}{\longrightarrow} 0, \qquad [0,T] \subset I. \tag{A.86}$$

For smooth initial data, the convergence also holds for every spatial derivative uniformly on $[0,T]\times B_R$, for each $R < \infty$.

The logarithmic Hessian and Poincaré inequality. The logarithmic Hessian is

$$\nabla^2(-\log \mu_t) = \nabla^2 U_\pi - \nabla^2 z_t. \tag{A.87}$$

For the small global solution, decrease δ_0 so that $\sup_{t\ge 0}\|\nabla^2 z_t\|_{L^\infty}$ is less than one quarter of $\beta\lambda_A$. Then (A.2) and (A.87) imply uniform convexity of $-\log\mu_t$ outside a fixed ball, independently of $t \ge 0$. Applying the bounded modification and Poincaré perturbation argument of lemma A.1 gives a Poincaré constant independent of time.

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For a larger local solution, continuity of $t \mapsto z_t$ in $\mathcal{X}^m$ and (6.1) give

$$\lim_{R\to\infty} \sup_{0\le t\le T_0} \sup_{|x|\ge R} |\nabla^2 z_t(x)| = 0. \tag{A.88}$$

Indeed, the image of $[0,T_0]$ is compact in $\mathcal{X}^m$, so a finite cover reduces this assertion to (6.1). Hence $-\log \mu_t$ is uniformly convex outside a ball with radius independent of $t \in [0,T_0]$. By (A.84) and the thermal Hessian bounds, the same radius, enlarged if necessary, and convexity constant work for μ_t^ε, $0 \le t \le T_0$, $0 \le \varepsilon \le \varepsilon_*$. The Poincaré constant is therefore independent of both time and cutoff on this interval.

Drift and continuity velocity. The first two drift derivatives can be estimated without differentiating the singular force:

$$\nabla^j B_t = -\nabla^{j+1} V - \nabla \mathsf{g} * \nabla^j \mu_t \qquad (j = 1, 2). \tag{A.89}$$

The confinement derivatives are bounded, and (A.11) applies to the Gaussian-decaying density derivatives. This proves (6.6).

Finally, the stationary thermal equation cancels the confinement in the continuity velocity:

$$u_t = -\beta^{-1} \nabla z_t - \nabla \mathsf{g} * (\mu_t - \pi). \tag{A.90}$$

The first-order version of (A.30), together with (A.11), yields

$$\|u_t\|_{W^{1,\infty}} \le C \|z_t\|_{\mathcal{X}^m}. \tag{A.91}$$

This proves (6.3). The same estimates with constants independent of $\varepsilon \in [0, \varepsilon_*]$, the force convergence from lemma 2.3, and (A.84) give

$$\sup_{0\le t\le T} \|u_t^\varepsilon - u_t\|_{W^{1,\infty}} \underset{\varepsilon\downarrow 0}{\longrightarrow} 0, \qquad [0,T] \subset I. \tag{A.92}$$

Together with (A.84) and (A.86), this proves (6.8) and completes the proof of proposition 6.1. □

Appendix B. Finite-particle well-posedness and noncollision

This appendix constructs the singular particle law used in lemma 6.3. The localization and noncollision arguments are standard (cf. [RS23, §4] for singular Riesz-type particle systems and [FJ15, Lemma 15] for the cluster-dispersion argument in an attractive planar system). We include the details for the interactions and initial configurations considered here. For strong solutions of infinite systems with singular interactions, see also [OT20].

We first construct a maximal strong solution by truncating the force, then exclude collapse of a cluster by descending induction on its cardinality. Here, the quadratic dispersion has constant Hessian, so no term involving $\Delta \mathsf{g}$ occurs, while the repulsive internal force has a nonnegative contribution.

Proposition B.1 (Global strong well-posedness and noncollision)**.** *Let $N \ge 2$, $\mathsf{d} \ge 2$, $0 \le \mathsf{s} < \mathsf{d}/2$, and $\beta > 0$. Suppose $V \in C^2_{\mathrm{loc}}(\mathbb{R}^{\mathsf{d}})$ and $\|\nabla^2 V\|_{L^\infty} < \infty$. If the initial configuration $X_N(0)$ is almost surely pairwise distinct and independent of the Brownian motions $(B_i)_{i\in[N]}$, then (1.3) admits a pathwise unique global strong solution. Moreover, for every $T < \infty$,*

$$\mathbb{P}\left(\min_{0\le t\le T} \min_{1\le i<j\le N} |x_i(t) - x_j(t)| > 0 \right) = 1. \tag{B.1}$$

We first take the initial configuration $X_N(0)$ to be deterministic and pairwise distinct. Set

$$F := -\nabla \mathsf{g}, \qquad \mathfrak{D}_N := \{X_N \in (\mathbb{R}^{\mathsf{d}})^N : x_i = x_j \text{ for some } i \ne j \text{ in } [N]\}. \tag{B.2}$$

The particle generator on the collision complement $(\mathbb{R}^{\mathsf{d}})^N \setminus \mathfrak{D}_N$ is

$$\mathcal{L}_N := \frac{1}{\beta} \sum_{i=1}^N \Delta_{x_i} + \sum_{i=1}^N \left(\frac{1}{N} \sum_{j\ne i} F(x_i - x_j) - \nabla V(x_i) \right) \cdot \nabla_{x_i}. \tag{B.3}$$

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Lemma B.2 (Maximal strong solution)**.** *There are a stopping time $\tau_\star \in (0,\infty]$ and a pathwise unique strong solution of (1.3) on $[0,\tau_\star)$ such that, on $\{\tau_\star < \infty\}$,*

$$\liminf_{t\uparrow\tau_\star} \min_{1\le i<j\le N} |x_i(t) - x_j(t)| = 0. \tag{B.4}$$

Proof. Choose smooth odd, globally Lipschitz vector fields F_ε on $\mathbb{R}^{\mathsf{d}}$ satisfying $F_\varepsilon = F$ on $\{|x| \ge \varepsilon\}$. These force truncations are used only in this construction; they need not be the heat truncations from Equation (2.12). Since ∇V is globally Lipschitz, the regularized system has a pathwise unique global strong solution (see, e.g., [Øks03, §5.2]). Set $\varepsilon_n := 2^{-n}$ and let $X_N^{(n)}$ be the corresponding solution, with the same initial configuration and Brownian motions for every n. For all sufficiently large n, the initial pair distances exceed ε_n. For $m \ge n$, set

$$\tau_{m,n} := \inf\{t \ge 0 : \min_{i<j} |x_i^{(m)}(t) - x_j^{(m)}(t)| \le \varepsilon_n\}, \qquad \tau_n := \tau_{n,n}. \tag{B.5}$$

For $m > n$, both processes solve the same locally Lipschitz equation on $(\mathbb{R}^{\mathsf{d}})^N \setminus \mathfrak{D}_N$ up to $\tau_n \wedge \tau_{m,n}$. Stopped pathwise uniqueness and continuity therefore give

$$\tau_{m,n} = \tau_n, \qquad X_N^{(m)} = X_N^{(n)} \text{ on } [0,\tau_n], \qquad \tau_m \ge \tau_n. \tag{B.6}$$

By (B.6), the process defined by $X_N(t) := X_N^{(n)}(t)$ for $t < \tau_n$ is independent of the choice of n and solves (1.3) on $[0,\tau_\star)$, where $\tau_\star := \lim_n \tau_n$. The drift is locally Lipschitz on $(\mathbb{R}^{\mathsf{d}})^N \setminus \mathfrak{D}_N$, so the same localization proves pathwise uniqueness there. If $\tau_\star < \infty$, then at times $\tau_n \uparrow \tau_\star$ some pair has distance ε_n; after passing to a subsequence, one fixed pair does so. This proves (B.4). □

For $I \subset [N]$ with cardinality $k \ge 2$, define

$$\overline{x}_I := \frac{1}{k}\sum_{i\in I} x_i, \qquad Q_I(X_N) := \sum_{i\in I} |x_i - \overline{x}_I|^2 = \frac{1}{k}\sum_{\substack{i,j\in I\\ i<j}} |x_i - x_j|^2. \tag{B.7}$$

For $j \notin I$, one has

$$Q_{I\cup\{j\}} = Q_I + \frac{k}{k+1}|x_j - \overline{x}_I|^2. \tag{B.8}$$

Lemma B.3 (Cluster dispersion equation)**.** *Before $\tau_\star$,*

$$\mathrm{d}Q_I = \left(\frac{2\mathsf{d}(k-1)}{\beta} + \mathcal{A}_I + \mathcal{R}_I\right)\mathrm{d}t + \mathrm{d}\mathcal{M}_I, \qquad \mathrm{d}\langle \mathcal{M}_I\rangle_t = \frac{8}{\beta} Q_I\,\mathrm{d}t, \tag{B.9}$$

where

$$\mathcal{A}_I = -\frac{2}{N}\sum_{\substack{i,j\in I\\ i<j}} (x_i - x_j)\cdot \nabla \mathsf{g}(x_i - x_j) \ge 0. \tag{B.10}$$

For every $\delta > 0$, there are $q_{I,\delta} > 0$ and $C_{I,\delta} < \infty$ such that

$$\mathcal{R}_I \ge -C_{I,\delta} Q_I \tag{B.11}$$

whenever $Q_I < q_{I,\delta}$ and

$$Q_{I\cup\{j\}} \ge \delta \qquad \text{for every } j \notin I. \tag{B.12}$$

For the full cluster, (B.12) is vacuous and the constants do not depend on δ.

Proof. The elementary identities

$$\nabla_{x_i} Q_I = 2(x_i - \overline{x}_I), \qquad \sum_{i\in I} \Delta_{x_i} Q_I = 2\mathsf{d}(k-1), \qquad \sum_{i\in I} |\nabla_{x_i} Q_I|^2 = 4Q_I \tag{B.13}$$

give the diffusion drift in (B.9) by Itô's formula. The martingale is

$$\mathcal{M}_I(t) := 2\sqrt{\frac{2}{\beta}}\sum_{i\in I}\int_0^t (x_i(r) - \overline{x}_I(r))\cdot \mathrm{d}B_i(r), \qquad t < \tau_\star. \tag{B.14}$$

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Independence of the Brownian motions and (B.13) give its quadratic variation. Pairing the two orientations of each internal interaction gives (B.10). For $\mathsf{s} > 0$,

$$z \cdot \nabla \mathsf{g}(z) = -|z|^{-\mathsf{s}}, \tag{B.15}$$

whereas for $\mathsf{s} = 0$ the same quantity equals -1. Thus, $\mathcal{A}_I \geq 0$.

For an exterior index $j \notin I$, subtract the common value $F(\overline{x}_I - x_j)$ and use $\sum_{i\in I}(x_i - \overline{x}_I) = 0$. Its contribution is

$$\frac{2}{N}\sum_{i\in I}(x_i - \overline{x}_I) \cdot (F(x_i - x_j) - F(\overline{x}_I - x_j)). \tag{B.16}$$

If (B.12) holds and $Q_I \leq \delta/16$, then (B.8) gives $|\overline{x}_I - x_j| \geq \sqrt{\delta - Q_I}$. Thus, the entire segment used to compare the two force values stays away from the origin:

$$|\overline{x}_I - x_j + \theta(x_i - \overline{x}_I)| \geq \sqrt{\delta - Q_I} - \sqrt{Q_I} \geq \frac{\sqrt{\delta}}{2}, \qquad i \in I, \quad 0 \leq \theta \leq 1. \tag{B.17}$$

Consequently,

$$\left|\frac{2}{N}\sum_{i\in I}(x_i - \overline{x}_I) \cdot (F(x_i - x_j) - F(\overline{x}_I - x_j))\right| \leq \frac{2}{N}\sup_{|z|\geq\sqrt{\delta}/2}|\nabla F(z)|\, Q_I. \tag{B.18}$$

Subtracting $\nabla V(\overline{x}_I)$ in the confinement contribution similarly gives

$$-2\sum_{i\in I}(x_i - \overline{x}_I) \cdot (\nabla V(x_i) - \nabla V(\overline{x}_I)) \geq -2\|\nabla^2 V\|_{L^\infty} Q_I. \tag{B.19}$$

Summing the exterior and confinement contributions proves (B.11). □

Lemma B.4 (Noncollapse of a cluster)**.** *Let I have cardinality $k \geq 2$ and suppose $\mathsf{d}(k-1) \geq 2$. Assume that every one-point strict enlargement $J \supsetneq I$ satisfies*

$$\mathbb{P}\left(\inf_{0\leq t<T\wedge\tau_\star} Q_J(X_N(t)) = 0\right) = 0 \qquad \text{for every } T < \infty. \tag{B.20}$$

Then the same assertion holds for I.

Proof. Fix $\delta > 0$ and stop at

$$\eta_{I,\delta} := \inf\left\{t < \tau_\star : \min_{j\notin I} Q_{I\cup\{j\}}(X_N(t)) \leq \delta\right\}, \tag{B.21}$$

with $\eta_{I,\delta} = \infty$ for the full cluster. Choose $0 < q_0 < \min\{q_{I,\delta}, 1\}$ as in lemma B.3. Let $\Phi \in C^2_{\mathrm{loc}}((0,\infty))$ be nonnegative and nonincreasing, equal to $-\log q$ on $(0, q_0/4]$, and constant on $[q_0/2, \infty)$. On $\{Q_I \leq q_0/4\}$, Itô's formula and lemma B.3 give

$$\mathcal{L}_N\Phi(Q_I) = \frac{4 - 2\mathsf{d}(k-1)}{\beta Q_I} - \frac{\mathcal{A}_I + \mathcal{R}_I}{Q_I} \leq \frac{4 - 2\mathsf{d}(k-1)}{\beta Q_I} + C_{I,\delta} \leq C_{I,\delta}. \tag{B.22}$$

The coefficient of Q_I^{-1} is nonpositive also when $\mathsf{d}(k-1) = 2$. In the transition region $q_0/4 \leq Q_I \leq q_0/2$, the inequality $\Phi' \leq 0$ and (B.11) give $\Phi'\mathcal{R}_I \leq C_{I,\delta}Q_I|\Phi'|$, while $\Phi'\mathcal{A}_I \leq 0$ and $Q_I|\Phi''|$ is bounded. Hence (B.22), with a larger constant, holds before $\eta_{I,\delta}$.

For $0 < a < \min\{Q_I(X_N(0)), q_0/4, e^{-1}\}$, set

$$\sigma_{I,a} := \inf\{t < \tau_\star : Q_I(X_N(t)) \leq a\}. \tag{B.23}$$

Apply Itô's formula up to $T \wedge \eta_{I,\delta} \wedge \sigma_{I,a} \wedge \tau_n$. The martingale in this formula is square integrable, since

$$\frac{8}{\beta}\int_0^{T\wedge\eta_{I,\delta}\wedge\sigma_{I,a}\wedge\tau_n} Q_I(X_N(r))\left|\Phi'(Q_I(X_N(r)))\right|^2 \mathrm{d}r \leq \frac{8T}{\beta}\sup_{q\geq a} q|\Phi'(q)|^2 < \infty. \tag{B.24}$$

Taking expectations and using (B.22) gives

$$\mathbb{E}[\Phi(Q_I(X_N(T \wedge \eta_{I,\delta} \wedge \sigma_{I,a} \wedge \tau_n)))] \leq \Phi(Q_I(X_N(0))) + C_{I,\delta}T. \tag{B.25}$$

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On $\{\sigma_{I,a} \le T \wedge \eta_{I,\delta} \wedge \tau_n\}$, continuity gives $Q_I(X_N(\sigma_{I,a})) = a$. Since $\Phi \ge 0$, (B.25) implies

$$\mathbb{P}(\sigma_{I,a} \le T \wedge \eta_{I,\delta} \wedge \tau_n) \le \frac{\Phi(Q_I(X_N(0))) + C_{I,\delta}T}{|\log a|}. \tag{B.26}$$

The events on the left increase to $\{\sigma_{I,a} \le T \wedge \eta_{I,\delta},\ \sigma_{I,a} < \tau_\star\}$. Therefore,

$$\mathbb{P}(\sigma_{I,a} \le T \wedge \eta_{I,\delta},\ \sigma_{I,a} < \tau_\star) \le \frac{\Phi(Q_I(X_N(0))) + C_{I,\delta}T}{|\log a|}. \tag{B.27}$$

Letting $a \downarrow 0$ excludes approach of Q_I to zero on $[0, T \wedge \tau_\star)$ unless $\eta_{I,\delta} < T \wedge \tau_\star$. By the induction hypothesis,

$$\mathbb{P}(\eta_{I,\delta} < T \wedge \tau_\star) \le \sum_{j \notin I} \mathbb{P}\Big(\inf_{0 \le t < T \wedge \tau_\star} Q_{I \cup \{j\}}(X_N(t)) \le \delta \Big) \xrightarrow[\delta \downarrow 0]{} 0. \tag{B.28}$$

Letting $\delta \downarrow 0$ completes the proof. □

Proof of proposition B.1. For the full cluster, lemma B.4 applies without an induction hypothesis. Descend in the cardinality of I. Since $\mathsf{d} \ge 2$, every cluster with $|I| \ge 2$ satisfies $\mathsf{d}(|I| - 1) \ge 2$, and therefore

$$\mathbb{P}\left(\inf_{0 \le t < T \wedge \tau_\star} Q_I(X_N(t)) = 0 \right) = 0 \qquad \text{for every } T < \infty. \tag{B.29}$$

If $\tau_\star$ were finite, (B.4) and the finiteness of the set of pairs would contradict (B.29) for a two-particle cluster. Hence $\tau_\star = \infty$ almost surely.

Since $\tau_n \uparrow \infty$ almost surely, for every finite T and almost every sample there is an n such that $T < \tau_n$. By the construction in lemma B.2, $X_N(t) = X_N^{(n)}(t)$ for all $t \in [0, T]$. The regularized solution is global, so X_N cannot explode in space. Continuity and the absence of collisions then give (B.1). Local pathwise uniqueness on $(\mathbb{R}^{\mathsf{d}})^N \setminus \mathfrak{D}_N$, as in lemma B.2, gives global pathwise uniqueness.

For a random pairwise-distinct initial configuration $X_N(0)$ independent of the Brownian motions $(B_i)_{i \in [N]}$, apply the deterministic construction conditionally on $X_N(0)$. Equivalently, use the measurable solution maps of the regularized equations and their pathwise stabilization in lemma B.2. This gives the asserted global strong solution and completes the proof. □

Appendix C. Proof of the iid counterexample

We prove proposition 1.6 in five steps. After constructing the stationary measure and its confinement, we establish entropy and fixed-particle density bounds for the singular dynamics. These bounds justify an adjoint fluctuation formula from independent initial data. We then estimate the amplification of a particular test and compare the resulting Gaussian variance with its product-reference variance. The final subsection deduces the chi-square divergence and the marginal entropy bound.

C.1. **The stationary measure and confinement.** Throughout this appendix, $\mathsf{d} = 4$ and $0 \le \mathsf{s} < 1$, and g is the whole-space kernel (1.1). Let $\gamma_a^{(m)}$ denote the centered Gaussian density on $\mathbb{R}^m$ with covariance $a^2\mathrm{Id}$, and write $x = (r, z) \in \mathbb{R} \times \mathbb{R}^3$. For $0 < \delta < 1/2$, set

$$\eta_\delta := e^{-\delta^{-2}}, \qquad \beta_\delta^{-1} := \delta^6, \qquad T_\delta := \sqrt{\delta}, \tag{C.1}$$

$$\mu_\delta(r, z) := \frac{1 - \eta_\delta}{2} \sum_{\sigma = \pm 1} \gamma_\delta^{(1)}(r - \sigma)\gamma_1^{(3)}(z) + \eta_\delta \gamma_2^{(4)}(r, z). \tag{C.2}$$

The first two components of μ_δ concentrate near the parallel hyperplanes $r = \pm 1$. The third component, although negligible as $\delta \downarrow 0$, gives a Gaussian lower bound at every fixed δ and determines the confinement at infinity. As elsewhere, we use the same symbol for a density and its probability measure. Define

$$h_\delta := \mathsf{g} * \mu_\delta, \qquad V_\delta := -\beta_\delta^{-1} \log \mu_\delta - h_\delta. \tag{C.3}$$

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The definition of V_δ makes μ_δ stationary; an additive constant in V_δ has no effect on the dynamics. We first obtain the entropy and fluctuation estimates at fixed δ, then let $\delta \downarrow 0$ to select the counterexample. Constants in this latter limit may depend on s and the fixed test cutoffs, but not on δ. Once δ has been chosen, it remains fixed as $N \to \infty$.

Lemma C.1 (Stationary measure and confinement)**.** *The density in* (C.2) *is smooth, positive, and even in* r*. For every fixed* δ*, it has a Gaussian upper bound, a bounded logarithmic Hessian, a one-body Poincaré inequality, and finite logarithmic or Riesz energy. Moreover,*

$$\|\nabla^2 \log \mu_\delta\|_\infty \le C\delta^{-4}, \tag{C.4}$$

and

$$\sup_{\substack{|r|\le R\\ z\in\mathbb{R}^3}} |\partial_r \log \mu_\delta(r,z)| \le C_R \delta^{-2}, \qquad R>0. \tag{C.5}$$

For $j=1,2$*,*

$$\sup_{0<\delta<1/2} \sup_x \int |\nabla^j \mathsf{g}(x-y)|\, \mu_\delta(\,\mathrm{d}y) < \infty. \tag{C.6}$$

The confinement in (C.3) *satisfies* (1.36)*, and* μ_δ *is its thermal equilibrium measure at inverse temperature* β_δ*. The stationary trajectory satisfies the intrinsic one-body and static hypotheses of theorem 1.1.*

Proof. We write

$$\mu_\delta = \eta_\delta \gamma_2^{(4)}(1+R_+ + R_-), \qquad R_\pm(r,z) := \frac{8(1-\eta_\delta)}{\eta_\delta \delta} \exp\left(-\frac{(r\mp 1)^2}{2\delta^2} + \frac{r^2}{8} - \frac{3|z|^2}{8}\right). \tag{C.7}$$

For fixed δ, each $R_\pm$ and all its derivatives decay rapidly at infinity. Consequently, $\log(1+R_+ + R_-)$ and all its derivatives have the same property. In particular, the logarithmic derivatives of order at least two are bounded.

There are constants $0 < c_\delta \le C_\delta < \infty$ such that $c_\delta \gamma_2^{(4)} \le \mu_\delta \le C_\delta \gamma_2^{(4)}$. This gives the Gaussian upper bound and, by comparison with the Gaussian Poincaré inequality,

$$\mathrm{Var}_{\mu_\delta}(f) \le 4(C_\delta/c_\delta) \int |\nabla f|^2 \,\mathrm{d}\mu_\delta. \tag{C.8}$$

The bounded density and Gaussian upper bound imply every static integrability condition used below, including square integrability of the centered potential and finiteness of the logarithmic energy.

Logarithmic derivative bounds. For the estimate uniform in δ, write w_0, w_+, w_- for the three mixture summands divided by μ_δ, and ℓ_0, ℓ_+, ℓ_- for their logarithms before division. Direct differentiation gives

$$\nabla^2 \log \mu_\delta = \sum_i w_i \nabla^2 \ell_i + \frac{1}{2}\sum_{i,j} w_i w_j (\nabla \ell_i - \nabla \ell_j)\otimes(\nabla \ell_i - \nabla \ell_j). \tag{C.9}$$

The Gaussian Hessians have size at most $C\delta^{-2}$. The difference of the two narrow-component scores is the constant vector $(2\delta^{-2}, 0)$. For a narrow and the broad component, the score difference is $\nabla \log R_\pm$.

Completing the square yields $R_\pm = e^{a_\delta - Q_\pm}$, where $0 < a_\delta \le C\delta^{-2}$ and $Q_\pm$ is a nonnegative quadratic form centered at a fixed point of bounded size, with largest quadratic coefficient at most $C\delta^{-2}$. Therefore,

$$|\nabla \log R_\pm|^2 \le C\delta^{-2} Q_\pm, \qquad w_0 w_\pm \le \min\{1, R_\pm\}. \tag{C.10}$$

The elementary inequality

$$\sup_{u\ge 0} u \min\{1, e^{a-u}\} \le a+1 \qquad (a\ge 0) \tag{C.11}$$

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proves (C.4) from (C.9). The first-coordinate score is a convex combination of $-(r \mp 1)/\delta^2$ and $-r/4$, which proves (C.5).

Uniform convolution bounds. The three transverse variables provide the integrability needed uniformly as the first-coordinate variance vanishes. Indeed, $|\nabla^j \mathsf{g}(x)| \le C_{\mathsf{s}}|x|^{-\mathsf{s}-j}$ for $j = 1, 2$, and $\mathsf{s} + 2 < 3$. Thus, for $j = 1, 2$ and $\sigma \in \{1, 2\}$,

$$\sup_{b,z} \int_{\mathbb{R}^3} (b^2 + |z - y|^2)^{-(\mathsf{s}+j)/2} \gamma_\sigma^{(3)}(y)\, \mathrm{d}y \le \|\gamma_\sigma^{(3)}\|_\infty \int_{|y|<1} |y|^{-\mathsf{s}-j}\, \mathrm{d}y + 1 < \infty. \tag{C.12}$$

Integrating the remaining one-dimensional probability density proves (C.6).

Confinement and stationarity.

For fixed δ, every derivative of μ_δ is a Schwartz function. The force $\nabla \mathsf{g}$ is locally integrable and bounded outside every neighborhood of the origin, with limit zero at infinity. Differentiating its convolution by putting the additional derivatives on μ_δ shows that ∇h_δ is smooth and every derivative is bounded and tends to zero at infinity. By (C.7),

$$\nabla V_\delta(x) = \frac{\beta_\delta^{-1}}{4} x - \beta_\delta^{-1} \nabla \log(1 + R_+(x) + R_-(x)) - \nabla h_\delta(x). \tag{C.13}$$

This proves (1.36); in particular, the second and third derivatives of V_δ are bounded.

The defining identity for the confinement gives

$$B_t = -\nabla V_\delta - \nabla h_\delta = \beta_\delta^{-1} \nabla \log \mu_\delta, \qquad u_t = B_t - \beta_\delta^{-1} \nabla \log \mu_\delta = 0. \tag{C.14}$$

Hence $\beta_\delta^{-1} \Delta \mu_\delta - \mathrm{div}(B_t \mu_\delta) = 0$. The density comparison above, (C.4), and (C.8) supply the one-body bounds. The first two derivatives of B_t are bounded for fixed δ by (C.7). The partition-function bounds and exponential transport-commutator estimates follow from [DGR26a, Theorems 2.2 and 2.8], respectively.

The identity (C.3) is precisely (1.28). Since (C.13) verifies (1.23)–(1.24), lemma A.1 identifies μ_δ as the unique thermal equilibrium measure. Its logarithmic perturbation relative to that equilibrium measure is zero. □

C.2. **Particle laws and entropy.** Fix $\delta \in (0, 1/2)$ throughout this subsection and suppress its subscript on β, V, μ, and h. Set

$$\Gamma_N := \mathbb{Q}_{N,\beta}(\mu). \tag{C.15}$$

By (C.3) and (1.6), Γ_N is the invariant canonical Gibbs measure of (1.3) with confinement V. The particle measure started from $\mu^{\otimes N}$ is generally nonstationary. Since $\nabla^2 V$ is bounded and the initial configuration is almost surely pairwise distinct, proposition B.1 gives the global noncolliding strong solution and the compact-time separation (B.1).

For the entropy and density estimates, use convolution with the heat kernel p_ε of covariance $2\varepsilon \mathrm{Id}$, and set

$$\mathsf{g}_{(\varepsilon)} := \mathsf{g} * p_\varepsilon, \qquad h_{(\varepsilon)} := \mathsf{g}_{(\varepsilon)} * \mu, \qquad V_{(\varepsilon)} := -\beta^{-1} \log \mu - h_{(\varepsilon)}. \tag{C.16}$$

This convolution mollification is distinct from the heat-parameter truncation g_ε in (2.12). The symbols $\mathsf{F}_{N,(\varepsilon)}$, $\mathsf{K}_{N,b,(\varepsilon)}$, and $\mathbb{Q}_{N,b,(\varepsilon)}$ denote (1.6)–(1.7) with g replaced by $\mathsf{g}_{(\varepsilon)}$. Write $\Gamma_{N,(\varepsilon)} := \mathbb{Q}_{N,\beta,(\varepsilon)}(\mu)$ and $P_{N,t}^{(\varepsilon)}$ for the law of the smooth particle system with $V_{(\varepsilon)}$, $\mathsf{g}_{(\varepsilon)}$, and the same iid initial law. The target confinement remains the fixed V; $V_{(\varepsilon)}$ is used only in this auxiliary construction. At every cutoff,

$$-\nabla V_{(\varepsilon)} - \nabla h_{(\varepsilon)} = \beta^{-1} \nabla \log \mu. \tag{C.17}$$

The mollified particle drift is globally Lipschitz at fixed N, ε; see, e.g., [Øks03, Section 5.2]. Since $\nabla \mathsf{g}$ is locally integrable and the derivatives of μ decay at Gaussian rates, $\nabla h_{(\varepsilon)} \to \nabla h$ locally with its first derivatives. The pair force and its first derivatives converge uniformly on compact sets away

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from the diagonal. Coupling the initial configurations identically and using the same Brownian motions, the localization argument (6.39)–(6.40) therefore applies. At each fixed N, T, it gives

$$P^{(\varepsilon)}_{N,\cdot} \longrightarrow P_{N,\cdot} \quad \text{as path laws on } C([0,T];(\mathbb{R}^4)^N), \qquad \varepsilon \downarrow 0. \tag{C.18}$$

Here, the positive minimum pair separation is supplied by proposition B.1. This argument uses neither a Gibbs initial law nor a uniform-in-N path-convergence estimate.

Lemma C.2 (Uniform relative entropy)**.** *For the initial law* (1.38)*,*

$$\sup_{N\geq 2}\sup_{t\geq 0}\Big\{\mathsf{H}(P_{N,t}\mid \Gamma_N)+\mathsf{H}(P_{N,t}\mid \mu^{\otimes N})\Big\}<\infty. \tag{C.19}$$

Proof. We first prove estimates uniform in the mollification parameter ε, then pass to the singular dynamics. The auxiliary diffusion has invariant measure $\Gamma_{N,(\varepsilon)}$, and its initial entropy is

$$\mathsf{H}(\mu^{\otimes N}\mid \Gamma_{N,(\varepsilon)}) = \log \mathsf{K}_{N,\beta,(\varepsilon)}(\mu) - \beta I_{(\varepsilon)}, \qquad I_{(\varepsilon)} := \frac{1}{2}\iint \mathsf{g}_{(\varepsilon)}\,\mathrm{d}\mu^{\otimes 2}. \tag{C.20}$$

Partition bounds uniform in the mollification parameter. Set $\mu_{(\varepsilon)} := \mu * p_{\varepsilon/2}$. Let $y_1,\dots,y_N$ have density $p_{\varepsilon/2}$ independently of one another and of $X_N \sim \mu^{\otimes N}$, and set $Y_N := (y_1,\dots,y_N)$. Conditional expectation gives

$$N\mathsf{F}_{N,(\varepsilon)}(X_N,\mu) = \mathbb{E}_Y\big[N\mathsf{F}_N(X_N+Y_N,\mu_{(\varepsilon)})\mid X_N\big]. \tag{C.21}$$

The pair term uses the difference of two independent heat increments; the one-body term uses one increment and the convolution defining $\mu_{(\varepsilon)}$. Their variances add to 2ε in both cases. Thus

$$\mathsf{K}_{N,b,(\varepsilon)}(\mu) \leq \mathsf{K}_{N,b}(\mu_{(\varepsilon)}), \qquad b>0. \tag{C.22}$$

For $0\leq\varepsilon\leq 1$, the mollified background is the Gaussian mixture

$$\mu_{(\varepsilon)}(r,z) = \frac{1-\eta_\delta}{2}\sum_{\sigma=\pm 1}\gamma^{(1)}_{\sqrt{\delta^2+\varepsilon}}(r-\sigma)\gamma^{(3)}_{\sqrt{1+\varepsilon}}(z) + \eta_\delta\gamma^{(4)}_{\sqrt{4+\varepsilon}}(r,z). \tag{C.23}$$

For fixed δ, the means are fixed and the covariance matrices in (C.23) range over a compact subset of the positive-definite matrices. Consequently, the family has common L^∞ and Gaussian-moment bounds. Its Riesz-potential bounds and, when $\mathsf{s}=0$, its absolute logarithmic-energy bounds are uniform as well: each relevant integral is a finite continuous function of the covariance parameters on this compact set. For $0<\mathsf{s}<1$, nonnegativity of g and the uniform potential bound also control the exponential negative-potential term in the partition estimate. For $\mathsf{s}=0$,

$$h_{\mu_{(\varepsilon)}}(x) \geq -\log(1+|x|) - C_\delta, \qquad 0\leq\varepsilon\leq 1, \tag{C.24}$$

so the uniform Gaussian moments control that exponential term as well. In particular, for each fixed $b>0$,

$$\sup_{0\leq\varepsilon\leq 1}\left\{\|\widetilde{\mathsf{g}}_{\mu_{(\varepsilon)}}\|_{L^2((\mu_{(\varepsilon)})^{\otimes 2})} + \|h_{\mu_{(\varepsilon)}} - 2I_{\mu_{(\varepsilon)}}\|_{L^2(\mu_{(\varepsilon)})} + \int e^{-b(h_{\mu_{(\varepsilon)}}-2I_{\mu_{(\varepsilon)}})}\,\mathrm{d}\mu_{(\varepsilon)}\right\}<\infty. \tag{C.25}$$

Here, $\widetilde{\mathsf{g}}_{\mu_{(\varepsilon)}}$ is the kernel centered with respect to $\mu_{(\varepsilon)}$ as in (2.5). Tracking the constants in [DGR26a, Appendix A], the large-N threshold and the partition bound in [DGR26a, Theorem 2.2(i)] may therefore be chosen uniformly for $0\leq\varepsilon\leq 1$, both for $b=\beta$ and $b=2\beta$. For the finitely many particle numbers below that threshold, continuity of $\varepsilon\mapsto \mathsf{K}_{N,b}(\mu_{(\varepsilon)})$ on $[0,1]$, obtained by dominated convergence, gives the same uniform conclusion. Also, $I_{(\varepsilon)}\to I_\mu$, and Jensen gives $\mathsf{K}_{N,b,(\varepsilon)}\geq e^{bI_{(\varepsilon)}}$. Combining these facts with (C.22), we obtain, for every fixed $b>0$,

$$1 \leq e^{-bI_{(\varepsilon)}}\mathsf{K}_{N,b,(\varepsilon)}(\mu) \leq C_{\delta,b}, \qquad N\geq 2, \quad 0<\varepsilon\leq 1. \tag{C.26}$$

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Entropy contraction and change of reference. Entropy contraction under the smooth Markov semigroup, as in [BGL14, Section 5.2], gives

$$\mathsf{H}(P_{N,t}^{(\varepsilon)} \mid \Gamma_{N,(\varepsilon)}) \le \log \mathsf{K}_{N,\beta,(\varepsilon)} - \beta I_{(\varepsilon)}. \tag{C.27}$$

For $G_{N,(\varepsilon)} := \mathrm{d}\Gamma_{N,(\varepsilon)}/\,\mathrm{d}\mu^{\otimes N}$, the partition-function identity is

$$\int G_{N,(\varepsilon)}^2 \,\mathrm{d}\mu^{\otimes N} = \frac{\mathsf{K}_{N,2\beta,(\varepsilon)}(\mu)}{\mathsf{K}_{N,\beta,(\varepsilon)}(\mu)^2}. \tag{C.28}$$

Applying the Donsker–Varadhan lemma to $\log G_{N,(\varepsilon)}$ with reference $\Gamma_{N,(\varepsilon)}$, and using (C.27)–(C.28), gives

$$\mathsf{H}(P_{N,t}^{(\varepsilon)} \mid \mu^{\otimes N}) \le \log \mathsf{K}_{N,2\beta,(\varepsilon)}(\mu) - 2\beta I_{(\varepsilon)}. \tag{C.29}$$

For fixed δ, these bounds are uniform in $t \ge 0$, $N \ge 2$, and $0 < \varepsilon \le 1$.

Passage to the singular dynamics. At fixed N, the normalized Gibbs densities converge in total variation. For $\mathsf{s} > 0$, nonnegativity of the mollified pair potential and boundedness of the one-body convolution give a common multiple of $\mu^{\otimes N}$. For $\mathsf{s} = 0$,

$$\mathsf{g}_{(\varepsilon)}(x) \ge -C - \log(1 + |x|), \qquad 0 < \varepsilon \le 1, \tag{C.30}$$

by Jensen applied to $\log|x + Y|$. This gives a fixed polynomial times a Gaussian as a majorant of the unnormalized density. Off-diagonal convergence, the nullity of the collision set under the product measure, and the positive lower normalizer bound prove the claim. Together with (C.18), joint weak lower semicontinuity of relative entropy [vEH14, Theorem 19] gives, for every fixed N and t,

$$\mathsf{H}(P_{N,t} \mid \Gamma_N) + \mathsf{H}(P_{N,t} \mid \mu^{\otimes N}) \le \liminf_{\varepsilon \downarrow 0} \Big(\mathsf{H}(P_{N,t}^{(\varepsilon)} \mid \Gamma_{N,(\varepsilon)}) + \mathsf{H}(P_{N,t}^{(\varepsilon)} \mid \mu^{\otimes N}) \Big) \le C_\delta. \tag{C.31}$$

The constant is independent of N and t, so taking their suprema proves (C.19). □

Lemma C.3 (Finiteness at fixed particle number)**.** *For fixed δ, N and $T < \infty$, there is a finite constant $C_{\delta,N,T}$ such that*

$$P_{N,t} \le C_{\delta,N,T}\, (\gamma_2^{(4)})^{\otimes N}, \qquad 0 \le t \le T. \tag{C.32}$$

Consequently, the fixed-N chi-square divergences relative to $\mu_\delta^{\otimes N}$ are finite.

Proof. For $0 \le \mathsf{s} < 1$ and $\mathsf{d} = 4$,

$$\Delta \mathsf{g}(x) = (\mathsf{s} - 2)|x|^{-\mathsf{s}-2} \le 0 \quad (x \ne 0). \tag{C.33}$$

This identity also holds distributionally. Use the approximation (C.16) and set

$$\widetilde{V}_{(\varepsilon)}(x) := V_{(\varepsilon)}(x) - \frac{|x|^2}{8\beta}. \tag{C.34}$$

For fixed δ, uniformly in $0 < \varepsilon \le 1$,

$$\|\Delta \widetilde{V}_{(\varepsilon)}\|_\infty + \|x \cdot \nabla \widetilde{V}_{(\varepsilon)}(x)\|_\infty \le C_\delta. \tag{C.35}$$

For the logarithmic-density term this follows from the rapid decay in (C.7). For the convolution term, bounded derivatives follow from local integrability of $\nabla^2 \mathsf{g}$ and the common Gaussian bounds on $\mu * p_\varepsilon$. For the factor of x, use

$$x \cdot (\nabla \mathsf{g} * \rho)(x) = ((y \cdot \nabla \mathsf{g}(y)) * \rho)(x) + \int y \cdot \nabla \mathsf{g}(x - y)\rho(y)\, \mathrm{d}y. \tag{C.36}$$

The first convolution is $-|x|^{-\mathsf{s}} * \rho$, or -1 when $\mathsf{s} = 0$. The second is bounded by splitting the force into its local L^1 part and bounded far part and using the uniform Gaussian bounds on $|y|\rho(y)$.

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A Gaussian supersolution. Set $q := \gamma_2^{(4)}$ and compare the smooth particle density with $q^{\otimes N}$. Applying the forward generator to this reference density and dividing by $q^{\otimes N}$ gives the scalar source

$$S_{N,(\varepsilon)}(X_N) := \sum_i \left(\Delta \widetilde{V}_{(\varepsilon)}(x_i) - \frac{1}{4} x_i \cdot \nabla \widetilde{V}_{(\varepsilon)}(x_i) \right) + \frac{1}{N} \sum_{i<j} \left(2\Delta \mathsf{g}_{(\varepsilon)}(x_i - x_j) - \frac{1}{4}(x_i - x_j) \cdot \nabla \mathsf{g}_{(\varepsilon)}(x_i - x_j) \right). \tag{C.37}$$

The Ornstein–Uhlenbeck part cancels exactly because q has covariance $4\mathrm{Id}$ and the quadratic part of $V_{(\varepsilon)}$ is $|x|^2/(8\beta)$.

There is a finite constant C_{s} such that, off zero,

$$2\Delta \mathsf{g}(x) - \tfrac{1}{4} x \cdot \nabla \mathsf{g}(x) = 2(\mathsf{s}-2)|x|^{-\mathsf{s}-2} + \tfrac{1}{4}|x|^{-\mathsf{s}} \le C_{\mathsf{s}}. \tag{C.38}$$

When $\mathsf{s} = 0$, the last factor $|x|^{-\mathsf{s}}$ equals one. Gaussian integration by parts gives

$$x \cdot \nabla \mathsf{g}_{(\varepsilon)} = (x \cdot \nabla \mathsf{g}) * p_\varepsilon - 2\varepsilon \Delta \mathsf{g}_{(\varepsilon)}. \tag{C.39}$$

Since $\Delta \mathsf{g} \le 0$ distributionally, (C.39) gives

$$2\Delta \mathsf{g}_{(\varepsilon)} - \tfrac{1}{4} x \cdot \nabla \mathsf{g}_{(\varepsilon)} = \left(2\Delta \mathsf{g} - \tfrac{1}{4} x \cdot \nabla \mathsf{g} \right) * p_\varepsilon + \tfrac{\varepsilon}{2} \Delta \mathsf{g}_{(\varepsilon)} \le C_{\mathsf{s}}. \tag{C.40}$$

Thus, (C.35) implies $S_{N,(\varepsilon)} \le C_\delta N$, uniformly in the mollification parameter.

At a fixed cutoff the quotient equation has a globally Lipschitz auxiliary drift, at most linear growth, and a bounded scalar source. Its initial datum is at most $(\sup \mu/q)^N$. The Feynman–Kac formula (see, e.g., [Øks03, Section 8.2]) gives

$$P_{N,t}^{(\varepsilon)} \le (\sup \mu/q)^N e^{C_\delta N t}\, q^{\otimes N}. \tag{C.41}$$

The Feynman–Kac solution is bounded with bounded first spatial derivative at each fixed cutoff: the initial quotient and the source have bounded first derivatives, and the auxiliary drift has bounded derivative. Multiplying it by $q^{\otimes N}$ and integrating the forward equation against expanding spatial cutoffs proves conservation of mass; the boundary terms vanish by Gaussian decay. Uniqueness for the smooth particle forward equation therefore identifies this solution with the actual quotient. The bound passes to the singular law by weak convergence and nonnegative continuous compactly supported tests. Finally $\mu \ge \eta_\delta q$, so the density ratio $P_{N,t}/\mu^{\otimes N}$ is essentially bounded at every fixed N, T. This proves the claim. No uniform bound in N is asserted by this lemma. □

C.3. **Fluctuations from independent initialization.** Continue to fix δ and suppress it from the notation in this subsection. Set

$$Lf := \beta^{-1}(\Delta f + \nabla \log \mu \cdot \nabla f), \qquad \mathcal{B}f(x) := \int \nabla \mathsf{g}(x-y) \cdot \nabla f(y)\, \mu(\mathrm{d}y), \qquad \mathcal{A} := L + \mathcal{B}. \tag{C.42}$$

The operator $\mathcal{A}$ is the adjoint of the linearized mean-field equation at μ. We denote its evolution by U_t, so that $\partial_t U_t \phi = \mathcal{A} U_t \phi$ and $U_0 \phi = \phi$. The next proposition separates the fluctuations of the independent initial configuration from those generated by the Brownian motions. The required adjoint regularity will be verified for the test used in the counterexample in lemma C.6.

Proposition C.4 (Fluctuations from iid initial data)**.** *Suppose $f_r := U_{T-r}\phi$, $0 \le r \le T$, is bounded with bounded first and second spatial derivatives, jointly continuous in time and space, and solves the adjoint equation classically locally. At-most-linear growth of $\partial_r f_r$ suffices. Then the initial law* (1.38) *gives*

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (\phi(x_i(T)) - \mu(\phi)) \longrightarrow \mathcal{N}(0, W_T), \tag{C.43}$$

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where

$$W_T := \mathrm{Var}_\mu(U_T\phi) + 2\beta^{-1}\int_0^T \int |\nabla U_r\phi|^2 \,\mathrm{d}\mu\,\mathrm{d}r. \tag{C.44}$$

Proof. For a bounded test f, set $S_N^t(f) := \sqrt{N}(\mu_N^t(f) - \mu(f))$. The exact finite-particle calculation, including the confinement, gives

$$S_N^T(\phi) = S_N^0(f_0) + M_N(T) - \frac{1}{2\sqrt{N}}\int_0^T N\mathsf{A}_1(\nabla f_r, X_N(r), \mu)\,\mathrm{d}r, \tag{C.45}$$

$$M_N(t) := \sqrt{\frac{2}{\beta N}}\sum_i \int_0^t \nabla f_r(x_i(r))\cdot \mathrm{d}B_i(r), \tag{C.46}$$

where $\mu_N^t := N^{-1}\sum_{i=1}^N \delta_{x_i(t)}$ and A_1 is the transport commutator in (2.2). In particular, $N\mathsf{A}_1/2$ has the normalization C_N used in Section 2. This is the finite-particle adjoint computation of [DGR26a, Proposition 8.2], with the additional one-body drift $-\nabla V\cdot\nabla f$. The computation is valid for the independent initialization used here.

By the exponential commutator estimate [DGR26a, Theorem 2.8], the bounded Hessians of f_r, and lemma C.2,

$$\sup_{N,r\le T} \mathbb{E}_{P_{N,r}}[N|\mathsf{A}_1(\nabla f_r, X_N, \mu)|] < \infty. \tag{C.47}$$

Here the estimate for both signs first bounds the exponential of the absolute value under Γ_N; the Donsker–Varadhan lemma transfers it to $P_{N,r}$. Consequently,

$$\mathbb{E}\left[\left|\frac{1}{2\sqrt{N}}\int_0^T N\mathsf{A}_1(\nabla f_r, X_N(r), \mu)\,\mathrm{d}r\right|\right] \le \frac{C_{\delta,T}}{\sqrt{N}} \xrightarrow[N\to\infty]{} 0. \tag{C.48}$$

To justify (C.45), first apply Itô's formula up to exit from collision-free compact sets. The estimate (C.47) controls the time integral of the commutator in L^1. The at-most-linear terms are integrable by lemma C.2 and the Gaussian exponential moments of μ. Finally, boundedness of ∇f_r gives L^2 convergence of the stopped stochastic integrals. These estimates justify removing the stopping times.

The martingale and the initial fluctuation. For a bounded test ζ, the product-measure calculation (8.63)–(8.64) applies to the centered function $\zeta - \mu(\zeta)$ on $(\mathbb{R}^4, \mu)$. Combined with lemma C.2 and the Donsker–Varadhan lemma, it yields

$$\mathbb{E}_{P_{N,r}}[|\mu_N(\zeta) - \mu(\zeta)|] \le C_\delta\,\mathrm{osc}(\zeta)N^{-1/2}, \qquad r \ge 0. \tag{C.49}$$

Set $v_T := 2\beta^{-1}\int_0^T \mu(|\nabla f_r|^2)\,\mathrm{d}r$. Applying (C.49) to $|\nabla f_r|^2$ in the predictable quadratic variation of (C.46) gives

$$\mathbb{E}[|\langle M_N\rangle_T - v_T|] \le 2\beta^{-1}\int_0^T \mathbb{E}[|\mu_N^r(|\nabla f_r|^2) - \mu(|\nabla f_r|^2)|]\,\mathrm{d}r \le \frac{C_{\delta,T}}{\sqrt{N}} \xrightarrow[N\to\infty]{} 0. \tag{C.50}$$

We also have the deterministic bound $\langle M_N\rangle_T \le 2\beta^{-1}T\sup_{0\le r\le T}\|\nabla f_r\|_\infty^2$. Let $\mathcal{F}_0 := \sigma(X_N(0))$, and use the filtration generated by the initial configuration and the Brownian motions. For each real θ, the exponential $t \mapsto \exp(i\theta M_N(t) + \theta^2\langle M_N\rangle_t/2)$ is a bounded-in-modulus true complex martingale on $[0,T]$. Therefore,

$$\mathbb{E}\left[\exp\left(i\theta M_N(T) + \frac{\theta^2}{2}\langle M_N\rangle_T\right)\middle|\mathcal{F}_0\right] = 1. \tag{C.51}$$

For every bounded $\mathcal{F}_0$-measurable random variable Z_N with $|Z_N| \le 1$, (C.51) and (C.50) imply

$$\mathbb{E}[Z_N e^{i\theta M_N(T)}] - e^{-\theta^2 v_T/2}\mathbb{E}[Z_N] \longrightarrow 0. \tag{C.52}$$

The calculation in (8.78) applies with this $\mathcal{F}_0$-measurable multiplier and bounds the difference by $C\mathbb{E}[|\langle M_N\rangle_T - v_T|]$.

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The independent-sample central limit theorem gives variance $\mathrm{Var}_\mu(f_0)$ for the initial statistic. Taking $Z_N = e^{i\theta S_N^0(f_0)}$ in (C.52) factors the limiting characteristic function into the initial Gaussian factor and the Brownian Gaussian factor. Together with (C.48), this proves (C.43)–(C.44). □

C.4. **Amplification and variance comparison.** We now restore the dependence on δ: L_δ and $\mathcal{B}_\delta$ are the operators in (C.42) with $\mu = \mu_\delta$ and $\beta = \beta_\delta$, and U_t^δ is the corresponding evolution. We let $\delta \downarrow 0$. The test below has product variance of order one, whereas its image under $\mathcal{B}_\delta$ has $L^2(\mu_\delta)$ norm of order δ^{-1}. This difference produces the variance comparison at time $T_\delta = \sqrt{\delta}$.

Choose an even smooth cutoff χ equal to one on $[-1/4, 1/4]$ and supported in $(-1/3, 1/3)$. Set

$$\psi(r) := (r-1)\chi(r-1) + (r+1)\chi(r+1), \qquad \phi_\delta(r,z) := \delta^{-1}\psi(r). \tag{C.53}$$

The function ψ is odd. In each narrow component, ϕ_δ measures displacement within that component in units of its width; it does not count the distance between the two components. We have

$$\mathrm{Var}_{\mu_\delta}(\phi_\delta) \longrightarrow 1 \qquad \text{as } \delta \downarrow 0. \tag{C.54}$$

The convergence follows from the Gaussian mixture (C.2) and the Gaussian tail bound [BGL14, Section 5.4] in the two neighborhoods where $\psi(r) = r \mp 1$. The broad component contributes at most $C\eta_\delta/\delta^2 \to 0$. Symmetry gives zero mean.

Lemma C.5 (Interaction amplification)**.** *There is a constant $c_{\mathsf{s}} > 0$ such that*

$$\delta^2 \|\mathcal{B}_\delta \phi_\delta\|^2_{L^2(\mu_\delta)} \longrightarrow c_{\mathsf{s}}^2, \tag{C.55}$$

where

$$c_{\mathsf{s}}^2 := \int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} \frac{\gamma_1^{(3)}(y)\, \mathrm{d}y}{(4+|z-y|^2)^{(\mathsf{s}+2)/2}} \right)^2 \gamma_1^{(3)}(z)\, \mathrm{d}z > 0. \tag{C.56}$$

Moreover, uniformly in small δ,

$$\|\mathcal{B}_\delta f\|_\infty + \|\nabla \mathcal{B}_\delta f\|_\infty \le C_{\mathsf{s}} \|\nabla f\|_\infty. \tag{C.57}$$

Proof. The bounds in (C.57) follow from (C.6). Write

$$H_\delta(x) := \delta \mathcal{B}_\delta \phi_\delta(x) = \int \partial_1 \mathsf{g}(x-y)\psi'(y_1)\, \mu_\delta(\mathrm{d}y). \tag{C.58}$$

These functions and their gradients are uniformly bounded by (C.6). Define

$$H_0(r,z) := \frac{1}{2} \int (\partial_1 \mathsf{g}(r-1, z-y) + \partial_1 \mathsf{g}(r+1, z-y)) \gamma_1^{(3)}(y)\, \mathrm{d}y. \tag{C.59}$$

The mean value theorem and (C.12) with $j = 2$ control the error from replacing each narrow Gaussian by its center. Since $\psi' = 1$ for $|r - \sigma| \le 1/4$, $\sigma = \pm 1$, the remaining errors are bounded by the Gaussian tails and the weight of the broad component. Thus, for some $c > 0$,

$$\begin{aligned} \|H_\delta - H_0\|_\infty &\le C_{\mathsf{s}} \left[\sum_{\sigma = \pm 1} \int_{\mathbb{R}} (|r-\sigma| + \mathbf{1}_{\{|r-\sigma| \ge 1/4\}}) \gamma_\delta^{(1)}(r-\sigma)\, \mathrm{d}r + \eta_\delta \right] \\ &\le C_{\mathsf{s}} \left(\delta + e^{-c/\delta^2} + \eta_\delta \right) \underset{\delta \downarrow 0}{\longrightarrow} 0. \end{aligned} \tag{C.60}$$

Also, $\mu_\delta \rightharpoonup \frac{1}{2}(\delta_{-1} + \delta_1) \otimes \gamma_1^{(3)}$. The limit H_0 is bounded and continuous. On the two limiting hyperplanes,

$$H_0(1,z) = -\int (4+|z-y|^2)^{-(\mathsf{s}+2)/2} \gamma_1^{(3)}(y)\, \mathrm{d}y, \qquad H_0(-1,z) = -H_0(1,z). \tag{C.61}$$

Thus,

$$\int |H_\delta|^2\, \mathrm{d}\mu_\delta \longrightarrow \frac{1}{2} \int_{\mathbb{R}^3} \left(|H_0(-1,z)|^2 + |H_0(1,z)|^2 \right) \gamma_1^{(3)}(z)\, \mathrm{d}z = c_{\mathsf{s}}^2 > 0. \tag{C.62}$$

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The uniform convergence in (C.60) justifies replacing H_δ by H_0 before taking the weak limit of the measures. Since $H_\delta = \delta \mathcal{B}_\delta \phi_\delta$, this proves (C.55). □

Lemma C.6 (Backward evolution)**.** *For $0 \le t \le 1$, the adjoint evolution $U_t^\delta \phi_\delta$ exists in the regularity class of proposition C.4. There is a constant independent of small δ such that*

$$\|\nabla U_t^\delta f\|_\infty \le C\|\nabla f\|_\infty, \qquad \|\nabla(U_t^\delta \phi_\delta - \phi_\delta)\|_\infty \le Ct/\delta, \tag{C.63}$$

for the bounded smooth functions used below, and

$$\|U_t^\delta \phi_\delta - \phi_\delta - t\mathcal{B}_\delta \phi_\delta\|_{L^2(\mu_\delta)} \le C\left(t\delta^3 + \frac{t^2}{\delta} + \delta^2 t^{3/2}\right). \tag{C.64}$$

Proof. Let P_t^δ be the Markov semigroup of L_δ. By (C.4), its drift has Lipschitz constant at most $C\beta_\delta^{-1}\delta^{-4} \le C$. Differentiating the stochastic flow gives

$$\|\nabla P_t^\delta f\|_\infty \le e^{Ct}\|\nabla f\|_\infty. \tag{C.65}$$

The semigroup preserves μ_δ, is self-adjoint on $L^2(\mu_\delta)$, and has Dirichlet form $\beta_\delta^{-1}\int |\nabla f|^2 \,\mathrm{d}\mu_\delta$. Its spectral theorem gives

$$\|(P_t^\delta - \mathrm{Id})f\|_{L^2(\mu_\delta)} \le \sqrt{\beta_\delta^{-1} t}\,\|\nabla f\|_{L^2(\mu_\delta)}. \tag{C.66}$$

Indeed, $(1 - e^{-a})^2 \le a$ for $a \ge 0$.

Construction and regularity. The Volterra equation

$$U_t^\delta f = P_t^\delta f + \int_0^t P_{t-r}^\delta \mathcal{B}_\delta U_r^\delta f \,\mathrm{d}r \tag{C.67}$$

is solved on the concrete class needed below as follows. For fixed δ, equip $C_b^1(\mathbb{R}^4)$ with

$$\|f\|_{C_b^1} := \|f\|_\infty + \|\nabla f\|_\infty. \tag{C.68}$$

Starting from the bounded smooth test ϕ_δ in (C.53), define

$$u_t^{(0)} := P_t^\delta \phi_\delta, \qquad u_t^{(n+1)} := P_t^\delta \phi_\delta + \int_0^t P_{t-r}^\delta \mathcal{B}_\delta u_r^{(n)} \,\mathrm{d}r. \tag{C.69}$$

(C.57) and (C.65) imply, for $0 \le t \le \tau$,

$$\sup_{0\le t\le\tau} \|u_t^{(n+1)} - u_t^{(n)}\|_{C_b^1} \le C\tau \sup_{0\le t\le\tau} \|u_t^{(n)} - u_t^{(n-1)}\|_{C_b^1}. \tag{C.70}$$

Choosing τ so that $C\tau < 1$ gives convergence in the bounded-function plus bounded-gradient norm. Repeating the construction from the final value on each interval gives a unique mild solution on $[0, 1]$ and the first inequality in (C.63).

The test ϕ_δ is smooth, bounded, and compactly supported in the first coordinate. Stochastic-flow continuity gives $P_t^\delta \phi_\delta \to \phi_\delta$ locally uniformly, together with its first derivative, as $t \downarrow 0$. The integral term in (C.67) is $O(t)$ in the norm (C.68); hence the mild solution has the required initial trace on this concrete class.

For fixed δ, $\sup_{x\in\mathbb{R}^4} \int |\nabla^3 \mathsf{g}(x - y)|\,\mu_\delta(\mathrm{d}y) < \infty$, since $\mathsf{s} + 3 < 4$ and μ_δ is bounded. Thus, for the bounded smooth functions under consideration,

$$\|\nabla^2 \mathcal{B}_\delta f\|_\infty \le C_\delta \|\nabla f\|_\infty, \qquad \|\nabla^2 P_t^\delta f\|_\infty \le C_\delta\left(\|\nabla f\|_\infty + \|\nabla^2 f\|_\infty\right), \quad 0 \le t \le 1. \tag{C.71}$$

The second estimate follows by differentiating the stochastic flow twice; the first two derivatives of its drift are bounded at fixed δ. Applying these estimates to differences of successive Picard iterates gives

$$\sup_{0\le t\le\tau} \|\nabla^2(u_t^{(n+1)} - u_t^{(n)})\|_\infty \le C_\delta \tau \sup_{0\le t\le\tau} \|u_t^{(n)} - u_t^{(n-1)}\|_{C_b^1}. \tag{C.72}$$

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The right-hand side is summable in n by (C.70). Hence the second derivatives converge uniformly on each short interval. Iteration gives bounded second derivatives on $[0,1]$, with a bound allowed to depend on δ.

Passing to the limit in (C.69) and applying local parabolic regularity [Lun95, Chapter 5] gives a classical solution on $(0,1]\times\mathbb{R}^4$. The differentiated flow is locally continuous at time zero, and the integral term is $O(t)$ also in the bounded-second-derivative norm at fixed δ. Thus, the required spatial derivatives have their initial trace. The time derivative has at most linear growth by (C.42), giving the regularity in proposition C.4.

Estimates uniform in δ. Since ψ has fixed compact support and bounded derivatives of every order, (C.5) and (C.4) imply

$$\|L_\delta\phi_\delta\|_\infty \le C\delta^3, \qquad \|\nabla L_\delta\phi_\delta\|_\infty \le C\delta. \tag{C.73}$$

The identity $P_t^\delta\phi_\delta-\phi_\delta=\int_0^t P_r^\delta L_\delta\phi_\delta\,\mathrm{d}r$ follows from Itô; all terms on the right are bounded. Taking gradients and using (C.65) gives a bound $Ct\delta$. Subtract ϕ_δ in (C.67), apply (C.57), and use the already proved gradient estimate for $U_r^\delta\phi_\delta$. This proves the second inequality in (C.63).

Finally, subtract $\phi_\delta+t\mathcal{B}_\delta\phi_\delta$ in (C.67) to obtain

$$U_t^\delta\phi_\delta-\phi_\delta-t\mathcal{B}_\delta\phi_\delta=(P_t^\delta-\mathrm{Id})\phi_\delta+\int_0^t(P_{t-r}^\delta\mathcal{B}_\delta U_r^\delta-\mathcal{B}_\delta)\phi_\delta\,\mathrm{d}r. \tag{C.74}$$

Adding and subtracting $P_{t-r}^\delta\mathcal{B}_\delta\phi_\delta$ inside the integral separates the right-hand side into three terms. Their $L^2(\mu_\delta)$ norms are bounded, respectively, by $Ct\delta^3$, Ct^2/δ, and $C\sqrt{\beta_\delta^{-1}}\,t^{3/2}/\delta$. For the last bound, use (C.66) and (C.57); for the second, use L^2 contraction, (C.57), and (C.63). Since $\sqrt{\beta_\delta^{-1}}=\delta^3$, this is (C.64). □

Proposition C.7 (Variance comparison)**.** *Let W_δ be the variance (C.44) at $T=T_\delta$ for $\phi=\phi_\delta$. Then*

$$\lim_{\delta\downarrow0}\mathrm{Var}_{\mu_\delta}(\phi_\delta)=1, \qquad \lim_{\delta\downarrow0}\delta W_\delta=c_{\mathsf{s}}^2>0. \tag{C.75}$$

In particular, $W_\delta>2\,\mathrm{Var}_{\mu_\delta}(\phi_\delta)$ for every sufficiently small δ.

Proof. Reflection of the first coordinate preserves μ_δ, L_δ, and $\mathcal{B}_\delta$, so $U_t^\delta\phi_\delta$ is odd and has zero μ_δ mean. At $T_\delta=\sqrt{\delta}$, the right-hand side of (C.64) is bounded independently of small δ. Since $\sqrt{\delta}T_\delta\mathcal{B}_\delta\phi_\delta=H_\delta$,

$$\|\sqrt{\delta}\,U_{T_\delta}^\delta\phi_\delta-H_\delta\|_{L^2(\mu_\delta)}\le\sqrt{\delta}\,\|\phi_\delta\|_{L^2(\mu_\delta)}+C\sqrt{\delta}\xrightarrow[\delta\downarrow0]{}0. \tag{C.76}$$

Here, (C.54) bounds the norm of ϕ_δ, and lemma C.5 gives $\|H_\delta\|_{L^2(\mu_\delta)}\to c_{\mathsf{s}}$. Consequently, $\delta\,\mathrm{Var}_{\mu_\delta}(U_{T_\delta}^\delta\phi_\delta)\to c_{\mathsf{s}}^2$. The noise part of (C.44) is bounded by

$$2\beta_\delta^{-1}\int_0^{T_\delta}\|\nabla U_r^\delta\phi_\delta\|_{L^2(\mu_\delta)}^2\,\mathrm{d}r\le C\beta_\delta^{-1}T_\delta\delta^{-2}=C\delta^{9/2}. \tag{C.77}$$

This proves (C.75). □

C.5. **Completion and marginal entropy.**

Proof of proposition 1.6. The background and entropy assertions follow from lemmas C.1 and C.2 and proposition B.1, and fixed-particle chi-square finiteness follows from lemma C.3. Fix δ small enough that $W_\delta>2\,\mathrm{Var}_{\mu_\delta}(\phi_\delta)$. All subsequent limits are in N, with $\mathsf{s},\delta,\beta_\delta,V_\delta,\mu_\delta,T_\delta$, and the test ϕ_δ fixed.

The product reference satisfies the independent-sample central limit theorem with variance $\mathrm{Var}_{\mu_\delta}(\phi_\delta)$. By proposition C.4 and lemma C.6, the particle law satisfies the central limit theorem

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with variance W_δ. Apply the same statistic $X_N \mapsto N^{-1/2}\sum_i(\phi_\delta(x_i) - \mu_\delta(\phi_\delta))$ to both laws. Data processing and joint weak lower semicontinuity [vEH14, Theorem 19 and (6)] give

$$\liminf_{N\to\infty} \chi^2(P_{N,T_\delta} \mid \mu_\delta^{\otimes N}) \geq \chi^2(\mathcal{N}(0, W_\delta) \mid \mathcal{N}(0, \mathrm{Var}_{\mu_\delta}(\phi_\delta))). \tag{C.78}$$

For $W, v > 0$, direct substitution of the Gaussian densities gives

$$1 + \chi^2(\mathcal{N}(0, W) \mid \mathcal{N}(0, v)) = \frac{\sqrt{v}}{\sqrt{2\pi}\, W} \int_{\mathbb{R}} \exp\left[\left(\frac{1}{2v} - \frac{1}{W}\right) x^2\right] \mathrm{d}x. \tag{C.79}$$

With $W = W_\delta$ and $v = \mathrm{Var}_{\mu_\delta}(\phi_\delta)$, the coefficient of x^2 is positive because $W_\delta > 2v$. The integral, and hence the right-hand side of (C.78), is infinite. This proves divergence of the whole sequence in (1.40). □

Remark C.8 (Marginal entropy from independent initial data). The same example is locally chaotic in relative entropy, in the sense that

$$\sup_{t\geq 0} \mathsf{H}(P_{N,t}^{(k)} \mid \mu_\delta^{\otimes k}) \leq C_\delta \frac{k}{N}, \qquad 1 \leq k \leq N. \tag{C.80}$$

To see this, for fixed t, set $H_j := \mathsf{H}(P_{N,t}^{(j)} \mid \mu_\delta^{\otimes j})$ and $H_0 := 0$. The chain rule, data processing, and exchangeability give, for $2 \leq j < N$,

$$H_{j+1} - H_j = H_1 + \mathsf{H}(P_{N,t}^{(j+1)} \mid P_{N,t}^{(j)} \otimes P_{N,t}^{(1)}) \geq H_j - H_{j-1}. \tag{C.81}$$

For $j = 1$, the same inequality follows from $\mathsf{H}(P_{N,t}^{(2)} \mid (P_{N,t}^{(1)})^{\otimes 2}) \geq 0$. Thus, averaging these nondecreasing increments gives

$$\frac{H_k}{k} = \frac{1}{k}\sum_{j=1}^{k}(H_j - H_{j-1}) \leq \frac{1}{N}\sum_{j=1}^{N}(H_j - H_{j-1}) = \frac{H_N}{N}. \tag{C.82}$$

Therefore, (1.39) proves (C.80). In particular, every sequence $k_N = o(N)$ satisfies

$$\sup_{t\geq 0} \mathsf{H}(P_{N,t}^{(k_N)} \mid \mu_\delta^{\otimes k_N}) \leq C_\delta \frac{k_N}{N} \xrightarrow[N\to\infty]{} 0. \tag{C.83}$$

The example rules out a bound $\chi^2(P_{N,t}^{(k)} \mid \mu_\delta^{\otimes k}) \leq C_\delta k^2/N^2$ holding simultaneously for all $1 \leq k \leq N$, since it fails at $k = N$ and $t = T_\delta$. It does not settle the sharp chi-square or relative-entropy rate k^2/N^2, or the total-variation rate k/N, for fixed marginals or for sublinear growing marginals.

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